

Types of pseudo-scope: variation and non-local QR

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Abstract

This dissertation investigates the scope-taking abilities of universal quantifiers and the syntactic constraints that govern them. Building on two standard assumptions from the literature—that universal quantifiers take scope via scope-shifting mechanisms, e.g., Quantifier Raising (QR), and that scope-taking is clause-bounded—I explore two syntactic environments that have been argued to be scope islands for universal quantification: relative clauses and finite complement clauses. I look at examples from the literature where universal quantifiers are embedded within these syntactic environments and, yet, we still observe readings with apparent wide scope universal quantification. Crucially, these apparent wide scope universal readings are diagnosed through ‘variation’ of an indefinite or definite article. In this thesis, I challenge the implicit assumption that variation reliably diagnoses wide scope universal force. To this end, the central claim of the thesis is that variation-based arguments can be confounded by alternative semantic mechanisms and so variation, whether of indefinites or definites, cannot be straightforwardly equated with wide-scope universal force. If this line of thought is true, readings of sentences involving variation do not furnish straightforward arguments against locality constraints on scope-taking. This dissertation thus argues (i) for a more nuanced view of the scope-variation relationship, one that recognizes multiple compositional pathways capable of inducing variation, and (ii) for greater caution when using variation to argue against syntactic locality constraints on scope-shifting.

Abrégé

Cette thèse examine les capacités des quantificateurs universels à prendre de la portée, ainsi que les contraintes syntaxiques qui les régissent. En m'appuyant sur deux hypothèses standard de la littérature—que les quantificateurs universels acquièrent leur portée à travers des mécanismes de déplacement de portée, notamment la Montée des Quantificateurs (MQ), et que la prise de portée est bornée à la proposition—j'explore deux environnements syntaxiques qui ont été avancés comme des îlots de portée pour la quantification universelle : les propositions relatives et les propositions complétives finies. J'examine des exemples tirés de la littérature dans lesquels des quantificateurs universels sont enchâssés dans ces environnements syntaxiques et où l'on observe néanmoins des lectures avec une quantification universelle apparemment à portée large. Fait cruciale, ces lectures à portée large universelle apparente sont diagnostiquées par la « variation » d'un article indéfini ou défini. Dans cette thèse, je remets en question l'hypothèse implicite selon laquelle la variation diagnostique de façon fiable la force universelle à portée large. À cette fin, la thèse centrale défendue ici est que les arguments fondés sur la variation peuvent être remis en cause par des mécanismes sémantiques alternatifs, de sorte que la variation—qu'elle porte sur des indéfinis ou des définis—ne peut pas être directement assimilée à une force universelle à portée large. Si ce raisonnement est juste, les lectures à variation ne constituent pas des arguments directs contre les contraintes de localité sur la prise de portée. Cette thèse défend ainsi (i) une conception plus nuancée de la relation entre portée et variation, qui reconnaît plusieurs voies compositionnelles susceptibles d'induire de la variation, et (ii) une plus grande prudence dans l'utilisation de la variation comme argument contre les contraintes de localité syntaxiques sur le déplacement de portée.

Contribution

This thesis is the original work of the author. It is considered original scholarship and provides distinct contributions to knowledge. All of the chapters were solely authored, with extensive written feedback and comments from Bernhard Schwarz and Michael Wagner.

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Introduction

0.1 Quantifiers and scope

The syntactic and semantic representation of quantificational expressions is one of the most intensively studied topics at the syntax–semantics interface. In particular, determining a quantifier’s *scope*—the part of the sentence in which it makes its semantic contribution—is integral to assessing the truth conditions of a sentence.¹ A central and well-documented property of quantificational expressions is that, when two or more occur in a sentence, the sentence typically admits more than one interpretation, depending on their relative scope. This phenomenon is known as a *scope ambiguity*, and a canonical illustration is the sentence in (1), where a singular indefinite DP and a universal DP scopally interact.

- (1) A proofreader checked each manuscript.
- a. *surface scope reading*: there is a single proofreader who checked all the manuscripts
 - b. *scope-shifted reading*: for every manuscript *x*, there is a potentially different proofread who checked *x*

The reading in (1-a) is referred to as the ‘surface scope’ reading and it involves evaluating the quantificational expressions in the linear order in which they appear in the sentence: the universal DP is within the scope of the indefinite, and the sentence makes a claim about a single proofreader

¹More specifically, by ‘scope’, we mean the part of the sentence in which a quantificational expressions makes its semantic contribution. So when we say that a quantifier ‘takes wide scope’ over another, we mean that its semantic contribution is computed first, in terms of linear order, such that the other expression falls within its scope. This point can be clarified by looking at the paraphrases of the intended readings.

who checked all the manuscripts. The reading in (1-b), referred to as the ‘scope-shifted’ or ‘inverse scope’ reading, is more interesting: the universal DP, *each manuscript*, behaves as if interpreted above the subject indefinite so that the sentence could make a claim about multiple proofreaders. In other words, the two quantificational expressions are evaluated in the inverse order in which they appear in the sentence. The diagnostic for this reading is to see whether the sentence is intuitively true in a context where the proofreaders can vary by manuscript: we say that the universal quantifier *induces variation* of proofreaders. In simple transitive sentences like (1), this variation effect is standardly taken as a symptom of wide-scope universal quantification, and this phenomenon—whereby a structurally lower universal seemingly induces variation in some other expression (like the witness of an indefinite or the referent of a definite)—is a central concern of this dissertation.

Throughout the dissertation, when we move beyond simple transitive sentences, I will use the term ‘variation reading’ to refer, descriptively, to any reading of a sentence under which the witness of an indefinite seems to vary with a lower universal, as in (1). I will also use the term ‘varying definite reading’ to refer, descriptively, to any reading of a sentence under which the referent of a definite seems to vary with a lower universal. These terms are intentionally theory-neutral: they pick out an interpretive pattern (i.e., variation) without presupposing any particular compositional structure. We deliberately avoid the terms ‘inverse scope reading’ or ‘scope shifted reading’, since these labels carry a theoretical commitment, namely, to a scope-shifting mechanism, which is precisely what we aim to evaluate. A central question of the dissertation is what that compositional structure is in these kinds of cases.

To this end, the variation effect, as illustrated in (1), will play a central role throughout this dissertation. A standard assumption in the literature, which motivates a large body of work on quantifier scope, is that variation as in (1-b) reliably diagnoses wide-scope universal quantification: if we observe that the value of some expression co-varies with a lower universal, then we take this as evidence that the universal has taken scope over it. This dissertation asks whether that inference is always warranted. As I will argue, variation turns out to be a less reliable diagnostic for scope than is typically assumed, because there exist compositional mechanisms capable of pro-

ducing variation-like effects without any scope-shifting at all. The empirical investigation centers on two syntactic environments—relative clauses and finite complement clauses—which have long figured in debates about how quantifier scope is constrained. In both cases, the question is not simply whether universal quantifiers can scope out of these environments, but whether the perceived variation requires scope-shifting in the first place, or whether it can arise by other means.

0.2 Approaches to scope-taking

Standard mechanisms that account for scope ambiguities require complicating the relations between the syntax and the semantics of natural language: what syntactic/semantic strategies does the grammar employ to derive the reading in (1-b)? How can we capture the observation that the universal DP semantically behaves as if it appeared in a different position than its actual position in the sentence? For this reason, a number of analyses have been proposed to analyze scope-taking and capture scope ambiguities, differing primarily in whether they posit a syntactic or a purely semantic mechanism. One widely adopted approach is Quantifier Raising (QR), first proposed by Chomsky (1975) and further developed by May (1977). Under the QR approach, inverse scope arises when a quantificational DP covertly moves at Logical Form (LF) to a position that c-commands the other quantificational expression. The surface and inverse scope readings of (1), then, receive the LFs in (2).

- (2) a. *Surface scope LF*: [a proofreader] λ_2 [each manuscript] λ_1 [t_2 checked t_1]
 b. *Inverse scope LF*: [each manuscript] λ_1 [a proofreader] λ_2 [t_2 checked t_1]

Montague (1973) provided a different kind of syntactic treatment of quantifier scope ambiguities: Quantifying-in. This mechanism derives scope ambiguities through a syntactic substitution operation: rather than moving a quantificational expression, it takes a sentential expression containing a free pronoun and substitutes the relevant quantificational expression for that pronoun directly, allowing it to take scope over the expression it is substituted into. Unlike QR, which derives scope ambiguities by covertly moving a quantificational expression to a sentence node at a dedicated

level of Logical Form, Quantifying-in generates different scope construals within a single derivational step, without positing any movement operation or additional level of syntactic representation beyond the surface form.

Alternatively, various semantic theories derive inverse scope without positing any syntactic movement either. Cooper Storage (Cooper, 1975, 1979) allows quantificational meanings to be ‘stored’ and retrieved later in the compositional derivation, producing scopal flexibility from a single syntactic representation. A type-shifting approach, developed by Hendriks (1993) following Partee and Rooth (1983), derives scope ambiguities by allowing semantic types to be shifted at any node. Categorical approaches have also been proposed to analyze quantifier scope, where they try to achieve similar flexibility through the algebraic structure of the type system itself (cf. van Benthem 1986, 1991 building on the Categorical Grammars in Ajdukiewicz 1935; Lambek 1958; further refinements in this tradition include Oehrle 1994; de Groote 2001; Muskens 2003; Carpenter 1997, a.o.).²

These frameworks differ considerably in their technical implementation—some positing multiple syntactic representations, others deriving scopal flexibility from a single syntactic structure through semantic or type-theoretic means—but they share the common goal of providing a principled account of how a quantificational expression can make its semantic contribution at a point in the derivation other than its linear position in the sentence (see Ruys and Winter 2010 for a more comprehensive overview of these various approaches and how they relate). Throughout the thesis, I will use the term ‘scope-shifting mechanism’ as a cover term for this family of different approaches. I will also assume, neutrally with respect to the choice of mechanism, that universal DPs take scope via some such operation. Crucially, as far as I can tell, nothing in the thesis hinges on which specific mechanism is correct: the findings bear on the availability and interpretation of scope-shifted readings in general, not on the particular compositional mechanism that derives

²See also Barker 2002 for a detailed discussion of quantifier scope which is not, strictly speaking, categorical but quite similar in nature. Instead, this approach makes use of ‘continuations’ (a notion from computer science). Barker proposes a different kind of account concerning the mismatch between the semantic types of quantifiers and their corresponding syntax, leading to a theory of quantifier scope. For more recent proposals in the same spirit of this approach—but also with important refinements—see Charlow 2014; Bumford 2015, 2022; Bumford and Charlow 2025.

them.

Throughout, I will also contrast the above scope-shifting mechanisms with other compositional pathways through which the semantic effects of scope-shifted readings can arise. In particular, I will use the term ‘pseudo-scope’ to refer to any compositional mechanism that produces the interpretive signature of a scope-shifted reading (in our case, observing that the value of some expression co-varies with a lower universal) without itself involving any scope-shifting operation.³ Under this characterization, pseudo-scope mechanisms are distinguished from scope-shifting mechanisms not by their interpretive output, which may be indistinguishable on the surface, but by the compositional pathway through which that output is derived. Some of these are well-documented in the literature: branching quantification (Hintikka, 1973; Gabbay and Moravcsik, 1974; Schlenker, 2006), allows quantifiers to interact in a non-linear manner so that sentences can be analyzed using a ‘branching’ structure; cumulative quantification, allows two quantifiers to interact in a way that produces variation effects without either taking scope over the other (Scha, 1981; Krifka, 1989; Sternefeld, 1997; Winter, 2000; Beck and Sauerland, 2001), and quantification over Skolem/Choice functions has been proposed as an alternative to wide-scope movement for certain indefinites (Reinhart, 1992, 1997; Winter, 1997; Kratzer, 1998). Other pseudo-scope readings have been proposed to arise from the semantic properties of specific grammatical environments. For instance, Fox and Sauerland (1996) argue that in generic contexts, universal quantifiers can produce the interpretive signature of wide scope quantification out of finite complement clauses as an ‘illusion’ generated by the semantics of generic tense—without any actual scope-shifting taking place. The existence of such mechanisms raises an important methodological question: if a sentence appears to license a scope-shifted reading, how can we be sure that this reflects genuine scope-shifting rather than the operation of a pseudo-scope mechanism? This question is central to the dissertation as we will be concerned with mechanisms that can produce interpretations that mimic the effects of variation and wide-scope universal quantification without invoking a scope-shifting operation. As I argue, the failure to distinguish scope-shifted readings

³Again, this term is meant to be descriptively neutral: it picks out a class of alternative mechanisms rather than any single analysis.

from pseudo-scope readings has led to an overestimation of the empirical case for non-local scope-shifting, and a more careful assessment of the two reveals that variation is a less reliable diagnostic for scope than is typically assumed.

Before moving on, it is worth flagging that, for expository purposes, I will use QR as the representative scope-shifting mechanism throughout the thesis. This choice is not necessarily arbitrary. As Ruys and Winter (2010) observe, the QR framework is by far the most empirically comprehensive of the available approaches: the nature of the restrictions on quantifier scope—their sensitivity to syntactic configuration, quantifier type, and embedding environment—has been far more extensively studied and described within the QR literature than within any of the alternatives. Since the empirical questions this dissertation explores are precisely questions about the distribution and constraints of scope-shifted readings, the QR literature provides the most developed theoretical and empirical backdrop against which to situate them. With this in mind, we turn to one of the most influential constraints on scope-shifting that has been proposed in that literature: the clause-boundedness hypothesis.

0.3 The clause-boundedness of scope-shifting

An early and influential observation originating in Chomsky 1975 is that quantifier scope appears to be constrained by finite clause boundaries, giving rise to a widespread assumption in the literature that scope-taking is clause-bounded: a quantificational DP cannot scope out of a finite complement clause or a relative clause (Lakoff 1970; Chomsky 1975; Rodman 1976; May 1977; Cooper 1979; Farkas 1981; Sharvit 1999a, a.o.). Under this view, relative clauses and finite clauses are sometimes referred to as ‘scope islands’: syntactic configurations that block scope-taking. To this end, one of the research questions which underlies this thesis is whether scope-taking is, in fact, clause-bounded.

To appreciate both the motivation behind this research question and its theoretical status, it is useful to trace the line of reasoning that led to it.⁴

⁴The following discussion draws extensively on the overview provided by Ruys and Winter (2010). What follows is an extreme oversimplification of the complex literature that discusses these phenomena. This discussion is not meant

Scope-shifting as a mechanism for scope ambiguity. The postulation of scope-shifting mechanisms arose directly from the observation, discussed in Section 0.2, that sentences with multiple quantificational expressions admit inverse scope/scope-shifted readings that cannot be derived from their surface syntactic structures. What the various scope-shifting approaches share is the core insight that the grammar must be equipped with some operation (whether movement-based, storage-based, or type-theoretic) that allows a quantificational expression to make its semantic contribution at a point in the derivation other than where it is linearly located. Among these approaches, QR is distinctive in that it implements scope-shifting via the independently motivated apparatus of syntactic movement, the same operation responsible for overt displacement phenomena such as *wh*-extraction. Under this approach, the theory of scope ambiguity is subsumed under the broader theory of syntactic movement, and scope-shifting is not a *sui generis* semantic phenomenon but a reflex of the grammar’s general displacement machinery.⁵ This has significant empirical consequences that spurred a rich body of literature on quantifier scope: if scope-shifting is a form of covert movement, it should be subject to the same locality constraints that govern overt movement, and the distribution of scope-shifted readings across different syntactic environments becomes directly informative about those constraints. It is this consequence, rather than any commitment to QR as the correct mechanism for scope-taking, that motivates the discussion that follows. Regardless of which scope-shifting mechanism one opts for, the question of which syntactic environments license or block scope-shifted readings is an empirical one, and the QR framework, given its extensive development in this domain, provides a useful backdrop for framing it.

to be an exhaustive overview of the literature on QR and quantifier scope (and all the relevant data and proposals that have been presented), but rather a short introduction to motivate and situate the research questions that the thesis will build on.

⁵The parallel between overt and covert movement is not uncontroversial. May (1977) proposed that both overt movement and covert movement (in the form of QR) are subject to Chomsky (1973)’s Subjacency condition. Some theories treat QR as only loosely analogous to overt A’-movement, and it is not obvious that every constraint on overt movement carries over to QR without independent motivation (Huang 1982; Fox 2000; Cecchetto 2004; Sauerland 2005; Bianchi et al. 2010, a.o.). For example, Huang (1982) argued that QR is subject to the ECP but not to Subjacency, and Cecchetto (2004) proposes that QR is constrained by independent semantic conditions (along the lines of Fox 2000’s Scope Economy) rather than by the feature-driven mechanisms that govern overt movement. See Fox (2000) for an influential treatment of QR within minimalist syntax.

The parallel between scope islands and extraction islands. The strongest evidence that scope-shifting is subject to syntactic locality conditions comes from a structural parallel between the environments that constrain quantifier scope and those that constrain overt syntactic extraction. The study of *wh*-movement has identified a range of syntactic configurations that systematically block extraction: the Complex NP Constraint (CNPC) blocks extraction from NPs containing relative clauses; adjunct islands block extraction from clausal adjuncts such as *if*-clauses; the Coordinate Structure Constraint blocks extraction from within conjuncts; and specificity constraints block extraction from certain definite NPs (Ross, 1967; Chomsky, 1973, 1977). The same syntactic configurations have been argued to constrain quantifier scope. For example, it has been argued that a universal DP embedded inside a relative clause cannot take matrix-level scope (Rodman, 1976), in parallel with the ungrammaticality of extracting NPs from the same relative clause environment. Adjunct islands and coordinate structures that prevent *wh*-extraction seem to place similar constraints on scope-taking (Reinhart, 1997; Winter, 1997). More generally, as pointed out by Ruys and Winter (2010), evidence that the distribution of scope-shifted readings is governed by syntactic locality exists to the extent that generalizations on quantifier scope can be stated in terms of syntactic properties of the relevant constructions. This convergence between scope islands and extraction islands is theoretically significant regardless of which scope-shifting mechanism one adopts: it suggests that the distribution of scope-shifted readings is not determined purely by semantic factors, but is at least partially sensitive to the syntactic environments in which quantificational expressions are embedded.

Asymmetries and the move to clause-boundedness. The picture outlined above—that the distribution of scope-shifted readings is sensitive to syntactic structure, and in particular that it tracks the same configurations that constrain overt extraction—does not, on its own, entail that scope-shifting is specifically constrained by clause boundaries. Any theory that ties scope-taking to syntactic locality could in principle capture the scope island-extraction island parallel without committing to a clause-specific constraint. What matters, under such a view, is the structural parallel between the configurations that restrict scope-taking and those that restrict overt extraction, not the

presence of a clause boundary *per se*. The clause-boundedness hypothesis gained its independent theoretical status through a different line of argument. Chomsky (1975) responded to Rodman (1976)'s observation that quantifier scope is sensitive to CNPC islands by proposing that this sensitivity need not be explained by appeal to the structural parallel with extraction at all. Rather, Chomsky proposed that quantifier scope seems unable to escape finite clauses in general, meaning the inability to scope out of CNPC islands follows as a trivial consequence (since CNPC islands contain a finite clause). This proposal stemmed from examples like (3). Chomsky observed that (3) is unambiguous: the universal DP seems unable to take matrix scope, despite the finite complement clause not being an island for overt movement, as illustrated in (4), where the *wh*-expression can move out of the finite complement clause of *say*. Thus, there appears to be a mismatch between overt movement constraints and scope island constraints.

- (3) John said that everyone had left.
 - a. *available reading*: John said all the people had left (single statement)
 - b. *unavailable reading*: for each person *x*, John said *x* had left (multiple statements)
- (4) Who did John say _ had left?

Under this view, then, the relevant constraint on scope-taking is not sensitivity to extraction islands but sensitivity to finiteness, and the two reduce to the same thing only incidentally: CNPC islands, for instance, just happen to be scope islands because they contain a finite clause, not because they are islands for overt movement. The clause-boundedness hypothesis is therefore best understood as an independent empirical constraint on the scope-taking abilities of quantificational expressions—one that applies regardless of whether scope-shifting is derived via movement, storage, or type-theoretic means—and does not require a perfect correspondence with the locality conditions on overt movement.

Clause-boundedness as an independent empirical constraint is not without its own difficulties, however. It proves to be both too restrictive and too permissive as a constraint on quantifier scope. On the one hand, scope is sometimes more restricted than finite clauses: coordinate conjuncts

and certain definite NPs function as scope islands even though they do not introduce a clause boundary. On the other hand, it is not clear whether scope is, in fact, *always* constrained by clause boundaries: for some speakers, finite complement clause boundaries, as in (3), are at least partially permeable to scope (May, 1977; Reinhart, 1997; Hornstein, 1995). We will consider a few potential counterexamples put forth in the literature.

First, as observed by Reinhart (1997), the sentence in (5-a) has a *de re* interpretation where it entails that Lucie believes that Gary Bettman is corrupt. One standard way of generating this entailment is by scoping the universal DP out of the finite complement clause so that it takes scope above the predicate *believe*. Though, as pointed out by Reinhart, one can plausibly capture this entailment without necessarily positing that the universal DP scopes out of the finite clause, especially since it is very difficult to get a scope-shifted reading of (5-b), where *someone* is interpreted in the scope of *every sports commissioner*. Therefore, the example in (5-a) is somewhat inconclusive, given that the universal DP cannot actually outscope the matrix indefinite and there are alternative ways of capturing the *de re* interpretation.

- (5) a. Lucie believes that every sports commissioner is corrupt.
 b. Someone believes that every sports commissioner is corrupt.
 c. *Possible LF*: [someone] λ_1 [every sports commissioner] λ_2 [t_1 believes that t_2 is corrupt]

Pursuing this line of thought though, Reinhart proposes that the scope construal that is available for sentences like (5-a) and (5-b) has a syntactic configuration like (5-c), where *every sports commissioner* scopes out of the finite complement clause, capturing the *de re* interpretation, but below the indefinite. If true, this would still be a violation of the clause-boundedness constraint (though it does raise the question as to why the universal DP cannot outscope the indefinite).

Another example that Reinhart (1997) discusses is provided in (6).

- (6) A doctor will make sure that we give every new patient a tranquilizer.

Reinhart's observation is that the universal DP, *every new patient*, can seemingly scope over the

matrix subject to induce variation of doctors by patients. The example in (6) involves a potential confound though insofar as it contains future tense, which could enable variation without necessarily involving scope-shifting—similar to how Fox and Sauerland (1996) have argued that generic tense enables variation as a kind of ‘illusion’, without invoking scope-shifting. For this reason, it is also unclear whether (6) constitutes a genuine counterexample to the claim that universals cannot scope out of finite complement clauses and, in turn, a counterexample to the clause-boundedness hypothesis. To this end, a more concrete counterexample to the clause-boundedness hypothesis was provided by Farkas and Giannakidou (1996), who argue, contra Fox and Sauerland (1996), that scoping out of finite complement clauses is not always an ‘illusion’. This is illustrated in (7), which does not involve generic tense since it is an episodic sentence .

(7) A student made sure that every invited speaker had a ride.

The observation is that (7) licenses a reading where students can vary by invited speakers. They take this reading to suggest that universal DP is scoping above the indefinite DP, meaning universal DPs can sometimes scope out of finite complement clauses. Thus, the takeaway is that, if scope can sometimes escape finite complement clauses, this is particularly informative given that *wh*-phrases can also move out of such syntactic environments. Taken together, then, these observations suggest that clause-boundedness, while capturing something real, may be an oversimplification: we learn that there must be further conditions on quantifier scope; clause-boundedness, alone, is not sufficient. The imperfect match between the distribution of scope-shifted readings and what either extraction-island sensitivity or clause-boundedness predicts underscores a broader point: the empirical profile of scope-taking is complex enough that no simple structural generalization fully captures it, and a more nuanced understanding of the environments that license (or block) scope-taking is needed.

For the purposes of this dissertation, we simply note that the theoretical status of the clause-boundedness hypothesis is less robust than its widespread adoption might suggest: no principled account of why clause boundaries should specifically constrain quantifier scope has achieved con-

sensus. More importantly, the empirical picture is equally unsettled: the literature has documented many cases in which universal quantifiers embedded inside finite complement clauses or relative clauses appear to give rise to wide-scope universal readings, where the universal DP seemingly takes scope at the matrix level. These are sometimes described as cases of ‘exceptional scope’ or as involving ‘non-local QR’ (Farkas and Giannakidou 1996; Anderson 2004; Szabolcsi 2010; Syrett 2015; Wurmbrand 2018; Barker 2022; Hoeks et al. 2022; Wang 2023, 2024, a.o.), and they constitute the central empirical focus of this dissertation.

0.4 Developing the problem: variation as a diagnostic for scope

The theoretical interest of exceptional scope data stems from the fact that if a sentence genuinely licenses a variation/varying definite reading, then the grammar must make that reading available somehow—through some compositional pathway. This makes the empirical distribution of such readings a valuable diagnostic: the range of syntactic environments, quantifier types, and embedding predicates that license (or block) them is informative about which semantic mechanisms are operative. Relative clauses and finite complement clauses are thus not merely potential counterexamples to the clause-boundedness hypothesis—they are test beds for competing theoretical accounts, and the empirical distribution of variation/varying definite readings across these environments can help us to adjudicate between these accounts.

The focus on these two environments in particular is not arbitrary. As pointed out above, they are the two scope-island environments identified in Chomsky 1975 as the original motivation for clause-boundedness, and they have served as the primary testing ground for this hypothesis in subsequent work. More importantly, they differ in a theoretically revealing way. Finite complement clauses are not, in general, islands for overt extraction: a *wh*-phrase can move out of a finite complement clause in English without violating island constraints.⁶ Relative clauses, by contrast,

⁶It should be noted that extraction out of finite complement clauses is not entirely unrestricted in English. The ‘*that*-trace effect’ constrains subject extraction specifically: when the complementizer *that* is overt, extracting the subject of a finite complement clause produces ungrammaticality, as illustrated by the contrast between *who did you think would win* and **who did you think that would win* (Chomsky and Lasnik, 1977; Chomsky, 1981). This constraint is, however, standardly analyzed as a filter on the complementizer–trace configuration rather than as a clause-level island condition: it targets a specific structural configuration involving an overt complementizer followed by a trace,

are CNPC islands and do block overt *wh*-extraction. If the locality constraint on scope-taking simply mirrors that on overt movement, this asymmetry predicts an empirical divergence: scope should be at least partially available out of finite complement clauses (since overt extraction is) but not out of relative clauses (since overt extraction is not). If instead the relevant constraint is clause-boundedness—applying wherever a finite clause boundary is present, regardless of whether that boundary also constitutes an extraction island—the two environments are predicted to pattern together, and the CNPC status of the relative clause is irrelevant to its scope-island status. The distribution of variation/varying definites readings across these two environments is therefore indicative not only of the availability of scope-shifting but of the underlying character of the constraint on scope-taking: specifically, whether it is better modeled as a reflex of movement (and hence sensitive to the extraction-island distinction) or as an independent syntactic or semantic principle.

It is against this theoretical backdrop that the variation diagnostic takes on particular significance. Whether variation readings are available in finite complement clauses but not relative clauses, or whether they are constrained uniformly across both environments, bears directly on which of the two accounts of scope-islandhood outlined above is correct—but only if variation reliably diagnoses wide-scope universal force in the first place. That is, when we extend the variation diagnostic from simple transitive sentences to these more complex syntactic environments, there is an implicit reasoning that goes: if a sentence embeds a universal DP inside a putative scope island (i.e., a clause boundary) and yet licenses a variation/varying definite reading, the universal must have taken matrix scope via some non-local scope-shifting mechanism. A number of cases in the literature have been taken to instantiate this pattern (Farkas and Giannakidou 1996; Barker 2022; Hoeks et al. 2022, a.o.).⁷ However, the question this dissertation pursues is whether this inference—from variation to non-local scope—is warranted. As I argue, the empirical distribution of these readings involving variation contains systematic patterns and hints that point toward

and does not generalize to object extraction or to cases where *that* is absent. For our purposes, the relevant contrast with relative clauses, which constitute CNPC islands blocking extraction altogether regardless of the type of extracted element or the presence of a complementizer, therefore stands.

⁷As noted above, these readings are sometimes described in the literature as ‘exceptional scope’, ‘non-local scope’ or ‘long-distance scope’ readings. We use the term ‘variation reading’ throughout for the reasons given in fn. 2. We reserve the above terms for when referring specifically to scope-shifting mechanisms.

pseudo-scope sources of variation: compositional mechanisms that mimic the semantic effect of wide-scope universal quantification without involving any scope-shifting at all.

Two such patterns are particularly important. The first concerns relative clauses. Certain sentences where a definite DP is modified by a relative clause containing a quantifier (what I will refer to as ‘relativized DPs’), like *the supervisor that each volunteer reported to*, license readings which carry separate presuppositions of uniqueness and existence for each volunteer—what I have been referring to as ‘varying definite readings’.⁸ These readings have been analyzed as a case of exceptional scope (cf. Hulse and Sauerland 2006; Szabolcsi 2010; Barker 2022; Wang 2023. Barker 2022), with the universal taking wide scope over the definite determiner (through something like non-local QR). However, as I argue in Chapter 1, there are cases where we observe variation of the referent of a definite but where scope-shifting is not a viable source of this perceived variation: configurations in which positing non-local scope-shifting either delivers the wrong truth conditions or requires scoping out of a further embedded scope island within the relative clause. Instead, I argue that these cases motivate a functional interpretation of the relativized DP where it denotes an $\langle e, e \rangle$ function from individuals to individuals, deriving the observed variation through the type (and interpretation) of the definite DP itself rather than through scope-shifting. Therefore, zooming out from the varying definite reading itself, the logic is that a parsimonious account should cover all these cases of variation observed with relativized DPs, not just the varying definite reading that has been explored in the literature.

In Chapter 2, we discuss an ambiguity concerning these kinds of relativized DPs: they can give rise to a ‘functional reading’ and a ‘pair-list’ reading (Sharvit, 1999a). Descriptively, these two readings involve different kinds of variation: the functional reading involves a principled way of mapping individuals to individuals while the pair-list reading involves an arbitrary way of mapping individuals to individuals. I argue that analyzing relativized DPs as functional DPs offers a parsimonious analysis of this ambiguity—under a particular view of definiteness: if, following

⁸Varying definite readings have also been referred to as ‘pair-list readings’ of relative clauses (Sharvit, 1999a). We adopt the term ‘varying definite reading’ to highlight the descriptive observation—that the referent of the definite description can vary with the lower universal—without presupposing an analysis.

Von Stechow et al. (2014), the notion of maximality invoked by the definite article is based on semantic strength. As a result, this line of analysis not only offers an account of varying definite readings but also explains two different kinds of perceived variation (i.e., principled and arbitrary variation) through a single mechanism.

The second pattern concerns finite complement clauses. The example in (7), repeated in (8), licenses a variation reading: in addition to a reading about a particular student, (8) licenses a weaker reading where the student may vary by speaker.

(8) A student made sure that every invited speaker had a ride.

However, as pointed out in the literature (cf. Farkas and Giannakidou 1996; Barker 2022; Hoeks et al. 2022), the availability of variation readings in this environment is *predicate sensitive*: there are some embedding predicates that systematically license variation readings, and others that do not. If variation simply tracked whether an embedded universal DP could escape the clause to take scope, the embedding predicate should not matter: locality constraints on scope are not generally sensitive to the lexical properties of the embedding verb and accounts that posit this predicate sensitive restriction (cf. Barker 2022) do so in an admittedly ad-hoc way (though, see Wang 2024 for an account that attempts to derive this predicate sensitivity from the semantic type of different finite complement clauses). Thus, the fact that variation readings are predicate sensitive points toward an account grounded in the semantic properties of the embedding predicate, rather than in any syntactic property governing the availability of scope.

Moreover, we will also show that variation readings are not restricted to sentences with universal DPs embedded inside the finite complement clause. As I discuss in Chapter 3, negative DPs can also give rise to variation readings when embedded inside a finite complement clause. This is significant for two reasons. First, non-local scope-shifting of a negative quantifier delivers the wrong truth conditions in this case: even under charitable assumptions about scope-shifting (i.e., that negative DPs can scope-shift to derive an inverse scope reading), scope-shifting of a negative DP above the matrix subject yields an unattested reading. Second, these readings exhibit

a predicate sensitivity that parallels the predicate sensitivity observed for variation readings with embedded universal DPs—a parallel that would be unexpected if the two were derived by independent mechanisms. Together, these observations provide independent evidence that a pseudo-scope mechanism is needed, one that naturally captures the predicate sensitivity and applies uniformly to both types of embedded quantifier.

As a step towards determining what this pseudo-scope mechanism is, I will present evidence that the predicate sensitivity observed in variation readings (with universal or negative quantifiers) parallels the predicate sensitivity observed in instances of cumulative predication with plural subjects, what I will refer to as ‘non-local cumulation’. Thus, our central finding in Chapter 3 is the following generalization: an embedding predicate licenses variation readings if it licenses non-local cumulation with plural subjects. As a consequence, I argue that the proper compositional path to variation should share a key component with the compositional path to non-local cumulation.

A word on methodology is in order, since the dissertation draws on two different kinds of evidence. The claims about available interpretations in Chapters 1 and 2 rest on introspective judgments—my own, checked against those of other native speakers and, wherever possible, against judgments already reported in the literature. I take this to be appropriate given the character of the claims at issue: what is at stake in those chapters is whether a given reading is available at all in a fully specified scenario, and the contrasts are categorical enough that introspection can settle them; where a judgment is more delicate, I try to flag it as such. The claims in Chapter 3 are different in kind, and call for a different method. There, the central claim is a correlation—that an embedding predicate licenses variation readings if it licenses non-local cumulation with plural subjects—which holds across a set of 36 predicates and therefore cannot be assessed one contrast at a time. Nor are the relevant judgments as crisp: predicates do not divide cleanly into those that license variation and those that do not, but arrange themselves along a gradient, and the differences between neighboring predicates on that scale are small enough to be unreliable under introspection.⁹ Introspection was thus the source of the initial hypotheses about where individual predicates

⁹Philipp and Zimmermann (2025), who compare the availability of scope-shifted readings in English and German experimentally, illustrate this general point. The contrast between the availability of scope-shifted readings in English

fall, but establishing the generalization itself required acceptability ratings from experiments together with statistical models capable of detecting a gradient effect of this size. The two methods are complementary rather than in competition: the choice between them tracks the kind of claim being made.

Taken together, though, the above observations motivate the central methodological point of the dissertation. The implicit assumption that variation reliably diagnoses wide-scope universal force can lead to an overextension of the exceptional scope hypothesis: once the full range of pseudo-scope mechanisms is taken into account, the case for non-local scope-shifting is considerably weakened. Thus, the motto that underlies the investigations throughout the thesis (and the thing to keep in the back of your mind as you read it) is accordingly:

Motto

“All that varies is not scope.”

More concretely, this dissertation investigates the scope-taking abilities of universal quantifiers and the syntactic constraints that govern them, by examining varying definite readings in relative clauses and variation readings in finite complement clauses. In both environments, I challenge the implicit assumption that these readings reliably diagnose wide-scope universal force. It is worth clarifying that the thesis does not necessarily argue that scope is, in fact, clause-bounded, nor that exceptional scope never occurs. Rather, the central claim is a more modest one: that variation-based arguments for non-local scope are confounded by pseudo-scope mechanisms, and that variation cannot be straightforwardly equated with wide-scope universal force. Consequently, variation readings and varying definite readings do not, on their own, furnish arguments against locality constraints on scope. This dissertation thus argues (i) for a more nuanced view of the scope–variation relationship, one that recognizes multiple compositional pathways capable of inducing variation; and (ii) for greater caution when using variation to adjudicate between competing accounts of quantifier scope. More generally, the thesis explores the types of pseudo-scope mechanisms that

and German has long been reported as categorical on introspective grounds. Their experiments find the difference to be gradient, though, with German permitting scope-shifted readings merely less readily than English. Where a contrast is a matter of degree, introspection is liable to present it as sharper than it is—which is precisely the risk the experiments in Chapter 3 were designed to guard against.

grammar employs.

0.5 Structure of dissertation and summary of contributions

The remainder of the dissertation is structured as follows. Chapter 1 develops the functional pseudo-scope account of varying definite readings with relative clauses. Chapter 2 extends this analysis to provide a unified analysis of two different readings of functional definite DPs that have been observed in the literature. Chapter 3 presents an experimental investigation into the predicate sensitivity of variation readings with finite complement clauses, leading to an empirical generalization on variation readings and its theoretical implications. A concluding chapter synthesizes the results and reflects on the broader implications for the syntax–semantics interface and the status of locality constraints on scope-taking. To this end, this dissertation makes three main empirical and theoretical contributions, corresponding to its three chapters. Below is a more detailed summary of the contributions of each chapter:

Chapter 1 addresses varying definite readings in relative clauses. Sentences where a definite DP is modified by a relative clause containing a universal quantifier, like *the supervisor that each volunteer reported to*, license readings which carry separate presuppositions of uniqueness and existence for each volunteer. Barker 2022 argues that these readings involve the universal DP scoping out of the relative clause and above the definite determiner, proposing to analyze them using a non-local scope shifting mechanism, like Quantifier raising (QR). In this paper, I argue for a functional interpretation of the relativized DP and propose that these readings result from the definite DP denoting an $\langle e, e \rangle$ function, from volunteers to supervisors. I present several pieces of independent evidence for our functional approach, providing a viable alternative to Barker’s QR analysis, and hence potentially undermining the argument for non-local QR. Combining these arguments with the fact that the functional approach can also be extended to derive the varying definite readings, I conclude that non-local QR is not the only possible way to analyze these examples. As a result, these cases are compatible with the view that relative clauses are scope islands for universal quantifiers.

Chapter 2 focuses more specifically on the internal composition of functional relative clauses. In the literature on questions, it has been observed that questions with quantifiers can give rise to a ‘functional reading’ and a ‘pair-list’ reading (Engdahl 1980; Groenendijk and Stokhof 1984; Chierchia 1993, a.o.). Descriptively, the functional reading involves a principled way of mapping individuals to individuals while the pair-list reading involves an arbitrary way of mapping individuals to individuals. Building on this observation, it has been observed that definite DPs with relative clauses containing quantifiers can give rise to both kinds of readings as well (von Stechow, 1990; Jacobson, 1994; Sharvit, 1999a; Heim, 2018). This paper proposes that a parsimonious analysis of such definite DPs becomes available if, following Von Fintel et al. (2014), the notion of maximality invoked by the definite article is based on semantic strength. To make this case, I compare the functional reading (what I refer to as the ‘principled variation reading’) and the pair-list reading (what I refer to as the ‘arbitrary variation’ reading). I argue that both readings involve a functional parse of the relative clause that delivers a function from individuals to individuals, of type $\langle e, e \rangle$. While the two readings are, in fact, distinct, I propose that the ambiguity between them is not rooted in how the relative clause itself is parsed—contrary to what has been proposed in the literature (e.g., Sharvit 1999a). Instead, I argue that each reading results from a different way of enabling reference to a function. Functional readings show that context can help establish unqualified uniqueness. In contrast, I suggest that pair-list readings arise from reference to a unique *maximal* function, without the aid of context—provided maximality is understood as based in semantic strength, as proposed in Von Fintel et al. 2014.

Chapter 3 addresses variation readings in finite complement clauses. Finite complement clauses are often assumed to be scope islands for quantifier raising (QR) of universal quantifiers. However, as observed by Farkas and Giannakidou 1996; Barker 2022; Hoeks et al. 2022, a.o., there are apparent counterexamples to this claim, where a universal DP appears to scope out of a finite complement clause to take scope over a singular indefinite in the matrix clause; i.e., ‘variation readings’. Barker 2022 argues that the availability of variation readings is predicate sensitive, and attributes this predicate sensitivity to non-local QR. In this paper, based on experimental re-

sults, I challenge Barker’s conclusion. First, I demonstrate that variation readings are also possible when we replace the embedded universal quantifier with a negative indefinite, and non-local QR is insufficient to explain these cases. Second, I argue instead for the following empirical generalization: embedding predicates license variation readings if they also license a form of cumulativeness in relational predication. Building on this generalization, I argue that the predicate sensitivity of both phenomena is rooted in the inferential profile of the embedding predicate (Pasternak, 2018; Harada, 2023).

Part I

Relative clauses as scope islands

Pseudo-scoping out of relative clauses: varying definite readings and functional parses

Sentences where a definite DP is modified by a relative clause containing a universal quantifier, like *the supervisor that each volunteer reported to*, license readings which carry separate presuppositions of uniqueness and existence for each volunteer. Barker 2022 argues that these readings involve the universal DP scoping out of the relative clause and above the definite determiner, proposing to analyze them using a non-local scope shifting mechanism, like quantifier raising (QR). In this paper, we argue for a functional interpretation of the relativized DP and propose that these readings result from the definite DP denoting an $\langle e, e \rangle$ function, from volunteers to supervisors.

We present several pieces of independent evidence for our functional approach, providing a viable alternative to Barker's QR analysis, and hence potentially undermining the argument for non-local QR. Combining these arguments with the fact that the functional approach can also be extended to derive the varying definite readings, we conclude that non-local QR is not the only possible way to analyze these examples. As a result, these cases are compatible with the view that relative clauses are scope islands for universal quantifiers.

1.1 Introduction

This paper addresses whether universal quantifiers can covertly move out of relative clauses for scope-taking purposes. More specifically, we address the claim that relative clauses impose a locality constraint and block quantifier raising (QR) of universal quantifiers—henceforth, ‘non-local QR’ (for previous discussion, see Rodman 1976; May 1977; Cooper 1979; Sharvit 1999; Hulsey and Sauerland 2006). Due to mixed evidence, this claim has been met with diverging opinions. On the one hand, given examples like (1), it has often been assumed that universal quantifiers cannot QR out of relative clauses (Rodman, 1976; May, 1977; Cooper, 1979; Sharvit, 1999). Both examples in (1) do not license a reading where the woman/bone can vary by man/corner.

- (1) a. John has dated a woman who loves every man.
b. Guinevere has a bone that is in every corner of the house. (Rodman 1976)

On the other hand, given an array of sentences suggesting otherwise, it has been argued that relative clauses should not be considered as scope islands for universal quantifiers (Hulsey and Sauerland, 2006; Szabolcsi, 2010; Barker, 2022; Wang, 2023). Barker 2022, in particular, provides several naturally occurring examples that license a reading where an embedded universal DP appears to take scope outside the relative clause. A variation of one such example is provided in (2).¹

- (2) John included the supervisor that each volunteer reported to. (c.f. Barker 2022, 637)

Under one reading, (2) conveys a global presupposition of uniqueness and existence concerning supervisors: there is a single supervisor which all volunteers reported to. This is the reading that results from the universal DP taking scope within the relative clause. Under another reading—the reading we are interested in—the supervisor can vary by volunteer and conveys the sentence

¹It is worth noting that one common challenge for the claim that universals undergo QR out of relative clauses is that this is only possible when the universal quantifier is in the embedded subject position (as opposed to the object position). However, Barker 2022 also provides examples where the universal quantifier is in the embedded object position. One such example is provided in (i).

- (i) Reflecting and thinking about all of this, we separate the word [that most represents each sign of the zodiac]. (c.f. Barker 2022, 637)

meaning in (3-b), paraphrased as: for each volunteer x , John included the supervisor that x reported to—henceforth, ‘the varying definite reading’.² This reading carries separate presuppositions of uniqueness and existence for each volunteer (i.e., a universal uniqueness presupposition), as illustrated in (3-a).

(3) a. *Presupposed:*

$$\forall x[\text{volunteer}(x) \rightarrow \exists! y[\text{supervisor}(y) \wedge \text{report-to}(y)(x)]]$$

b. *Asserted meaning:*

$$\forall x[\text{volunteer}(x) \rightarrow \text{include}(\iota y[\text{supervisor}(y) \wedge \text{report-to}(y)(x)])(\text{John})]$$

Taking the varying definite reading as our central data point, one natural hypothesis is to assume that the embedded universal DP scopes out of the relative clause and above the definite determiner. This reading could be straightforwardly captured using a non-local scope shifting mechanism, like QR, as illustrated in (4).

(4) *LF*: [each volunteer] λ_1 [John included [the supervisor [that t_1 reported to]]]

Without an alternative analysis of (2) on the table, it seems reasonable to conclude that QR is our best bet for capturing the varying definite reading. After all, the varying definite reading is exactly what one obtains from the LF in (4): *each volunteer* undergoes QR out of the relative clause, to take wide scope, inducing variation of the definite and yielding a universal uniqueness presupposition.

Crucially, though, the varying definite reading in (2) is only conclusive evidence for non-local QR under the assumption that there is no alternative analysis that could explain this reading, and the perceived variation (either as well, or better than the QR approach). With this in mind, in this paper, we present an argument for the existence of a compositional pathway to varying definite readings that does not require non-local QR. In what follows, we call such an alternative a ‘pseudo-scope’ analysis, since it does not invoke QR.

Constructing a complete argument for a pseudo-scope analysis of (2) consists of two parts.

²It should be noted that this has also been referred to as a ‘pair-list’ reading of relative clauses in the literature (cf. Sharvit 1999).

The first part provides indirect evidence for the pseudo-scope analysis by motivating its necessary components through data involving structurally similar DPs to (2)—a DP headed by a definite determiner modified by a relative clause containing an embedded universal DP (henceforth, ‘relativized DPs’)—but with different matrix subjects and different matrix predicates. We will observe two things. First, going beyond the example in (2), we present cases where we observe variation of the definite—just as we do in (2)—but which, we argue, cannot be due to QR. Second, and relatedly, we contend that each data point provides independent motivation for a different ingredient that is needed for a pseudo-scope approach to (2). To this end, we propose that varying definite readings can be analyzed as a kind of functional reading where we re-analyze the variation in examples like (2) not through QR but through a functional interpretation of the relativized DP. We show that a functional pseudo-scope account can not only capture the other cases of variation that we bring into the fold, but also mimic the effects of non-local QR in (2). The second part of the argument provides direct evidence for our pseudo-scope analysis in two forms: first, an undergeneration challenge showing that not all varying definite readings can be captured through QR; and second, data directly supporting functional parses of varying definite readings, suggesting that a functional analysis is not merely available but necessary in at least some cases. Given these two forms of evidence, the pseudo-scope analysis that we propose virtually comes for free, and therefore there is no strong argument for non-local QR stemming from varying definite readings.

The paper is structured as follows. In Section 1.2, we present an undergeneration challenge for a QR approach, suggesting that an alternative, pseudo-scope approach may be needed in some cases. In Section 2.2, we provide an overview of the main claims of the paper and situate our discussion in the broader context of functional relative clauses. In Sections 1.4–1.6, we give content to our pseudo-scope approach by presenting data that serves to motivate the different components of our alternative analysis. In Section 1.7, we show that, once we put together all the components, a pseudo-scope account of the varying definite reading in (2) falls out for free. We also bring the preceding arguments together to provide a direct argument for a functional parse of varying definite readings. In Section 2.5, we offer some concluding remarks.

1.2 An argument for a pseudo-scope approach

Concerning the varying definite reading in (2), in this section, we present a limitation of the QR analysis of varying definite readings. More specifically, in this section, we argue that there are varying definite readings that pose a challenge for a QR based approach, and propose that, in these cases, an alternative route to variation should be explored, one which does not invoke QR.

The example in (2) involves a universal DP embedded in a relative clause. The argument presented in this section is as follows: in the crucial data point, we further embed the universal DP in an environment where we have independent evidence that QR is not licensed (i.e., out of finite complement clauses). If a varying definite reading is still possible, this data point would further complicate the QR approach by requiring that the scope islandhood of finite complement clauses is context dependent—it is lifted if the finite complement is inside a relative clause, in which case QR can escape the finite complement and the relative clause at the same time. To this end, first consider the example in (5-a), which provides an independent baseline suggesting that universal DPs cannot freely QR out of finite complements.

- (5) a. A student claimed each invited speaker was wrong.
b. *Available interpretation:* A single student that claimed all the invited speakers were wrong.
c. *Unavailable interpretation:* For each invited speaker x, a (potentially) different student claimed x was wrong.

In (5-a), the student cannot vary by speaker; the only interpretation is provided in (5-b). If *each volunteer* were to QR out of the finite complement clause of *claim*, then there should be an interpretation of (5-a) as paraphrased in (5-c)—contrary to fact.³ With this baseline in mind, consider the example in (6-a).

- (6) a. *[At the community center, each volunteer reports to a different supervisor. John is*

³Even Barker (2022), who argues that finite complements are not scope islands for QR per se, assumes that the complement of *claim* is.

supposed to keep track of all this in a file but he's clueless regarding who reports to who. However, he remembers that Mary claimed that Alice reports to Bill, Chris reports to Diane and Eric reports to Francine. After writing these supervisor/volunteer pairs in the file, Mary looks it over. She looks puzzled so John replies:]

John (to Mary): What? **I included the supervisor that you claimed each volunteer reported to.**

- b. *Available interpretation:* For each volunteer x, I included the supervisor you claimed x reported to

The target sentence in (6-a) differs from (2) insofar as the island has, in a sense, been ‘strengthened’ by adding another layer of embedding: in the relative clause, there is the finite complement clause of *claim*. This configuration is shown in (7).

(7) *Structural configuration in (6-a):* [$\dots \iota$ [$_{RC} \dots$ [$_{TP} \dots \forall \dots$]]]

‘RC’ = relative clause boundary; ‘TP’ = finite complement clause boundary

What we observe is that, parallel to (2), the target sentence can still receive a varying definite reading, as paraphrased in (6-b).

In order to generate the intended interpretation under the QR approach, we would need to posit an LF like (8-a), which contains an instance of the unavailable LF for the control in (5-a), in (8-b).

(8) a. *LF:*

[each volunteer] λ_1 [I included [the supervisor [$_{RC}$ that you claimed [$_{TP}$ t_1 reported to]]]]

b. *LF:* [each invited speaker] λ_1 [a student claimed [$_{TP}$ t_1 was wrong]]

The LF in (8-a) requires an instance of QR where the DP *each volunteer* covertly moves out of the finite complement and out of the relative clause. In principle, this kind of QR may be available in the grammar, if neither relative clauses nor finite complement clauses impose locality constraints on QR. For the sake of the argument, let us assume that relative clauses do not impose locality

constraints on QR. But above, we observed that the finite complement clause of *claim* does seem to block QR of universal DPs. This leaves us with two options: (i) either posit that universal DPs can QR out of finite complements of *claim*, but only when the finite clause is itself embedded in a relative clause, or (ii) conclude that the varying definite reading in (6-a) is not derived through QR, but another mechanism. While in principle possible, at present, the first option would need to be stipulated, as it does not seem to follow from any known principles. Therefore, we take the availability of a varying definite reading in (6-a) as evidence that we may need something other than QR to derive the perceived variation of the definite.

The argument presented in this section raises the question of whether we can construct other examples where the universal DP is embedded inside a scope island for QR that is itself embedded inside a relative clause. Not all scope islands work though. To construct these kinds of examples, we need a particular configuration, which we can state through the following generalization: varying definite readings are available even in cases where the universal DP is embedded in a further scope island that also includes the *wh*-trace from relativization—and that scope island is itself *not* an island for *wh*-movement (so that relativization does not lead to ungrammaticality). This profile rules out other possible configurations, like antecedents of conditionals, which are considered to be scope islands for QR but are also islands for overt movement. This makes finite complement clauses the perfect test bed since they can be scope islands for QR and yet still permit *wh*-extraction. For this reason, we have focused on finite complement clauses.⁴

⁴That being said, other embedding verbs can be used to make the same argument that was made with *claim*. Consider the baseline in (i) and the example in (ii) with the embedding verb *know*.

- (i) a. A professor knew each student was wrong.
b. *Unavailable interpretation:* For each student *x*, a (potentially) different professor knew *x* was wrong.
- (ii) a. [*At the community center, each volunteer works with a different supervisor. John is in charge of matching volunteers with supervisors so he asked each volunteer who they want to work with: Al wants to work with supervisor 1, Chris wants to work with supervisor 2, and Diane wants to work with supervisor 3. After writing these supervisor/volunteer pairs in the file, Mary looks it over and John says:*]
John (to Mary): **I wrote down the supervisor that I knew each volunteer wanted to work with.**
b. *Available interpretation:* For each volunteer *x*, I wrote down the supervisor that I knew *x* wanted to work with

An anonymous reviewer also provided us with a naturally occurring example from COCA, which we minimally modified to get (iii).

In what follows, we refer to our alternative account as a ‘pseudo-scope’ approach since, as we will see, it does not invoke QR of the universal DP out of the relative clause. In the next section, we explore what pseudo-scope is, and what an alternative pseudo-scope account looks like.

1.3 A pseudo-scope account: the functional approach

1.3.1 Overview of main claims

In the previous section, we identified a limitation of the QR analysis of varying definite readings: we presented an undergeneration challenge for the QR analysis by observing that similar varying definite readings are possible, even if the universal DP is placed within a scope island for QR. As a result of this argument, we believe that there is a need for a non-QR analysis of varying definite readings. We propose a pseudo-scope analysis of (2), where variation is captured as a functional reading, that extends to (6-a), hence does not undergenerate in this way. In the remainder of the paper, we give content to said pseudo-scope account by taking steps towards explaining what exactly pseudo-scope amounts to.

Under our view, capturing the varying definite reading as a kind of functional reading involves three assumptions. These three assumptions are illustrated in the schematic LF in (9) where (i) the relativized DP denotes an $\langle e, e \rangle$ function, (ii) a covert pronoun saturates this function, and (iii) a covert universal attaches at the matrix level to supply wide scope universal force and bind the pronoun.⁵

- (9) *Schematic LF*: $\forall_1$ [John included $\underbrace{[[\text{the supervisor that each volunteer reported to}]]}_{\langle e, e \rangle \text{ function}} (\text{pro}_1)$]]

- (iii) The halls were still lined with the biographies she said each grade should write.

As pointed out by the reviewer, the example isn’t perfect. Under one possible interpretation, there is one biography per student, not one biography per grade (meaning that there is some metonymy going on), but there is still variation driven by the universal DP that is embedded inside the finite complement clause of *say*.

- (iv) a. A teacher said each student should write a biography.
b. *Unavailable interpretation*: For each student x, a (potentially) different teacher said x should write a biography.

⁵We are not the only one who makes these, or similar, assumptions. The most comprehensive account of varying definite readings, found in Sharvit 1999, also makes similar assumptions, albeit through a different implementation.

The first assumption, which we refer to as the ‘Function Assumption’, is that relativized DPs, like *the supervisor that each volunteer reported to*, can denote $\langle e, e \rangle$ functions (mapping volunteers to supervisors). The second, which we refer to as the ‘Covert Saturation Assumption’, is that the $\langle e, e \rangle$ function can be saturated in predication. The third, which we refer to as the ‘Universal Closure Assumption’, is that the $\langle e, e \rangle$ function can be universally closed, covertly. The Function Assumption serves to capture the perceived variation, as we will see in Section 1.4. The Covert Saturation Assumption is needed because the argument of the $\langle e, e \rangle$ function can be filled to feed predication, as we will see in Section 1.5. The Universal Closure Assumption is needed because, as we will see in Section 1.6, it is possible to covertly ‘close’ the $\langle e, e \rangle$ function, even in the absence of a universal DP.

The schematic LF in (9) is not the only way of cashing out these assumptions, but merely one way. In particular, even though we make use of covert pronouns in (9), to saturate the argument of the $\langle e, e \rangle$ function, this is just *one* way of doing so. As we will see, the crucial assumption is that, one way or another, we need a way to combine the $\langle e, e \rangle$ function with the matrix predicate (which, as we show in Section 1.5, in turn allows a matrix level operator to bind the argument of the $\langle e, e \rangle$ function).⁶ Similarly, there are alternative ways of capturing the Universal Closure assumption that do not involve a covert universal operator.⁷

In fact, the above three assumptions are not necessarily novel (cf. Sharvit 1999), but the evidence we present for them is. For this reason, our goal is to further motivate each of these assumptions. If our arguments are successful, one consequence is that a line of approach towards analyzing (2) falls out for free, with no added theoretical cost. Again, we stress that the details con-

⁶Alternatively, this can also be done in variable free semantics by z-shifting the matrix predicate (Jacobson, 1994, 2018).

⁷Alternatively, one can adopt a convention in for evaluating an individual predicate as it is applied to a function, as done in Hulsey and Sauerland 2006. This would involve a meaning for the definite article as in, (i-a), and the rule in (i-b). As far as we can tell, nothing we say here hinges on the assumption that a covert pronoun combines with the relativized DP.

- (i) a. $\llbracket \text{the} \rrbracket (P_{((ee)t)}) = f$, where f is the unique function such that
 - i. $\text{dom}(f) = \min\{\text{dom}(g) \mid g \in \text{dom}(P)\}$
 - ii. $P(f) = 1$
- b. $P_{(et)}(f_{(ee)}) = 1$ iff $\forall x \in \text{dom}(f) : P(f(x)) = 1$

cerning each assumption, and how it is formally implemented, is a matter of debate—independent of the empirical arguments that we will present. As long as *some* version of these ingredients are at our disposal, we can capture the varying definite reading.

In Section 1.3.2, we briefly situate our proposal within the literature on ‘functional relative clauses’. In Sections 1.4–1.6, we present independent evidence for our functional approach, particularly each of the three ingredients needed to analyze (2). We present data points which also involve variation of a definite, but we argue that a semantic mechanism other than QR must be the source of this variation in each case. In view of these data points, we contend that the perceived variation in all these cases is the result of a functional reading, not QR.

1.3.2 Background: functional relative clauses

In addition to the varying definite reading, it has been observed that the kind of relativized DPs that we have been considering give rise to what has been called a ‘functional reading’.⁸ This reading has primarily been discussed in relation to copular clauses (von Stechow, 1990; Jacobson, 1994; Sharvit, 1999). To illustrate, consider the example in (10-a), where we equate the relativized DP, *the supervisor that each volunteer reported to*, with the functional expression, *their best friend*.

- (10) a. The supervisor that each volunteer reports to is their best friend.
 b. “The unique (natural) function that maps each volunteer to the supervisor that they report to is the best friend function”

Under a functional reading of the relativized DP, the sentence in (10-a) can be paraphrased as in (10-b) (Stechow, 1984; Jacobson, 1994; Sharvit, 1999). Crucially, these kinds of functional relative clauses involve a distinct functional expression which is co-extensional with the relativized DP; in this case, the expression *their best friend*.⁹ For the relativized DPs discussed in this paper—so

⁸Terminology: we refer to the two readings as a ‘functional reading’ and a ‘pair-list’ or ‘varying definite reading’ (we use ‘pair-list’ and ‘varying definite’ interchangeably). Sometimes, we refer more narrowly to the interpretation of the relative clause itself; in this case, we use the terms ‘(purely) functional relative clause’ and ‘pair-list’ or ‘varying definite relative clause’.

⁹Jacobson 1994; Heim 2018 also discuss the same kind of functional relative clause in non-copular sentences, as illustrated in (i).

called ‘varying definite relative clauses’—there is no co-extensional functional expression that can be equated with the relativized DP.

That being said, as with pair-list and functional questions (cf. Groenendijk and Stokhof 1984; Chierchia 1993; Krifka 2001; Bumford 2015, 2017; Xiang 2023 and references therein), varying definite/pair-list readings and functional readings of relative clauses appear to be genuinely distinct readings. Functional relative clauses involve *natural* relations (e.g. *his mother*, *his father*, *his best friend*, etc.) and can be formed with a wide range of quantifiers, whereas varying definite/pair-list readings are restricted to *every* and *each* and allow arbitrary mappings or functions (see Sharvit 1999 for further discussion and a detailed comparison of the two readings). By an arbitrary mapping, we simply mean an extensional specification of the pairs that make up the mapping (following the literature on questions, Sharvit refers to varying definite readings as ‘pair-list readings’). With functional readings, the mapping is, in a sense, intensionally specified as it is co-extensional with a distinct functional expression, like *their best friend*. While we focus exclusively on varying definite readings, the presence of this second reading of relativized DPs in no way affects the main claim defended in this paper, which is that varying definite readings can also involve a functional parse, not only a QR one.

Much of the literature on functional relative clauses (e.g. Jacobson 1994; Heim 2018) focuses exclusively on cases like those in (10-a); there is not much discussion in the literature concerning the pair-list/varying definite cases, and how the two readings compare. By contrast, Sharvit 1999 discusses both readings and argues that, despite their distinct empirical distributions, both still underlyingly involve functional dependencies, meaning functional parses are more widespread than previously assumed.¹⁰ One consequence of our proposal is to provide further support for this claim: although varying definite/pair-list and functional relative clauses differ in important

-
- (i) No third grade boy invited the woman that every fourth grade boy invited (namely, his mother).

In (i), the ‘namely continuation’ containing the functional expression *his mother* supplies the relevant co-extensional function, and the sentence as a whole is understood to convey that no third grade boy invited his (own) mother.

¹⁰Sharvit’s argument that pair-list relative clauses denote $\langle e, e \rangle$ functions comes primarily from the assumption that QR is clause-bounded. Since this is precisely what we are discussing in this paper, we do not share this assumption and so we provide a novel argument. According to Sharvit, another argument that pair-list relative clauses denote $\langle e, e \rangle$ functions comes from subject-object asymmetries. We briefly return to this issue in the conclusion.

ways, both admit a functional interpretation of the relative clause. In this respect, our proposal is in the same spirit as Sharvit’s, in that it analyzes varying definite relative clauses via a functional parse and thereby strengthens the conclusion that functional parses are not only found in (purely) functional relative clauses.

At the same time, we remain somewhat agnostic regarding the exact compositional pathway to varying definite readings, and more specifically, how functional pair-list/varying definites are constructed. With regard to the internal composition of the relative clause, the implication of our central argument is merely that some compositional pathway to functional pair-list/varying definites must exist, strengthening a conclusion previously reached in Sharvit (1999). Importantly, our argument does not bear on what the proper compositional pathway to these definites may be. The difference between our proposal and Sharvit’s is then more one of taste, since both proposals involve similar ingredients. In particular, Sharvit implicitly makes the same three assumptions that we made above. I am simply spelling out a proposal where some of these assumptions correspond to separate parts of the LF and, as a result, they are implemented in different parts of the syntax (i.e., a covert universal operator and covert pronoun). In contrast, Sharvit builds them into a single semantic operator. While our goal in this paper is not to advocate for our proposal over Sharvit’s, we do believe that one benefit of ours is that we motivate each of the necessary components through independent evidence, as we will see next.¹¹ Under this line of thought, a functional analysis of varying definite readings comes at no real added theoretical cost.¹²

While we remain agnostic concerning what the *right* compositional pathway to functional varying definites is, for concreteness, in Section 1.4.2 we do outline one possible internal composition

¹¹In contrast, as we understand it, Sharvit’s analysis relies on two key components. The first is a rule of ‘absorption’, following Chierchia 1993’s analysis of pair-list questions. This rule converts the relative operator and the quantifier into a single constituent. The second component is a semantic rule (i.e., a type-shifting operator) that combines together the universal quantifier, the rest of the relative clause, the head of the relative clause, the determiner and the rest of the sentence. As part of its semantic contribution, this rule extracts the minimal witness set of the universal quantifier and feeds it into a distributivity operator—which is also built into the semantics of the rule—to mimic the effects of wide scope QR. We will not go into the details of Sharvit’s analysis here, but simply note that her analysis differs substantially from ours in certain regards. We leave a proper comparison of both proposals to future research since the main claim being made in this paper is simply that varying definite readings can be captured through a functional interpretation of the relativized DP; we are not necessarily advocating for our proposal over Sharvit’s.

¹²A proper comparison of the functional relative clauses found in copular clauses and those found in varying definite readings is beyond the scope of this paper, but we view this as important work for future research.

of the relative clause. The compositional structure that we propose is based largely on shared background assumptions from the literature on functional relative clauses/questions (function variables of type $\langle e, e \rangle$, binding of function variables by the *wh*-trace, etc.).

1.4 Function Assumption

1.4.1 A novel argument

We begin by making the argument that, in certain cases, the relativized DP in (2) requires a functional parse. This argument comes from examples like (11), involving the same relativized DP as (2) but a different matrix predicate. The intended interpretation is provided below.

(11) The supervisor that each volunteer reported to varied.

Intended: different volunteers had different supervisors that they reported to.

Like our initial example in (2), the sentence in (11) permits variation of supervisors: the value of *the supervisor that each volunteer reported to* can vary by volunteer.¹³ Furthermore, variation of supervisors is permitted even though the truth conditions for (11) do not involve wide scope universal quantificational force. This suggests that that we need a source of variation that is not the result of the embedded universal DP taking wide scope. To clearly establish this point, we can

¹³It is worth noting that sentences with the predicate *vary* are not isolated examples. There are other examples which illustrate the same point, as shown in (i).

- (i)
 - a. John sorted the time that each participant took (in descending order).
 - b. The time that each trip took fluctuated.
 - c. The weight that each assignment was given was uneven.
 - d. The output that each input is mapped to follows a simple rule.
 - e. The meaning that each word receives is systematic.
 - f. The route that each delivery driver took changed frequently.
 - g. Mary compared the proposal that each committee member submitted.
 - h. The temperature that each sensor recorded diverged significantly.
 - i. The strategy that each team adopted differed.
 - j. The amount that each donor contributed ranged from \$10 to \$1000.
 - k. The grade that each student received depended on the grader.

For some of the examples in (i), QR would deliver a sensible interpretation. For example, the QR interpretation of (i-d) is: for each input *x*, the output that *x* is mapped to follows a simple rule. However, this interpretation is compatible with a different rule applying for each input and, furthermore, does not capture the intuition that one rule governs the entire input–output mapping. For these reasons, we also believe that the observed variation in these kinds of examples should not be captured through QR, but another mechanism.

look at what happens when the embedded universal DP undergoes QR out of the relative clause, as illustrated by the LF in (12-a) with the resulting truth conditions in (12-b).

- (12) a. *LF*: [[each volunteer] λ_1 [[the supervisor [that t_1 reported to]] [varied]]]
 b. *Interpretation*: for each volunteer x , the supervisor that x reported to varied

In contrast to what we saw for (2), QR of *each volunteer* derives the wrong truth conditions in this case. As (12-b) indicates, the interpretation that results from QR of the universal DP does not correspond to the intended truth conditions for (11). Instead, we get an interpretation where, for any given volunteer, their respective supervisor changes, presumably over time. In other words, one natural way to make sense of the interpretation in (12-b), is to implicitly assume that each volunteer's supervisor varies over time. Crucially though, this kind of time dependence delivers the wrong kind of variation (and the wrong interpretation) in (11). To further clarify this point, consider what happens when we use a slightly different relative clause, as in (13-a).

- (13) a. The supervisor that each volunteer first reported to varied.
 b. *QR interpretation*: for each volunteer x , the supervisor x first reported to varied

The difference in (13-a) is that the predicate in the relative clause, *first reported to*, cannot vary over time: each volunteer has a unique supervisor that they *first* reported to. As a result, the interpretation that results from QR, paraphrased in (13-b), is not a sensible interpretation. However, (13-a) can still receive the same interpretation as (11), where different volunteers had different supervisors that they first reported to. Therefore, we conclude that we need an alternative route to capturing the observed variation in (11) and (13-a)—it cannot be due to QR.

What we would really like to capture is that variation takes place across volunteers in these examples. That is, (11) conveys that the volunteers don't all have the same supervisor. Intuitively then, the predicate *vary* makes a claim about the volunteer-supervisor pairings: it requires at least two different volunteers to report to different supervisors. This suggests that *vary* is a predicate that operates over functions from individuals to individuals, taking a functional argument of type $\langle e, e \rangle$. We can capture these intuitions through the denotation for *vary* in (14).

$$(14) \quad \llbracket \text{vary} \rrbracket = \lambda f_{\langle e, e \rangle}. \exists y, z \in \text{dom}(f)[f(y) \neq f(z)]$$

In (14), *vary* takes an $\langle e, e \rangle$ function as argument and asserts that there are two inputs of that function which map to different outputs.

Returning to our example in (11), to get the right truth conditions, we then need the relativized DP to denote an $\langle e, e \rangle$ function. Furthermore, this function should map volunteers to supervisors. To this end, we assume the domain of the function is determined by the NP restrictor of the embedded universal (i.e., the set of volunteers), and the output is the unique supervisor that each volunteer reports to. Glossing over the internal composition of the relativized DP for now, we simply assume it denotes the $\langle e, e \rangle$ function in (15).

$$(15) \quad \llbracket \text{the supervisor that each volunteer reported to} \rrbracket \\ = \text{the function that maps each volunteer to the supervisor that they reported to}$$

Since the relativized DP in (15) denotes an $\langle e, e \rangle$ function, we can compose it directly with *vary*, which is of type $\langle \langle e, e \rangle, t \rangle$. This results in the desired truth conditions, shown in (16).

$$(16) \quad \llbracket \text{the supervisor that each volunteer reported to varied} \rrbracket = 1 \text{ iff} \\ \exists y, z \in \text{dom}(f)[\text{the sup. that } y \text{ reported to} \neq \text{the sup. that } z \text{ reported to}] \\ = \exists y, z \in \llbracket \text{volunteer} \rrbracket [\text{the sup. that } y \text{ reported to} \neq \text{the sup. that } z \text{ reported to}]$$

In sum, the example in (11) provides evidence for a functional interpretation of the relativized DP. This is because one straightforward way of capturing the intended interpretation makes direct use of the $\langle e, e \rangle$ function in the semantic composition by predicating something of this function. Positing that the universal DP undergoes QR out of the relative clause does not even help in this case as it leads to the wrong truth conditions.

At this stage, we would like to make an observation which constrains the $\langle e, e \rangle$ function that we make use of in the semantic composition, in this case. We implicitly assumed that the domain of the function in (15) ranges over the set of volunteers. It is important to note that this implicit assumption is in fact necessary: the set of volunteers must exhaust the domain of the function.

If the domain merely included the volunteers, and also contained other non-volunteer individuals, this would lead to the wrong truth conditions in (16), as the resulting truth conditions would be too weak. The sentence would be predicted to be true in the following scenario: assume the domain of the function is the set of volunteers plus Mary, and all the volunteers report to the same supervisor, but Mary reports to a different supervisor. In this scenario, such a function would lead to the truth conditions in (17), where y could be any volunteer and z could be Mary.

- (17) $\llbracket \text{the supervisor that each volunteer reported to varied} \rrbracket = 1$ iff
 $\exists y, z \in \text{dom}(f) [\text{the sup. that } y \text{ reported to} \neq \text{the sup. that } z \text{ reported to}]$
 $= \exists y, z \in \llbracket \text{volunteer} \rrbracket \cup \{\text{Mary}\} [\text{the sup. that } y \text{ reported to} \neq \text{the sup. that } z \text{ reported to}]$

That (17) is predicted to be true in the above scenario does not align with speaker intuitions concerning the meaning of (11), and so we must make sure to rule out such scenarios. This suggests the following restriction on our $\langle e, e \rangle$ function (in section 1.4.2, we will discuss a mechanism which ensures that the domain of the function is, in fact, the minimal one, capturing this constraint):

- (18) **Restriction on $\langle e, e \rangle$ function:** the NP restrictor of the universal quantifier must exhaust the domain of the function (i.e., the function has the *minimal* domain supplied by the NP restrictor of the universal quantifier).

Before moving on, we would like to discuss a potential alternative analysis of (11). The *vary* data could be accounted for by an alternative QR analysis. Under this analysis, the relativized DP in (11) would have an inverse-linking style LF where the universal DP takes scope at the edge of the whole DP, and the wide-scoping universal quantifier is interpreted as generalized sum-forming conjunction, creating a plurality of individuals. This kind of analysis is proposed in Bumford 2022 and is motivated by examples like (19), where the DP, *two Senators from each State*, is clearly interpreted as a sum of 100 senators.

- (19) The Senate of the United States shall be composed of two Senators from each State.

How would this apply to the example in (11)? If the relativized DP is instead interpreted via

inverse linking and sum-forming universal quantification, we have an alternative way of capturing the intended truth conditions of (11). This is because the relativized DP will be semantically plural, and the example will receive an interpretation along the same lines as *the supervisors (the volunteers reported to) varied*. Under this analysis, *vary* would instead take a plural argument and assert that two of the atomic members of said plurality are different. In other words, a plurality would vary if it is non-atomic, as illustrated in (20). These truth conditions are very similar to the ones in (16).¹⁴

$$(20) \quad \llbracket \text{vary} \rrbracket = \lambda X_e. \exists y, z \sqsubseteq X[y \neq z]$$

However, we are skeptical that this kind of analysis, in fact, extends to the cases of interest in (11). For starters, consider (21), which are variants of the example in (19).

- (21) a. The senate shall be composed of the delegate from each state.
b. #The senate shall be composed of the delegate that each state sent.

In (21-a), once again, the DP can be interpreted as a sum of 50 delegates, similar to (19). In contrast, our judgment is that (21-b) sounds degraded and the DP cannot be interpreted as a sum of 50 delegates. In fact, even Bumford (2022) claims that this kind of inverse-linking LF is island sensitive and would not apply to cases like (21-b). Therefore, given the contrast in (21), we do not believe that this alternative analysis is better suited to handle the example in (11) than the analysis outlined above.¹⁵

¹⁴As a reviewer points out, what's required by *vary* is quite vague, and so, maybe it is not too surprising if it can take different kinds of arguments: a functional DP, of type $\langle e, e \rangle$, and a plural argument of type e . We are simplifying for our purposes here. See also footnote 15.

¹⁵Relatedly, another reviewer points out another possible objection to the proposed argument against QR. The reviewer notes that the problem that arises for QR, illustrated by (11), is a much more general problem where universal quantification fails to license distributive inferences. To illustrate, consider the example in (i), which was brought to our attention by the reviewer.

- (i) Each state varies in its custodial rulings.

The intended interpretation of (i) is one where the custodial rulings are not all the same across states. As a result, from the sentence in (i), we cannot conclude that, for example, Louisiana varies in its custodial rulings. In other words, it seems like the issue with QR that was identified for (11) is the same issue that arises with (i). Therefore, a solution to (i) might provide a solution for (11). For instance, as the reviewer points out, one solution to (i) could draw on the literature on sentence-internal *different*, where a similar failure of distributivity inferences is observed (Brasoveanu,

1.4.2 A note on composing functions

Our goal in this section has simply been to make an empirical argument that relativized DPs can denote $\langle e, e \rangle$ functions. If this argument is accepted, then we have evidence for the Function Assumption. That is, we learn that there must be *some* compositional pathway available that delivers an $\langle e, e \rangle$ function—but what exactly this compositional pathway consists of, and *how* it comes to denote this function is not necessarily our concern, but a separate issue. To this end, for the discussion at hand, the details concerning the internal composition of the relative clause are not our central focus, all that matters for our purposes is that the whole relativized DP ends up denoting an $\langle e, e \rangle$ function. That being said, we can still ask if this is compatible with what we know about semantic composition and, as far as we know, there is no evidence that there *cannot be* a possible compositional pathway to $\langle e, e \rangle$ functions. In fact, Sharvit 1999 spells out a proof of concept that this is compositionally possible—though, nothing we have said relies on Sharvit’s particular

2011). Drawing on this kind of analysis, we would be able to derive a reading like ‘each state varies in its custodial rulings from the other states’. While the problem illustrated by (i) may be related to the problem concerning (11), we do not believe they are exactly the same. In (11), the definite DP is headed by a singular definite. To create a minimal pair with (i), though, we need to use a plural definite, as illustrated in (ii).

- (ii) a. The states varied in their custodial rulings.
- b. ?The state varied in its custodial rulings.

In fact, in (iii) below, we observe that we can even create a variant of (11) where we remove the relative clause but keep the singular definite and the same interpretation is still possible (Palucci, 2024).

- (iii) The supervisor varied (across volunteers).

Further evidence that the relativized DPs we are discussing do not pattern like regular universal DPs, as in (i), is provided in (iv). When we look at other predicates of functions, like *fluctuate*, we see that we can replicate the problem illustrated in (11), even though there does not appear to be a minimal pair with a matrix universal DP.

- (iv) a. The time that each trip took fluctuated.
- b. ??Each trip fluctuated in time.

Furthermore, in our discussion, we have simplified the semantics of *vary*. However, we believe that a further exploration suggests that *vary* can take two different kinds of arguments: a functional one, as proposed here, and a plural one—similar to sentence-internal *different*. In examples like (i), we propose that *vary* takes a plural argument, as it does in (ii-a) as well.

Therefore, we hypothesize that, while superficially similar, (11) and (i) may actually involve different solutions. That being said, it is very possible that the example in (i) can be analyzed using the kind of sum-forming universal quantification proposed in Bumford 2022, while the example in (11) involves a functional parse of the relativized DP. This would explain the differences between the two cases, and why the universal quantification in cases like (i) behaves more similarly to overt plural definites, like (ii-a). We leave this hypothesis to future research, though.

composition being correct. For completeness, we now illustrate another possible way—but certainly not the only way—of spelling out the internal composition, one that draws on elements from Jacobson 1994; Sharvit 1999; Heim 2018. We stress that we are not committed to the particular details in this section, but are merely illustrating how we can derive an $\langle e, e \rangle$ function.

Under this approach, we assume that the *wh*-trace has a complex syntax: it translates as a variable, *f*, over functions of type $\langle e, e \rangle$ applied to a variable over individuals (Groenendijk and Stokhof, 1984; Engdahl, 1980; Heim, 2018), which we represent syntactically as a pronoun. This variable over individuals gets bound by the embedded quantifier while the variable *f* is bound by the *wh*-trace.

As we have already seen, letting the relativized DP denote an $\langle e, e \rangle$ function which maps supervisors to volunteers allows us to capture the variation of supervisors by volunteers. This means that the embedded universal DP takes scope locally within the relative clause and contributes to the formation of our $\langle e, e \rangle$ function by supplying the domain of the function through its NP restrictor (Groenendijk and Stokhof 1984; Chierchia 1993; Sharvit 1999; Heim 2018, a.o.). Even with this assumption, we still permit variation of supervisors due to the fact that the resulting function can map each volunteer to a different supervisor.¹⁶ Now, consider the LF in (22).

$$(22) \quad LF: \text{the } [\text{Min } [\text{supervisor } [\text{wh } \lambda f. [\text{each volunteer }] \lambda_1. t_1 \text{ reported to } f(\text{pro}_1)]]]$$

In addition to the function variable of type $\langle e, e \rangle$ and the covert pronoun, the LF in (22) also makes use of a polymorphic entry for the definite determiner, (14), and a ‘Min’ operator, which we discuss below.

$$(23) \quad \llbracket \text{the} \rrbracket = \lambda P_{\sigma t}: \exists! x_{\sigma} [P(x)]. \iota x_{\sigma} [P(x)]; \sigma = \langle e, e \rangle \text{ in this case}$$

We begin with (90).

$$(24) \quad \llbracket \lambda_1. t_1 \text{ reported to } f(\text{pro}_1) \rrbracket^g \\ = \lambda x: x \in \text{dom}(g(f)). \text{reported-to}(g(f)(x))(x)$$

¹⁶This is one of the main reasons for appealing to $\langle e, e \rangle$ functions in the first place: they generally allow us to replace wide-scope universals, like $\forall x \exists y [R(x, y)]$, with narrow-scope ones, like $\exists f \forall x [R(x, f(x))]$ (Hintikka, 1986).

In (90), we assume that a partial function arises from a presupposition associated with the complex trace, namely, that x is in the domain of the function, f . Next, the embedded universal DP binds the individual variable and the *wh*-trace binds the function variable, (25).

$$(25) \quad \llbracket \text{wh } \lambda f. [\text{each volunteer}] \lambda_1. t_1 \text{ reported to } f(\text{pro}_1) \rrbracket^g \\ = \lambda f_{\langle e, e \rangle}: \forall x[\text{volunteer}(x) \rightarrow x \in \text{dom}(f)]. \\ \forall x[\text{volunteer}(x) \rightarrow \text{reported-to}(f(x))(x)]$$

In (25), the embedded universal DP scopes inside the relative clause but we still get variation of supervisors due to the function, f . Furthermore, abstracting over the function variable, f , and assuming universal projection of the presupposition associated with the complex trace, this yields a predicate of functions that i) presupposes that every volunteer is in the domain of the function, f , and ii) maps every volunteer to someone they reported to. The idea behind the universal presupposition in (25) is that, since the complex trace presupposes that x is in the domain of f , when x gets bound by the universal quantifier, the presupposition associated with the complex trace projects universally, leading to the presupposition that every volunteer is in the domain of f (cf. Heim 1983; Chemla 2009; Sudo et al. 2012; Sudo 2014; Heim 2018).

Following Jacobson (1994), we also assume a classic head external analysis of the relative clause where the head noun of the relative clause gets interpreted in its surface position (see e.g. Heim and Kratzer 1998). This means that, for it to combine with the rest of the relative clause, we must assume that it can be type-shifted from a predicate of individuals, of type $\langle e, t \rangle$ to a predicate of functions, of type $\langle \langle e, e \rangle, t \rangle$ (Jacobson, 1994). As a result, the head noun, *supervisor*, denotes the set of supervisor-valued functions, (26).

$$(26) \quad \llbracket \text{supervisor} \rrbracket^g = \lambda f_{\langle e, e \rangle}. \forall x \in \text{dom}(f)[\text{supervisor}(f(x))]$$

Since the head noun and the relative clause both denote predicates of function, we can now simply combine (25) and (26) intersectively to derive the following predicate of functions.

$$(27) \quad \llbracket [\text{supervisor} [\text{wh } \lambda f. [\text{each volunteer}] \lambda_1. t_1 \text{ reported to } f(\text{pro}_1)]] \rrbracket^g$$

$$= \lambda f_{\langle e, e \rangle}: \quad \forall x[\text{volunteer}(x) \rightarrow x \in \text{dom}(f)].$$

$$\forall x \in \text{dom}(f)[\text{supervisor}(f(x))] \wedge \forall x[\text{volunteer}(x) \rightarrow \text{reported-to}(f(x))(x)]$$

In (27), we now have a predicate of functions that i) presupposes that every volunteer is in the domain of the function, f , and ii) maps every volunteer to a supervisor they reported to. Through presupposition projection, then, we see that the embedded NP restrictor of the universal constrains the domain of the function by ensuring that the domain includes the set of volunteers. At this stage, though, we recall what we saw in the previous subsection: that there is a further constraint on our $\langle e, e \rangle$ function insofar as it must be the *minimal* function from volunteers to supervisors.^{17,18} That is, we saw independent evidence that the function's domain does not only include the NP restrictor of the universal quantifier—i.e., the set of volunteers—but it is *exactly* this set, and nothing more.

This brings us to the one component of our LF in (22) that is novel insofar as it is not part of the background literature on functional relative clauses. One way of capturing this observation is to posit a silent minimization operator in the object language which takes a predicate of $\langle e, e \rangle$ functions and returns a set of $\langle e, e \rangle$ functions, as defined in (28). We emphasize that this way of capturing the minimality requirement is not the only way of capturing the observation that the $\langle e, e \rangle$ function ranges over the minimal domain, it is just one that has been appealed to in other cases where the same issue arises, as in pair-list questions for instance (cf. Pafel 1999; Preuss 2001; Grosu and Krifka 2007; Fox 2012; Xiang 2023).

$$(28) \quad \llbracket \text{Min} \rrbracket (K_{\langle \langle e, e \rangle, t \rangle}) = \{f \in K : \forall f' \in K[f \subseteq f']\}$$

¹⁷In Section 1.6, we will discuss examples which highlight another important point concerning the $\langle e, e \rangle$ function that is referenced in the composition. Above, we proposed that the domain of the $\langle e, e \rangle$ function is determined by the universal DP embedded in the relative clause. But the examples in Section 1.6 will show us is that, while this is a straightforward way of determining the domain of the function, it is not the only way; the context can also determine the function domain.

¹⁸The same issue arises with the varying definite reading as well. To foreshadow a bit, if the $\langle e, e \rangle$ function did not range over the minimal domain, we would derive truth conditions which are too strong. If the domain of the function merely contained the set of volunteers, but also contained other non-volunteer individuals, we would get different truth conditions since the covert universal is restricted to the same domain as the function. For example, assuming the domain of the function includes the set of volunteers and Mary, we would derive the truth conditions in (i).

(i) $\llbracket \forall_2 [\text{John included } [[\text{the supervisor that each volunteer reported to}] (\text{pro}_2)]] \rrbracket$
 $= 1 \text{ iff } \forall y[y \in \llbracket \text{volunteer} \rrbracket \cup \{\text{Mary}\} \rightarrow \text{included}(t_x[\text{supervisor}(x) \wedge \text{reported-to}(x)(y)])(\text{John})]$

For functions, we assume that the entailment relation at play in (28) is set inclusion concerning the extension of the function (viewing functions as sets of ordered pairs): if a function g is a subset of a function h , then the domain of h includes the domain of g , and h agrees with g on all members of the domain of g . For the case at hand, then, any function which maps each volunteer to a supervisor, but also maps non-volunteers to a supervisor, will be ruled out since it will be entailed by the minimal functions which map only the volunteers to a supervisor.

In (29), we apply the minimization operator to the predicate of functions in (25).

$$\begin{aligned}
 (29) \quad & \llbracket \text{Min} [\text{supervisor} [\text{wh } \lambda f. [\text{each volunteer}] \lambda_1. t_1 \text{ reported to } f(\text{pro}_1)]] \rrbracket^g \\
 & = \text{the set of minimal functions that map every volunteer to a supervisor that they report to} \\
 & = \{ f: \lambda x_e: \text{volunteer}(x) \wedge \exists! y [\text{supervisor}(y) \ \& \ f(x) = y]. \iota y [\text{reported-to}(y)(x)] \}
 \end{aligned}$$

This operator ensures that, not only is the set of volunteers included in the domain, but the set of volunteers exhausts the domain: the function is only defined over the minimal domain provided by the NP restrictor of the embedded universal.

In (29), we are assuming that every volunteer reports to a unique supervisor. But what justifies this assumption? We have seen that the Min operator rules out potential functions that extend the domain beyond the set of volunteers, but determining a unique function can also fail if any of the volunteers in the domain report to multiple supervisors (since then, multiple functions over a minimal domain can be constructed). To illustrate, let us assume that the set of volunteers is Ann, Bill and Carol and that Ann reported to supervisor 1, Bill to supervisors 2 and 4 and Carol to supervisor 3. In this situation, there are two possible functions over the minimal domain of volunteers, shown in (30).

$$\begin{aligned}
 (30) \quad & \text{a. } f_1 = \{ \langle \text{Ann}, \text{supervisor}_1 \rangle, \langle \text{Bill}, \text{supervisor}_2 \rangle, \langle \text{Carol}, \text{supervisor}_3 \rangle \} \\
 & \text{b. } f_2 = \{ \langle \text{Ann}, \text{supervisor}_1 \rangle, \langle \text{Bill}, \text{supervisor}_4 \rangle, \langle \text{Carol}, \text{supervisor}_3 \rangle \}
 \end{aligned}$$

This means that, for there to be a unique function that the definite determiner can reference, a further constraint needs to be satisfied: that each volunteer reported to a single supervisor. We simply take this to be a condition that constrains the availability of varying definite readings to

satisfy the uniqueness requirement of the definite determiner (see Sharvit 1999 for more on this uniqueness requirement). Since the definite determiner presupposes that there is a unique function, each volunteer must report to a unique supervisor: a unique function over a minimal domain implies that each element in the domain—i.e., each volunteer—is mapped to a *single* output—i.e., supervisor. Therefore, under the assumptions that (i) a silent minimization operator in the object language applies at LF and (ii) the definite determiner carries a presupposition that there is a unique function, the uniqueness presupposition falls out.¹⁹

As a result, the definite determiner picks out the unique function from the set in (29), leading to (31).

$$\begin{aligned}
 (31) \quad & \llbracket \text{the} \llbracket \text{Min} \llbracket \text{supervisor} \llbracket \text{wh } \lambda f. \llbracket \text{each volunteer} \rrbracket \lambda_1. t_1 \text{ reported to } f(\text{pro}_1) \rrbracket \rrbracket \rrbracket^g \\
 & = \text{the unique function, } f, \text{ mapping each volunteer, } x, \text{ to the sup. they reported to, } f(x) \\
 & = \iota f_{\langle e, e \rangle}: \text{dom}(f) = \llbracket \text{volunteer} \rrbracket. \\
 & \quad \forall x \in \text{dom}(f) [\text{supervisor}(f(x))] \wedge \forall x [\text{volunteer}(x) \rightarrow \text{reported-to}(f(x))(x)]
 \end{aligned}$$

Having presented one way of illustrating the internal composition of the relative clause, we can also now revisit the generalization from Section 1.2 concerning the examples where the universal DP is embedded inside a scope island within the relative clause. We proposed that these examples must have a particular profile that is hard to come by. We can now be more explicit in stating this generalization and why it comes about. The configurations needed for these examples must satisfy the following three requirements: (i) the universal DP must be in a scope island (e.g. under *claim*), (ii) the complex *wh*-trace must be within that same scope island, so that the universal DP can bind

¹⁹It is worth pointing out that the picture is more complicated when the head of the relative clause contains plural marking, as in (i).

- (i) John included the supervisors that each volunteer reported to.

Consider (i) in a scenario where, again, Ann, Bill and Carol are the volunteers. Ann reported to supervisors 1 and 2, Bill to supervisors 3 and 4, and Carol to supervisors 5 and 6. The varying definite reading of (i) can be paraphrased as: for each volunteer *x*, John included both of the supervisors that *x* reported to. However, in this scenario, even after appealing to a minimization operator, there is no unique function mapping volunteers to supervisors. Instead, what we are intuitively after is the maximally informative function mapping volunteers to supervisors, as this would be the function that maps each volunteer to *both* of their supervisors. This suggests that, in this case, the semantics of the definite determiner needs to be ‘richer’ insofar as it should reference maximal informativity (Von Stechow et al., 2014), instead of a Fregean definite which simply picks out the unique member of the NP extension. We leave a further investigation of this to future research.

into the *wh*-trace, that is, bind the argument of the function variable, and (iii) the *wh*-trace must not be on an island for *wh*-movement (since relativization will then lead to ungrammaticality). This kind of configuration is illustrated in (32), where we have a complex *wh*-trace consisting of a function variable, *f*, and an individual argument, *pro*₁ and the universal DP binds the argument, *pro*₁. At the same time, the object of *reports to* is still able to overtly move out of the finite complement clause since this is not an island for overt movement.

(32) [_{DP} the supervisor that [you claimed [[each volunteer] λ_1 *t*₁ reports to *f*(*pro*₁)]]]

This is the reason we focused on finite complement clauses, since they meet all three requirements; unlike other scope islands for universal DPs.^{20,21}

²⁰An anonymous reviewer raises the data point in (ii), which is a prediction of our analysis. We begin with the simpler example in (i) though—due to the complexity of (ii), especially the pronoun *them*. In (i), the universal DP cannot scope over the relativization trace, since the trace is outside the finite complement of *claim*. As a result, our analysis predicts that (i) should not have a varying definite reading.

(i) John included the supervisor that _ claimed each volunteer reported to me.

According to the author's intuitions, this prediction is borne out: (ii) does not seem to license a varying definite reading. As pointed out by the reviewer, the judgment concerning (ii) is more subtle and not unequivocal, and we agree.

(ii) John included the supervisor that _ claimed each volunteer reported to them.

In any case, it is worth pointing out that the judgment concerning (ii) does at least seem to go in the predicted direction: it is more difficult to get the varying definite reading in (ii) than in (6).

²¹An anonymous reviewer also points out that nothing in our analysis of varying definite readings restricts the nature of the embedded quantifier. Furthermore, given that the matrix level universal quantification comes from an independent operator, we might expect a varying definite reading to be available for the sentence in (i), contrary to fact.

(i) Mary contacted the woman that no boy decided to invite.

This raises two related points worth unpacking. First, while the matrix level operator provides the universal force in our analysis of varying definite readings, it is not exclusively tied to varying definite readings. To illustrate, consider (ii), where the relative clause receives a purely functional interpretation (not a varying definite one), and yet we still observe matrix level universal force since the intended reading can be paraphrased as: for every boy *x*, Mary contacted *x*'s sister. Presumably, the intended reading of (ii) can be captured through the same silent universal operator.

(ii) Mary contacted the woman that no boy decided to invite (namely, his sister) to let them all know.

Second, it is true that nothing in our account, as formulated, restricts the nature of the embedded quantifier, when it comes to varying definite readings. This is because the domain of the function is specified in the presuppositional content, as outlined above, and negative quantifiers also project their presuppositions universally (Chemla, 2009; Sudo et al., 2012; Sudo, 2014). As a result, applying Min to the predicate of functions in (iii) leads to a function compatible with a scenario where every boy gets mapped to the unique woman he didn't invite.

(iii) $\llbracket \llbracket \text{woman} [\lambda f. [\text{no boy}] \lambda_1. t_1 \text{ invited } f(\text{pro}_1)] \rrbracket \rrbracket^g$

Having presented an argument for the Function Assumption, in the next section, we present another piece of data in support of this. We also go one step further and argue that, not only do we need a functional parse of relativized DPs, but there is also evidence that these kinds of functional DPs need to be saturated, in order to feed predication. This data provides independent evidence for the Covert Saturation Assumption.

1.5 Covert Saturation Assumption

In this section, we present another attested reading which involves variation of a definite. Crucially, though, we contend that the source of this variation is once again distinct from QR, instead involving a functional parse. Furthermore, we argue that the reading discussed in this section provides support for the Covert Saturation Assumption, since it involves saturating the functional definite DP with a covert argument that can seemingly be bound by another operator higher in the structure. The resulting reading is then akin to a bound variable reading, when the relative clause contains an overt pronoun.

We begin by considering the dialogue in (33). The target sentence in (33-b) involves a structurally similar object DP as (2)—a definite DP modified by a relative clause containing a universal DP—but the matrix subject is changed.

- (33) a. A (to B and C): What did you guys do in the library last night?
b. B: We revised the paper that each of us wrote for the seminar.

Given the context, the first person plural pronoun in (33-b) refers to the individuals B and C. Given the dialogue in (33), B's response in (33-b) can be understood as conveying the following interpretation: for each x in $\{B, C\}$, x revised the paper that x wrote. We refer to this interpretation as the

$$= \lambda f_{\langle e, e \rangle}: \quad \forall x[\text{boy}(x) \rightarrow x \in \text{dom}(f)]. \\
\quad \forall x \in \text{dom}(f)[\text{woman}(f(x))] \wedge \neg \exists x[\text{boy}(x) \wedge \text{invited}(f(x))(x)]$$

Therefore, we would expect that, in contexts where each boy didn't invite exactly one woman, a varying definite reading should be possible with an embedded negative quantifier, contrary to fact. This is a problem with the analysis, as formulated here. However, if one adopts the more involved internal composition in Sharvit 1999, this problem can be avoided. There are also ways of modifying the internal composition presented here to address this issue. This would take us too far afield, though, as it is not directly relevant to the arguments being made in this paper. We note that this is also a well-known puzzle in the realm of functional questions.

‘dependency reading’. We observe that (33-b) is similar to the varying definite reading insofar as the intended reading of (33-b) also permits variation of papers by individuals: B and C revised their own papers (the dependency reading also conveys a universal uniqueness presupposition since it presupposes that B and C each wrote a single paper). Furthermore, the paraphrase of the truth conditions also involves wide scope universal force. Unlike the varying definite reading though, the matrix subject is not singular, but plural.²²

In fact, the truth conditions for the dependency reading parallel the truth conditions of a bound variable reading where a universal matrix subject binds a variable in the relative clause. We can highlight this parallel by looking at the bound variable reading in (34).

- (34) a. Each student revised the paper that she wrote for the seminar.
 b. *LF*: [each student] λ_1 [t_1 revised the paper that she₁ wrote for the seminar]

This parallel with bound variable readings suggests a possible way of analyzing the dependency reading: let the matrix subject in (33-b) get interpreted distributively and have the distributive subject bind a pronoun in the relative clause. The challenge is that, in (33-b), there is no overt singular pronoun within the relative clause to be bound. In contrast, in (34), we have an overt singular pronoun, *she*, inside the relative clause. Effectively, what we want then is for the relativized DP in (33-b) to be interpreted as if it contained a pronoun, in order to establish the desired dependency between the matrix and embedded subjects. One possible way of achieving this, drawing on Heim 2018, is as follows: assume the relativized DP denotes an $\langle e, e \rangle$ function, as in (35-a), and feed the whole DP a covert pronoun, as illustrated in (35-b).

- (35) a. $\llbracket$ the paper that each of us wrote $\rrbracket$
 = the function that maps each of B and C to the paper that they wrote
 b. $\llbracket$ [the paper that each of us wrote] (pro_y) $\rrbracket$

²²We note that the dependency is also available with other distributive matrix subjects, like a distributive definite plural subject, (i), and a universal quantifier, (ii).

- (i) Yesterday in class, the students revised the paper that each of them wrote for the seminar.
 (ii) Yesterday in class, each of the students revised the paper that each of them wrote for the seminar.

= the paper that y wrote (for any y)

In (35-b), the covert pronoun saturates the argument of the $\langle e, e \rangle$ function and returns an e type output: y's paper. In doing so, it also supplies a variable that the matrix subject can bind.

Positing a covert pronoun to saturate the functional DP also bears on a related compositional challenge that arises, concerning functional DPs in argument positions more generally. The compositional challenge is that the functional DP in object position is of type $\langle e, e \rangle$, but the matrix predicate, *revised*, expects an argument of type e , as in (36). Thus, somehow, we need a way to turn something of type $\langle e, e \rangle$ into something of type e . Feeding a covert pronoun into the $\langle e, e \rangle$ function enables the DP to compose with the matrix predicate by resolving this type-mismatch (we note that the exact same problem arises with the matrix predicate, *include*, in the varying definite reading of (2)). The output in (35-b) is the paper that y wrote (assuming y is the value of the covert pronoun that serves as input to our function), a suitable argument for the lexical entry in (36). In a sense then, the need to saturate the argument of the functional DP—to feed matrix predication—gives us a possible bindee for the matrix subject.

$$(36) \quad \llbracket \text{revised} \rrbracket = \lambda x_e. \lambda y_e. \text{revised}(x)(y)$$

Pursuing this line of analysis, a possible LF for the dependency reading in (33-b) is given in (37-a).

- (37) a. *LF*: We D [λ_1 [t_1 revised [[the paper that each of us wrote] (pro_1)]]]
(where 'D' is a distributive operator)
- b. *Resulting interpretation*: for each y in {B, C}, y revised the paper that y wrote

In (37-a), the covert pronoun pro_1 that saturates the function provides a variable that gets bound by the matrix subject, the first person plural pronoun. This results in the truth conditions in (37-b). Effectively, the use of a covert pronoun, and the subsequent binding by the matrix subject, creates the desired dependency between the matrix and embedded subjects, and allows us to capture the desired truth conditions. In fact, under this analysis, the dependency reading is just a bound vari-

able reading, exactly like (34), but the pronoun that gets bound is located syntactically outside the DP. The result, though, is that we end up with the same truth conditions in both cases.

Having seen how dependency readings can be captured under the functional approach, we can now see what would be predicted if QR were the source of variation of the definite. Crucially, having the embedded universal DP undergo QR to a position outside the relative clause does not capture the intended reading. The LF after QR is provided in (38-a), and the resulting truth conditions in (38-b).

- (38) a. $[[\text{each of us}] \lambda_1 [[\text{We}] D \lambda_2 [t_2 \text{ revised } [\text{the paper that } t_1 \text{ wrote}]]]]$
 (where ‘D’ is a distributive operator)
- b. *Resulting interpretation:* for each x in $\{B, C\}$ and for each y in $\{B, C\}$, y revised the paper that x wrote.

The challenge is that the truth conditions in (38-b) involve the cross product of $\{B, C\}$ with itself. That is, in this case, we get a scopal interaction—which is not what the intended interpretation involves. Instead, as observed above, the truth conditions that we are trying to capture for (33-b) involve the diagonal only, not the full cross-product: each individual y revised the paper that y wrote. Therefore, the dependency reading in (33-b) is another instance where we observe variation of the definite, but the source of variation cannot be due to QR, as QR leads to the wrong truth conditions. Once again then, we have independent evidence for assumptions that enable an alternative to a QR analysis of variation, namely, the Function Assumption and the Covert Saturation Assumption.

We note that, in some sense, the dependency reading of (33-b) is rather unexpected since the embedded universal does not scopally interact with the distributive DP in the matrix subject position—its quantificational force is effectively bleached. Instead, what we observe is a dependency between the matrix subject and the embedded subject, which we propose to capture as a binding relation. Crucially though, this binding relation is made possible by feeding a covert pronoun to the functional DP. Furthermore, under a functional approach, the role of the embedded

universal DP is simply to contribute to the construction of an $\langle e, e \rangle$ function by supplying the domain—it does not scope out of the relative clause. Therefore, under this view, there is no reason to expect a scopal interaction between the embedded universal DP and the distributive subject.

In sum: dependency readings provide another argument that something other than QR is needed as a source of variation of the definite. In addition, they also provide evidence for the Covert Saturation Assumption. That is, the dependency reading is essentially just a bound variable reading—even though there is no overt pronoun in the relative clause to supply a variable. Instead, we can achieve the same truth conditions by having the relativized DP denote an $\langle e, e \rangle$ function, supplying this function with a covert pronoun, and then letting the matrix subject bind the covert pronoun. Again, it is worth reiterating that, even though we made use of covert pronouns to saturate the relativized DP, this is just *one* way of capturing the dependency reading and not the *only* way. The crucial takeaway from this discussion is that we need a way of capturing the observation that the e type argument of the $\langle e, e \rangle$ function can be filled to feed predication—in one way or another—and this, in turn, allows the matrix universal DP to bind the argument of the function.

In the next section, we present independent evidence for our third assumption: the Universal Closure Assumption. We present evidence that the wide scope universal force comes from an external operator. This evidence comes from cases where we observe universal quantificational force, even in the absence of a universal DP; that is, we present other instances of varying definite readings which do not involve an embedded universal DP.

1.6 Universal Closure Assumption: varying definite readings without universal DPs

Under the QR approach, the crucial assumption is that the varying definite reading involves the DP, *each volunteer*, undergoing QR to take wide scope over the definite determiner. This suggests that variation of the definite is available insofar as there is an embedded universal DP that can take matrix scope and induce variation. Under our proposed analysis, the Universal Closure Assumption involves wide scope universal force coming from a covert operator (though see fn. 7 for

an alternative way of implementing the Universal Closure Assumption). In this section, we look at a variant of the sentence in (2), but without an embedded universal DP in the relative clause. The logic is as follows: if a varying definite reading is still licensed in this case, this suggests that we may need a source of variation distinct from QR—since there is no longer a universal DP to undergo QR and derive variation. This would provide motivation for suspecting that the universal force, in fact, comes from an external operator (and possibly that the universal quantification in (2) is not due to *each* itself, but this covert universal operator, since both examples give rise to the same reading).

First, we look at a variant of the sentence in (2), but without the relative clause containing the universal DP (henceforth, the ‘bare DP’ case). To this end, consider (39), where the bolded target sentence is just (2) without the relative clause.

- (39) *[John is updating the student files with each student’s supervisor and student ID. Since John is a sloppy worker, Mary is skeptical that he included these pieces of information on each file. Bill says:]*

Bill (to Mary): He forgot to include the student ID, but at least **John included the supervisor.**

Like (2), variation is permitted: *the supervisor* in (39) does not refer to a single individual as the intended interpretation in (39) would not be satisfied if John included a single supervisor. Instead, the supervisor can vary according to some contextually determined set of individuals. In the case at hand, the relevant set of individuals is the set of students. It is worth noting that, unlike cases where we proposed that the universal DP determined the domain of the $\langle e, e \rangle$ function, the domain of the relevant function in (39) is contextually determined. In addition, like in (2), we feel the presence of universal quantification even though no universal quantifier is present in (39): the target sentence is true iff for each contextually relevant student x , John included the supervisor of x .

What we observe then is that the target sentence in (39) still licenses a varying definite reading, just like (2). As a result, we want to derive the target presupposition in (40-a) and the target

sentence meaning in (40-b). In this case though, there is no universal DP to undergo QR, take wide scope, and induce variation. A possible LF is provided in (41).

(40) a. *Presupposed:*

$\forall x[\text{student}(x) \rightarrow \exists! y[\text{supervisor}(y)(x)]]$

b. *Asserted meaning:*

$\forall x[\text{student}(x) \rightarrow \text{include}(\iota y[\text{supervisor}(y)(x)])(\text{John})]$

(41) *LF:* $\forall_1 [\text{John included } [\underbrace{\text{the supervisor } (\text{pro}_1)}_{\langle e, e \rangle \text{ function}}]]$

Looking beyond the bare DP cases, we can also come up with examples which are more similar to the original example in (2) insofar as they also contain a relativized DP (instead of a bare DP), but no embedded universal. To this end, consider (42-a), with the bolded target sentence in (42-c).²³

(42) a. *[John and Mary each created a file with the graduate students and their supervisors.*

The file needs to include each student's supervisor. John simply copies Mary's file and shows it to the department chair, Bill, who looks puzzled.]

b. Bill (to John): These are all wrong.

c. John (to Bill): What do you mean? **I included the supervisor that Mary included.**

d. *Available interpretation:* For each student x, John included the supervisor that Mary included.

Again, like (2), variation is permitted: *the supervisor* in (42-a) does not refer to a single individual as the intended interpretation would not be satisfied if John included a single supervisor. Instead, the supervisor can vary according to the contextually determined set of students. Also, like (2), we feel the presence of matrix level universal quantification even though no universal quantifier

²³ Another example which illustrates the same point is provided in (i).

(i) a. *[John, Bill and Mary are playing a game where they have to guess how many pieces of candy are in a jar. There are three different jars with candy. Miraculously, John and Mary both got them all right.]*

b. Bill (to John): How did you get them all right?

c. John (to Bill): **I guessed the number that Mary guessed.**

d. *Available interpretation:* For each jar x, John guessed the number that Mary guessed.

is present in (42-a): the target sentence is true iff for each contextually relevant student x , John included the supervisor of x , according to Mary. What we observe then is that the target sentence in (42-a) still licenses a varying definite reading, just like (2) and (39).

So even without a universal DP inside the relative clause, we still observe the same reading that we did in (2). Given that there is no universal DP to induce variation, a QR approach appears to be a non-starter. This provides evidence for the claim that the universal quantification in these examples comes from a covert operator, since it cannot come from an embedded universal DP (more generally, this provides evidence that the $\langle e, e \rangle$, in one way or another, can be universally closed).²⁴ In view of this, we maintain that the source of variation and the source of universal quantification may very well come from distinct mechanisms: the former is due to the relativized DP denoting an $\langle e, e \rangle$ function while the latter is due to a covert operator.

Before moving on, it is also worth noting that the kind of covert universal operator that we propose has been independently argued for, to capture other kinds of wide scope distributive readings (Chatain, 2019).

Chatain (2019) discusses a number of examples which exhibit exceptionally wide universal scope, similar to the varying definite reading. To illustrate these kinds of wide scope distributive readings, consider (43).

- (43) a. *[Alex is a guard at the nuclear powerplant facility. The powerplant employees have different work hours and do not all arrive at the same time.]*
- b. When these employees arrived, Alex smiled. (But she didn't when those ones did.)
- c. *Available Interpretation:* for each of these employees x , when x arrived, Alex smiled.

In (43), the plural DP, *these employees*, is embedded in a scope island, but the universal force

²⁴A reviewer points out that another connection in the literature could provide a potential justification for covert universal quantification: so called telescoping sentences (Sells et al., 1985; Roberts, 1989), (i).

- (i) a. Every chess set comes with a spare pawn. It is taped at the back. (Sells et al., 1985)
- b. Every actor came to the stage. He took his prize and made a speech.

The reviewer proposes that these kinds of sentence can be argued to provide evidence for a mechanism of covert universal quantification that could parallel what we are arguing for. We thank the anonymous reviewer for bringing this data to our attention; we leave a detailed exploration of this connection to future research.

associated with the distributive reading takes scope over the whole sentence. In this sense, it parallels the varying definite reading which we have been discussing. While a proper discussion of Chatain’s data and analysis is orthogonal to the discussion at hand, we do note that he converges on a similar conclusion as us: he proposes that these wide scope distributive readings are the result of a covert universal operator that attaches at the matrix level. We find it reassuring that a similar account has been proposed for independent data, and we believe that this serves as further evidence that the kind of covert universal operator we appeal to for varying definite readings in fact has precedent in other kinds of wide scope readings.²⁵

Having provided independent evidence for our three assumptions of the functional approach, in the next section, we put these pieces together and also provide a more direct argument for a functional derivation of varying definite readings.

1.7 Putting the pieces together

Having provided independent evidence for both the Covert Saturation Assumption and the Universal Closure Assumption, we can now briefly illustrate how these two ingredients combine with the Function Assumption to deliver the truth conditions of the varying definite reading. To show that a functional analysis of varying definite readings is within reach, consider again our target sentence in (44).

(44) John included the supervisor that each volunteer reported to.

When we refer to a varying definite reading of (44), there are three desiderata that we aim to capture. First, the supervisor can vary by volunteer, and this variation can map volunteers to supervisors in an arbitrary manner, as described in Section 1.3.2. Second, there are separate presuppositions of uniqueness and existence for each volunteer: as illustrated by the target presupposition in

²⁵We leave it to future research to further explore the commonalities between the wide scope distributive readings that Chatain observed and the varying definite readings that we have been discussing, and also, to assess whether the same underlying mechanism derives both kinds of readings. While this is certainly an appealing thought, our goal at this stage is simply to point out the obvious parallel between both kinds of readings and suggest that the same covert operator may be at play in both cases.

(45-a), we get a universal uniqueness presupposition. Third, and related to both the first and second desiderata, there is wide scope universal quantificational force. This is illustrated by observing that the target sentence meaning in (45-b) makes a universal claim concerning who John included: he included each supervisor.

(45) a. *Presupposed:*

$$\forall x[\text{volunteer}(x) \rightarrow \exists! y[\text{supervisor}(y) \wedge \text{report-to}(y)(x)]]$$

b. *Asserted meaning:*

$$\forall x[\text{volunteer}(x) \rightarrow \text{include}(\iota y[\text{supervisor}(y) \wedge \text{report-to}(y)(x)])(\text{John})]$$

It is worth pointing out that these different components of the truth conditions can be easily conflated insofar as it is assumed that the permitted variation of supervisors (as well as the universal uniqueness presupposition) and the wide scope universal force are two sides of the same coin. One reason for this is that they are simultaneously captured by allowing the embedded universal DP to QR to a position out of the relative clause: the universal DP takes wide scope and induces variation of supervisors, yielding a universal uniqueness presupposition. Here, we opt for a different approach: we argue that the source of variation (and the universal uniqueness presupposition) comes from one mechanism, and the source of wide scope universal quantification comes from a different mechanism. Now, recall, we propose an LF as in (46).

$$(46) \quad LF: \forall_1 [\text{John included } \underbrace{[\text{the supervisor that each volunteer reported to}]}_{\langle e, e \rangle \text{ function}} (\text{pro}_1)]]$$

In (46), the relativized DP externally composes with the matrix predicate in the same way as in Section 1.5: a covert pronoun is fed into the $\langle e, e \rangle$ function.²⁶ Then, as in Section 1.6, the function is universally closed through a covert operator—which is where the wide scope universal force comes from. For interpretability, we assume that the covert matrix universal is pragmatically restricted to the set of all volunteers—the same domain of the $\langle e, e \rangle$ function.²⁷ In a sense then,

²⁶Again, appealing to covert pronouns is not the only way to resolve the type-mismatch; it is just *one* way.

²⁷Alternatively, the argument could be made that the domain of the universal is also, in a sense, semantically determined. The covert pronoun serves as the argument to our $\langle e, e \rangle$ function. This means that, when the universal

the contribution of the covert universal parallels that of the matrix subject in the case of dependency readings: just as the matrix subject gets interpreted distributively and binds the pronoun fed into the relativized DP in the case of dependency readings, the covert operator supplies universal quantification and binds the pronoun fed into the relativized DP in the case of varying definite readings.

The result is the truth conditions in (47), which are the desired truth conditions. Thus, the same machinery that we argued for in Sections 1.4–1.6 can be leveraged to capture the varying definite reading in (2).

$$(47) \quad \llbracket \forall_1 [\text{John included } [[\text{the supervisor that each volunteer reported to }] (\text{pro}_1)]] \rrbracket^g \\ = 1 \text{ iff } \forall y [\text{volunteer}(y) \rightarrow \text{included}(\iota x [\text{supervisor}(x) \wedge \text{reported-to}(x)(y)])(\text{John})]$$

Before proceeding, we briefly return to the example in (6-a), from Section 1.2, and note that the functional approach also straightforwardly accounts for this example through the LF in (48).

$$(48) \quad LF: \forall_1 [\text{I included } [[\text{the supervisor that you claimed each volunteer reports to}] (\text{pro}_1)]]$$

In (48), the relativized DP again denotes an $\langle e, e \rangle$ function and the wide scope universal force comes from the covert universal, $\forall_1$, which binds the covert pronoun: effectively, there is no difference between (2) and (6-a). Under the functional approach, it makes no difference how deeply embedded the universal DP is since the perceived variation comes from the denotation of the DP itself, not QR of the embedded universal.²⁸

Taking stock then, in the previous sections, we argued that we have independent evidence for our three assumptions: (i) the Function Assumption (ii) the Covert Saturation Assumption, and (iii)

binds the variable supplied by the pronoun, it will have to quantify over the values of the variable that the pronoun ranges over, which is simply the domain of the function.

²⁸It is logically possible that the pseudo-scope account needed to explain (6-a) is not the same functional pseudo-scope account which we have been arguing for. However, one piece of evidence that (6-a) involves the same functional pseudo-scope analysis as presented above is provided in (i).

- (i) The supervisor that you claimed each volunteer reports to varies.

Just as we saw with the example in (11) in Section 1.4, (i) permits variation of supervisors by volunteers. Given this variation, we take it to mean that the relativized DP in (i) can also denote an $\langle e, e \rangle$ function.

the Universal Closure Assumption. Nothing we have said up until now, though, leads us to believe that the varying definite reading in (2) actually involves these ingredients; we have not presented any direct evidence tying these things together. This begs the question: do we have any reason to believe that the relativized DP in (2) does not denote an $\langle e, e \rangle$ function? Similarly, do we have any reason to believe that an analysis of (2) does not invoke these three assumptions? In our view, there does not appear to be any evidence suggesting that these ingredients cannot be responsible for this varying definite reading and we now present a more direct argument that varying definite readings can involve functional parses, one that is based on intuited truth conditions in certain data.

The idea will be to leverage the argument from Section 1.4 concerning *vary* and create an example that properly contains the structure in (2). To this end, consider again our example in (11), repeated in (49).

(49) The supervisor that each volunteer reported to varied.

Intended: different volunteers had different supervisors that they reported to.

In our discussion of (49), we argued for a functional interpretation of the relativized DP. The argument, however, involved a different matrix predicate than (2) and observing that the predicate *vary* requires a functional argument of type $\langle e, e \rangle$. One objection to this argument could be that, even if examples like (49) require a functional parse, this does not mean that the initial example in (2) necessarily does too. It could be that the same relativized DP receives a functional parse in (49) and a QR parse in (2). To address this objection, we turn to examples like (50). We believe these kinds of examples strengthen the argument for a functional interpretation by providing direct evidence, and confirming, that there is indeed a functional path to the varying definite reading.

(50) *[The volunteers at the community center are each required to report to a different supervisor. Mary is compiling a list of the volunteers and their respective supervisors.]*

Mary included the supervisor that each volunteer reports to, which varied as required.

In (50), the first part of the sentence is the same as (2) and conveys the same interpretation: for each volunteer x , Mary included the supervisor that x reports to. In the second part of the sen-

tence, we again have the predicate *vary*, but now it combines with the relative pronoun, *which*, to convey that different volunteers report to different supervisors, as in (11). If the link between the relative pronoun and the relativized DP is anaphoric, and involves co-reference—as illustrated in (51)—this gives us a clue as to the semantics of the relativized DP (Arnold and Borsley, 2008). More specifically, if the anaphoric link holds, this amounts to both expressions having the same denotation since the relative pronoun would inherit its denotation from the relativized DP.

(51) Mary included [the supervisor that each volunteer reports to]₁, which₁ varied

As argued above, *vary* requires a functional argument. For the relative clause, *which varied*, to receive the right interpretation then, the relative pronoun must receive a functional interpretation, meaning that the relativized DP itself must receive a functional interpretation. Under these assumptions, the first part of the sentence would need to be interpreted as in (52-a), where the relativized DP receives a functional parse, so that the second part of the sentence gets interpreted as in (52-b).

- (52) a. $\llbracket \forall_2 [\text{Mary included } [[\text{the supervisor that each volunteer reports to}]_1 (\text{pro}_2)]] \rrbracket$
 $= 1 \text{ iff } \forall y [\text{volunteer}(y) \rightarrow \text{include}(\iota x [\text{supervisor}(x) \wedge \text{report-to}(x)(y)])(\text{Mary})]$
- b. $\llbracket \text{which}_1 \text{ varied} \rrbracket$
 $= \llbracket [\text{the supervisor that each volunteer reports to}] \text{ varied} \rrbracket$
 $= 1 \text{ iff } \exists y, z \in \llbracket \text{volunteer} \rrbracket [\text{the sup. that } y \text{ reports to} \neq \text{the sup. that } z \text{ reports to}]$

Therefore, examples like (50) can be used to detect what semantic mechanism is responsible for the varying definite reading in (2): to capture the right interpretation in (50), we need a functional parse of the relativized DP; as a consequence, there needs to be a route to the varying definite reading which involves a functional parse, and not QR. If this is one way of capturing the varying definite reading, this makes one wonder whether a QR account is needed for (2) at all.²⁹

²⁹We can also use dependency readings instead of *vary* data to make the same sort of argument. We argued that sentences like (33-b) involve a functional interpretation of the relativized DP. However, this argument involved examples which differed from the initial example in (2) by changing the matrix subject from a singular individual to a plural one, a move which may not be innocent as the plural matrix subject is an essential component of the dependency reading. For this reason, once again, the argument could be made that examples like (33-b) involve a functional parse while the

1.8 Conclusion

In this paper, we argued that there is a compositional pathway to varying definite readings that does not involve the embedded universal DP undergoing QR out of the relative clause. The proposed pseudo-scope account of varying definite readings effectively mimics non-local QR through a functional parse of the relativized DP and a covert universal operator. Furthermore, we independently motivated each of the components of our functional pseudo-scope account by arguing that there is variation without QR in some cases, suggesting an alternative compositional pathway to variation may be needed. To this end, we showed that the same kind of functional analysis that captures the perceived variation in these other cases can also derive the varying definite reading in (2).

We did not necessarily argue that a non-local QR parse is unavailable for (2) (though we did identify a limitation of the QR analysis), but rather, that there is no need for such a non-local parse, given that the independently motivated functional parse can explain the variation. This means that (2), and similar examples, do not necessarily constitute conclusive evidence in favor of non-local QR. Moreover, given the data that we have discussed, the argument can be made that a QR approach is, in a sense, myopic whereas a functional approach leads to a more parsimonious

initial example in (2) still involves a QR parse. To this end, consider (i), where we assume that the first person plural pronoun refers to the individuals B and C.

- (i) Mary revised the paper that each of us wrote, which we proofread beforehand.
Intended interpretation: For each x in $\{B, C\}$, Mary revised the paper that x wrote and x proofread beforehand

Just like in (50), the first part of (i) is similar to (2) and conveys a varying definite reading: for each x in $\{B, C\}$, Mary revised the paper that x wrote. The second part in (i) involves the relative pronoun *which* and conveys a dependency reading: for each x in $\{B, C\}$, x proofread the paper that x wrote. Again, if the link between the relative pronoun and the relativized DP is anaphoric, and involves co-reference as illustrated in (ii), this gives us a clue as to the semantics of the relativized DP. More specifically, if the anaphoric link holds, both expressions have the same denotation since the relative pronoun would inherit its denotation from the relativized DP.

- (ii) Mary revised [the paper that each of us wrote]₁, which₁ we proofread

Therefore, for the second part of the sentence to receive the right interpretation, the pronoun must receive a functional interpretation, meaning that the relativized DP would itself receive a functional interpretation. Under a QR approach, the second sentence would receive the wrong interpretation as we would end up looking at the cross product of the set $\{B, C\}$, which is intuitively wrong. Therefore, even in examples like (i), using the same logic surrounding the example in (50), a functional parse is required.

theory. This is because analyzing varying definite readings under a functional lens gives us a way to make sense of other data involving relativized DPs, and gives us the flexibility to capture other cases of variation which cannot involve QR. If we need an alternative route to capturing variation for these cases, then it is not clear that we need the QR account for (2) at all. Put differently: all else being equal, a theory which does not invoke non-local QR for (2) is not necessarily better; however, if a pseudo-scope account is capable of capturing all these different cases of variation—including (2)—then presumably this is preferable to an account which involves non-local QR *and* pseudo-scope variation. Therefore, under this view, a QR approach seems insufficiently general, and is only really well suited to an analysis of the kinds of examples demonstrated specifically in (2).

The empirical generalization that emerges from this discussion is that varying definite readings do not provide an argument for non-local QR, since these readings can also be captured through a viable alternative approach (i.e., a functional approach). As a result, this suggests that, given the examples we have looked at in this paper, we have no reason to abandon the claim that relative clauses are scope islands for QR after all, as apparent counterexamples can be explained through other semantic mechanisms, like a functional interpretation of the relativized DP. Again, to really make a case for non-local QR out of relative clauses, one would need to be cautious of the confound of functional readings, since they also deliver the perceived variation that QR is meant to capture.

It is worth reiterating that, throughout the paper, we have not focused on the internal composition of the relativized DP, for expositional purposes, but also because we believe it is somewhat orthogonal to the argument in the paper. Our main argument throughout the paper is that $\langle e, e \rangle$ functions are needed to capture certain readings of relativized DPs. How we generate those $\langle e, e \rangle$ functions is a different question (in fact, it is unclear if any of the data we have presented has any bearing on the internal composition of the relativized DP). We do note that, in the literature, there has been much discussion on this topic and several authors have proposed different accounts concerning the internal composition of relativized DPs (Jacobson, 1994, 2018; Sharvit, 1999; Barker, 2018; Heim, 2018). All of these analyses are equally compatible with our proposal about the data

at hand. So we did not attempt to adjudicate between them.

That being said, there is one particular point concerning the details of the internal composition that merits a brief discussion.³⁰ All of the analyses of functional relative clauses that we are aware of involve a complex relative *wh*-trace, composed of a function variable, *f*, and an individual variable, *x*, and the universal DP binding into this complex trace (Jacobson, 1994; Sharvit, 1999; Heim, 2018). Under one influential view of pronoun binding, as proposed in Reinhart (1983), for a quantifier to bind a pronoun, it has to c-command the pronoun in surface structure, meaning that the quantifier's trace has to c-command the pronoun at LF. If Reinhart's proposal extended to the covert argument variable in the complex trace, we could expect Weak Crossover effects throughout: in our examples, the universal DP would have to c-command the complex trace, otherwise, we run the risk of a Weak Crossover violation since the universal DP would have to covertly move above the individual variable that it binds. As a result, these analyses predict that object relatives are possible—since the universal DP in the embedded subject position c-commands the complex trace—but subject relatives are not—since, now, the complex traces c-commands the universal DP. The problem is that the data is not aligned with this expectation—as evidenced by (53), which can also receive a varying definite reading despite the universal DP not c-commanding the trace.³¹

(53) John included the supervisor that _ evaluated each volunteer.

From the perspective of the analyses put forth in Jacobson 1994; Sharvit 1999; Heim 2018, this relation between trace and universal DP is surprisingly free, as we see in (53). But this is not an argument against our proposal that the variation in the data we have considered comes from a functional parse of the relativized DP (see also Barker 2012 for further support for the point

³⁰We thank an anonymous reviewer for pushing us on this point.

³¹We do note that speakers do tend to be sensitive to the subject/object asymmetry concerning functional relative clauses and the position of the universal DP insofar as they find it easier when the universal DP is in the subject position. The claim we are defending, though, is that this seems to be a violable constraint and, furthermore, whenever we observe varying definite readings with universal DPs in object position, the corresponding purely functional reading is also possible. Ideally, more should be said about the asymmetry concerning universal DPs embedded in subject/object position, but we believe that this is a somewhat orthogonal issue to our main claim and, as such, is beyond the scope of this paper. Moreover, under a QR based approach, we might expect the asymmetry to go in the other direction. Since this approach requires covert movement, we would expect varying definite readings to be more acceptable when the universal DP is in object position—given that there is usually a ban on subject extraction (Chomsky, 1973, 1986).

that lack of c-command is not by itself expected to prevent the requisite variable binding in the composition). Especially since we see DPs of a parallel form for which there seems to be no alternative to the view that they are functional. For example, alongside (53), we have (54), which is a purely functional reading where the universal DP does not c-command the trace.

(54) The supervisor that _i evaluated each volunteer is their best friend.

Therefore, our analysis of varying definite readings does not incur any cost with regard to the relative structural position of the universal DP and the complex trace. That is, the lesson from (54) is that, in some way or other, the internal composition of functional relative clauses must allow for subject relatives where the universal DP is in object position. Any account of this will carry over to our data for free since our proposal is compatible with whatever solution turns out to be correct. There is one speculative hypothesis worth considering, though: it is logically consistent to propose that overt pronouns exhibit Weak Crossover effects while the covert pronouns required for functional parses do not. We do not explore this hypothesis here, however, but hope to address it in future research.³²

Finally, we would like to conclude by making an observation. Under the functional approach, we posited a covert universal operator in order to capture the intended truth conditions of the varying definite reading. Here, we present one piece of evidence that is compatible with this assumption. More specifically, we observe that sentences containing relativized DPs do not always convey wide scope universal force. That is, *include* does not always combine with the object DP with universal quantificational force. To this end, consider (55).³³

(55) a. *[There are 6 students. John was updating the student record. He was supposed to create a file and add the supervisors that the students work with. Unfortunately, John didn't complete his task: he only included 4 out of the 6 supervisor/student pairs.]*

At least **John mostly included the supervisor that each student works with.**

³²See also Krifka 2001 for further discussion on this issue, but in the domain of functional and pair-list questions.

³³The judgment concerning (55) is admittedly subtle. However, the authors of this paper find the target sentence to be true, given the provided context. In addition, several other native English speakers that were consulted reported the same judgment.

- b. *Intended*: John included most of the supervisor/student pairs

In (55), an overt adverbial quantifier, *mostly*, is present and we no longer observe universal quantificational force. Rather, the sentence as a whole gets the quantificational force of the adverbial quantifier, resulting in a ‘partial inclusion’ interpretation, as illustrated in (56).³⁴

- (56) $\llbracket \text{John mostly included the supervisor that each student reports to} \rrbracket = 1$ iff
for most $x \in \llbracket \text{student} \rrbracket$: $\text{include}(\text{the supervisor that } x \text{ works with})(\text{John})$

In other words, when we insert an overt matrix level quantifier, the covert universal operator is no longer necessary. The functional approach offers a rather straightforward way of capturing this interpretation, if we assume that *mostly* quantifies over students (Berman, 1991) and can also bind the covert pronoun that gets fed into the relativized DP. This is illustrated by the LF in (57).

- (57) *LF*: $\text{most}_2 [\text{John included} [[\text{the supervisor that each student works with}] (\text{pro}_2)]]$

In (57), instead of the pronoun being bound by a covert universal operator, we assume that it is bound by *mostly* itself. In contrast, it is not obvious to us how this partial inclusion interpretation could be captured under a QR approach. Consider the two LFs for (55) in (58).

- (58) a. *LF* 1:
[each student] $\lambda_1 [[\text{mostly}] [\text{John included the supervisor that } t_1 \text{ reports to}]]$
b. *LF* 2:
[mostly] [[each student] $\lambda_1 [\text{John included the supervisor that } t_1 \text{ reports to}]]$

In (58-a), *each student* undergoes QR to a position above *mostly*, resulting in the following interpretation: for each student x , for the most part, John included the supervisor that x reports to. In (58-b), *each student* undergoes QR to a position below *mostly*, resulting in the following interpretation: for the most part, for each student x , John included the supervisor that x reports to. Both interpretations do not quite capture the partial inclusion interpretation that is intended in (55).

³⁴Sometimes referred to as ‘quantificational variability’ in the literature on questions (Berman, 1991; Beck and Sharvit, 2002; Fox, 2012; Xiang, 2019, 2023).

Therefore, if wide scope universal force in (2) did come from *each* itself—as the QR approach predicts—one would need to explain why the truth conditions for (55) do not seem to involve wide scope universal quantification.³⁵

³⁵We also note a final observation which distinguishes the purely functional reading from the varying definite reading. We have discussed different possibilities concerning what can bind the covert pronoun: a distributive matrix subject (as in the dependency reading), a covert universal (as in the varying definite reading), and a quantificational adverb (as in the partial inclusion reading). But there is another obvious possible binder: a singular referential subject, as illustrated in (i) (where it is assumed that Mary is a volunteer and so she is in the domain of the function).

- (i) Mary included the supervisor that each volunteer actually reports to.
 - a. *LF*: Mary₁ included [[the supervisor that each volunteer reports to] (pro₁)].
 - b. *Unavailable interpretation*: Mary included her own supervisor.

What we observe in (i) is that this kind of binding relationship does not seem to be possible—at least not with the kind of (pair-list) function that we have been concerned with. Interpreting the LF in (i-a) would correspond to an interpretation where Mary included the supervisor that the input *x* gets mapped to. Assuming *x* denotes the individual Mary, the generated interpretation would be one where Mary included her own supervisor, which is not an available interpretation for (i).

In contrast, consider (ii), where now the relevant function is a purely functional one (as signaled by the ‘namely’ continuation which is meant to equate the relativized DP with the ‘primary supervisor’ function).

- (ii) Mary included the supervisor that each volunteer actually reports to (namely, her primary supervisor).
 - a. *LF*: Mary₁ included [[the supervisor that each volunteer reports to] (pro₁)].
 - b. *Available interpretation*: Mary included her own (primary) supervisor.

In (ii), the matrix subject, *Mary*, can now bind the covert pronoun, *pro*₁, leading to an interpretation where Mary includes her own primary supervisor. It is unclear what underlies the contrast between (i) and (ii), but we believe this to be an important step for future research, especially in terms of gaining a deeper understanding into the distribution and composition of functional readings more generally.

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Preface to Chapter 2

The first chapter of this dissertation argued that varying definite readings do not require the universal DP to undergo QR out of the relative clause. Instead, I proposed a pseudo-scope analysis where the relativized DP denotes a function from individuals to individuals, of type $\langle e, e \rangle$. This conclusion raises a natural follow-up question that the first chapter largely sets aside. In the literature on functional relative clauses, an ambiguity has been observed between two kinds of readings that relativized DPs can give rise to. Under one reading, typically called the ‘functional reading’, the variation is principled: the relativized DP picks out a ‘natural function’—like the “mother-of” function—that determines who each individual is mapped to. Under the other reading, typically called the ‘pair-list reading’, the variation is arbitrary. The first chapter’s empirical arguments bear primarily on the pair-list reading, where the universal quantifier inside the relative clause does not induce variation via a natural function but in an arbitrary manner. In making the case for a functional analysis of these readings, however, the first chapter does not address what gives rise to the distinction between principled and arbitrary variation in the first place, or whether the two readings differ in how the relative clause is internally composed. Having established that functional parses are available and necessary for varying definite readings, the second chapter takes up this question directly.

Taken together, the two chapters develop an integrated theory of varying definite readings in relativized DPs. The first chapter establishes that these readings involve functional parses. The second chapter takes this result as its starting point and develops a unified account of how functional parses give rise to both principled and arbitrary variation, locating the source of the ambiguity not in the internal composition of the relative clause, but in the semantics of definiteness. As a result,

we can maintain the same compositional structure for both readings that relativized DPs can give rise to. Beyond consolidating the analysis of pair-list/varying definite readings with the functional reading, the approach to definiteness that I appeal to also makes a range of additional empirical predictions that the second chapter shows to be correct.

One terminological point is worth flagging before moving on. In the first chapter, I described the constraint on the $\langle e, e \rangle$ function as a minimality requirement, stating it by way of a minimization operator: the domain of the function must be exactly the NP restrictor of the embedded universal, and nothing more. The second chapter captures this same requirement, but does so by appealing to a notion of maximal informativity. The shift in vocabulary is not a shift in substance. ‘Minimality’ names the observation—that the function ranges over the smallest domain available—whereas ‘maximal informativity’ names the notion I use to capture it, and the two align because a function with a smaller domain yields a stronger, and hence more informative, proposition. Selecting the maximally informative function just is selecting the minimal one.

It may help to see this logic at work in a simpler configuration. Compare *the primary supervisor varied (across volunteers)* with *the supervisor varied (across volunteers)*. These are not a departure from the cases at issue: on the view defended in the first chapter, both definite DPs denote $\langle e, e \rangle$ functions, the only difference being that here the domain of the function is supplied by context rather than by a universal DP inside a relative clause. What the pair illustrates is an entailment: if the primary supervisor varied across volunteers, then the supervisor varied, but not conversely (if the supervisor varied across volunteers, then it need not be the case that the primary supervisor varied—it could have been the secondary supervisor that varied). The reason is that *primary supervisor* is the narrower predicate, it is a subset of the predicate *supervisor*. A claim about a more narrowly specified function is a claim that says more; being more restrictive about which function one is talking about is thus a way of being more informative, not less. The requirement that the second chapter imposes is of exactly this kind, only with the restriction falling on the domain of the function rather than on its values: the function whose domain is exactly the volunteers is the most restricted of the candidates that satisfy the relative clause, and it is for that

reason the most informative. What looks like minimality when stated as a condition on the domain is maximality when stated in terms of informativity.

The broader picture that emerges is one in which what appears to be the signature of wide-scope universal quantification—in this case, variation of the referent of a definite—can arise through a distinct mechanism, which does not require the universal quantifier to scope-shift. This theme of scope-independent variation is further examined in the third chapter, where I turn to finite complement clauses.

Maximality for definiteness and the ordering of functions in pair-list readings

In the literature on questions, it has been observed that questions with quantifiers can give rise to a 'functional reading' and a 'pair-list' reading (Engdahl 1980; Groenendijk and Stokhof 1984; Chierchia 1993, a.o.). Descriptively, the functional reading involves a principled way of mapping individuals to individuals while the pair-list reading involves an arbitrary way of mapping individuals to individuals. Building on this observation, it has been observed that definite DPs with relative clauses containing quantifiers can give rise to both kinds of readings as well (von Stechow, 1990; Jacobson, 1994; Sharvit, 1999; Heim, 2018). This paper proposes that a parsimonious analysis of such definite DPs becomes available if, following Von Fintel et al. (2014), the notion of maximality invoked by the definite article is based on semantic strength.

To make this case, we compare the functional reading (what we refer to as the ‘principled variation reading’) and the pair-list reading (what we refer to as the ‘arbitrary variation’ reading). We argue that both readings involve a functional parse of the relative clause that delivers a function from individuals to individuals, of type (ee). While the two readings are, in fact, distinct, we propose that the ambiguity between them is not rooted in how the relative clause itself is parsed—contrary to what has been proposed in the literature (e.g., Sharvit 1999). Instead, we argue that each reading results from a different way of enabling reference to a function. Functional readings show that context can help establish unqualified uniqueness. In contrast, we suggest that pair-list readings arise from reference to a unique *maximal* function, without the aid of context—provided maximality is understood as based in semantic strength, as proposed in Von Stechow et al. 2014.

2.1 Introduction

The classic Fregean analysis of definiteness requires uniqueness: the contribution of the definite article is to pick out the unique member of the NP extension it combines with. However, as has been noted in the literature, this standard analysis of definiteness faces certain limitations, leading to theories of definiteness where the Fregean uniqueness condition has been proposed to be weakened with reference to a notion of maximality (Sharvy, 1980; Von Stechow et al., 2014). In this paper, we argue that strength-based maximality for definiteness, as proposed in Von Stechow et al. 2014, has utility in functional readings of definite DPs with relative clauses containing quantifiers. We make this case by examining a contrast between two readings of such DPs—the functional reading (cf. Engdahl 1980; Stechow 1984; von Stechow 1990; Jacobson 1994; Sharvit 1999; Heim 2018) and the pair-list reading (cf. Groenendijk and Stokhof 1984; Chierchia 1993; Sharvit 1999; Krifka 2001; Fox 2012; Xiang 2023)—and showing that the difference between them is not a difference concerning the internal composition of the relative clause. Instead, it is a difference in how the definite article achieves reference to a function. Once that difference is understood, we argue that the arbitrary variation reading provides evidence that the definite article in these constructions references a particular kind of maximality.

2.1.1 Functional definites: the different readings

Consider the DP in (1), which involves a definite DP modified by a relative clause containing a quantifier—henceforth, a ‘relativized DP’:

- (1) [DP the woman that each fourth grade boy invited]

Under one reading, this DP refers to some particular woman that all the fourth grade boys invited. To reinforce this intuition, consider the world described by the facts in (2), w_1 . The target sentence in (3) is intuitively true in w_1 .

- (2) **Facts in w_1 :** Every fourth grade boy invited Ms. Hakimi. No other woman was invited by

every fourth grade boy.

- (3) [The woman that each fourth grade boy invited] is Ms. Hakimi.

This reading of (1) is unsurprising in w_1 : the relative clause restricts the denotation of the NP, *woman*, to a singleton set of individuals (containing just Ms. Hakimi), and the sentence simply equates the relativized DP and Ms. Hakimi. We can call this the ‘individual reading’ of the relativized DP, and we can capture it by simply having the embedded universal DP scope locally within the relative clause, as illustrated in (4).

- (4) a. LF : [the woman [wh λ_2 [each fourth grade boy] λ_1 t_1 invited t_2]]
b. $\llbracket (4-a) \rrbracket = \iota \{y: \text{WOMAN}(y) \wedge \forall x[4GB(x) \rightarrow \text{INVITE}(y)(x)]\}$

But the DP in (1) is more complex than it appears. To see why, consider what conclusions we can draw in other scenarios. For instance, the target sentence in (6) is felicitous and true in the world described by the facts in (5), w_2 .

- (5) **Facts in w_2 :** Every fourth grade boy invited his own mother. Billy, a fourth grade boy, in addition invited Ms. Hakimi. No fourth grade boy invited anyone else.
(6) [The woman that each fourth grade boy invited] is his mother.

In w_2 , the relative clause no longer restricts the denotation of the NP, *woman*, to a singleton set of individuals: there is no one individual that every boy invited. Rather, different boys invited different women—Billy even invited two women. The important observation is that, in this case, the sentence permits variation of women by fourth grade boys. Crucially, though, this perceived variation is independently principled insofar as it is based on a "rule" external to the sentence: given any fourth grade boy x , this rule tells us who x invited. That is, as established by the first fact in w_2 , for each fourth grade boy, the rule tells us that the woman he invited is his mother.

This observation seems to reflect an ambiguity of the definite DP in (1): it has a ‘single individual’ reading, which can render (3) felicitous and true in w_1 , and a ‘multiple individual’ reading,

which can render (6) felicitous and true in w_2 .

When it comes to the reading outlined in (5), the intuition reported from the literature is that, under this reading, the relativized DP refers to a particular function—in w_2 , the "mother-of" function (cf. Engdahl 1980; Stechow 1984; von Stechow 1990; Jacobson 1994; Sharvit 1999; Heim 2018). Given this intuition, it is often assumed that the compositional structure of such readings, in some way or other, involves the relativized DP denoting a function from individuals to individuals, of type (ee). We will return to this in Section 2.2.

This second reading of the DP in (1) isn't the end of the story, though. A third reading has been countenanced in the literature, cf. Sharvit (1999). One way of seeing this reading is to consider a copular sentence, (8), in yet another type of possible world where it is felicitous, w_3 .

(7) **Facts in w_3 :** The fourth grade boys are Al, Bill and Chris. Al invited Carol; Bill invited Diane; Chris invited Mary. No fourth grade boy invited any other women.

(8) [The woman that each fourth grade boy invited] is the woman that each of them admires.

Similar to what was observed in (5), the relative clause doesn't restrict the NP *woman* to a singleton set of individuals in this case either: again, the woman varies by fourth grade boy. Unlike what was observed in (5), for each boy, the perceived variation is not independently principled but completely arbitrary: there is no natural relationship, like being someone's mother, that happens to connect each boy to the woman he invited. Al, Bill, and Chris each invited different women. This reading is usually characterized by the fact that it enables truth judgments in the absence of an external principle for variation.

Before proceeding, a note on terminology is in order. We will depart from the terminology typically used in the literature and refer to the functional reading as the 'principled variation reading' and the pair-list reading as the 'arbitrary variation reading'. The motivation for the original labels is to highlight the parallel to the literature on questions (cf. Sharvit 1999), where similar readings have been observed. The reason we opt for different terminology is that, as we'll see, under our view, both readings are "functional" readings insofar as they involve functional parses of the rel-

ative clause. For this reason, we find that the terminology from the literature can be misleading or, at the very least, confusing. We also note that, while we refer to the two readings as a ‘principled variation reading’ and an ‘arbitrary variation reading’, sometimes, we refer more narrowly to the interpretation of the relativized DP itself. In these cases, we will use the terms ‘principled functional DP’ and ‘arbitrary functional DP’, respectively.

While there has been a good deal of discussion (and consensus) concerning how to analyze principled variation readings, this is not quite the case for arbitrary variation readings. According to some authors (cf. Hulsey and Sauerland 2006; Barker 2022; Wang 2023), the intuited truth conditions for the arbitrary variation reading correspond to a reading where the universal takes inverse scope. Part of the reason is that much of the prior discussion of these readings does not involve copular clauses, as in (8), but instead centers around examples where the relativized DP sits in an argument position, as in (9). When it comes to sentences like (9), the central observation that these authors make is that the intuited truth conditions involve wide-scope universal quantification, suggesting that the universal DP inside the relative clause can take matrix level scope. This leads to an analysis as in (10), where the universal DP undergoes QR out of the relative clause.

(9) Mary contacted [_{DP} the woman that each fourth grade boy invited].

Intended: for each fourth grade boy x , Mary contacted the woman that x invited

(10) *LF:* [each fourth grade boy] λ_1 [Mary contacted [the woman that t_1 invited]]

However, Sharvit (1999) has argued, on the basis of scope locality (among other things), against a QR based approach to arbitrary variation readings (see also Palucci 2026 for further arguments against a QR based analysis of arbitrary variation readings like (9)). Moreover, the sentence in (8) also illustrates that a QR based approach has limitations. By inserting the relativized DP, *the woman that each fourth grade boy invited*, into a copular clause, where it is equated with another relativized DP, we get a reading that is difficult to capture through QR. To illustrate, assume that Al, Bill and Chris invited Carol, Diane and Mary, respectively—as in w_3 above. Given this, (8) conveys that Al, Bill and Chris also admire Carol, Diane and Mary, respectively. That is, what

(8) conveys is that each boy-woman pair is equivalent: the pre-copular relativized DP and the post-copular relativized DP pick out the same boy-woman pairings.¹

The above observations show that the definite DP in (1) is three-way ambiguous. It has an individual reading, which can render (3) true in world w_1 , since the relativized DP picks out a specific individual. It has a principled variation reading, which can render (6) felicitous and true in world w_2 . And, finally, it has an arbitrary variation reading, which can render (8) felicitous and true in world w_3 . The individual reading, however, is well-understood: it falls out straightforwardly from letting the embedded universal take local scope, as in (4), and raises no compositional challenges. The puzzle concerns the remaining two readings. Both the principled variation and arbitrary variation readings permit variation of women by fourth grade boys, yet they differ in a subtle but theoretically significant way: the principled variation reading appeals to an external principle (like being someone's mother), while the arbitrary variation reading does not. The central question of this paper is what drives this difference, and whether it requires a difference in how the relative clause itself is composed. We argue that it does not.

Following Sharvit (1999), in this paper, we will assume that both the principled variation and arbitrary variation readings involve a functional parse of the relative clause. As a result, the relativized DP in both cases denotes a type (ee) function from individuals to individuals. While we maintain that the two readings are, in fact, distinct readings, we propose that the ambiguity between these two readings is not rooted in how the relativized DP itself is parsed—contrary to what has been proposed in the literature (cf. Sharvit 1999). Instead, we argue that each reading results from a different way of enabling reference to a function. In particular, we claim that definiteness has a crucial role to play in functional DPs, as this will enable arbitrary variation readings. In turn, this claim requires a particular understanding of definiteness itself: as invoking maximality relative to a strength-based ordering (Von Stechow et al., 2014).

¹As far as we can tell, the only way to derive these truth conditions, under a QR based approach, is to assume a form of across-the-board extraction of the universal DP out of both the pre and post-copular relativized DPs—something which, to our knowledge, has no independent motivation.

(i) *LF*: [each fourth grade boy] λ_1 [The woman that t_1 invited is the woman that t_1 admires]

2.1.2 The proposal: definiteness as maximal informativity

The challenge we address in this paper can be articulated as follows: how is it that, in cases like (5) and (7), the definite article not only successfully refers to a function, but also to the right one? We contend that the two types of readings are crucially enabled by two different properties of definiteness. The first is a ‘salience’-based contextual domain restriction that restricts the NP denotation; here, a restriction to functions given by an external principle. This means that the function selected by the definite article is determined by the intersection of C (a domain restricting variable given by salience) with the denotation of the NP. The second is a filtering of the NP denotation by discarding non-maximal elements. Only one of the above two properties may actually play a role for successful reference in any given case. The first property cannot play a role if context does not furnish any external principles for variation (i.e., any ‘natural functions’). The second property cannot play a role if domain restriction yields a set of functions that are all equally maximal. Previous work on functional relative clauses agrees that the first property is instrumental in principled variation readings; here, we argue that the second property is instrumental in arbitrary variation readings: these fall out when maximality is construed as strength-based. That is, we show that it’s possible to assume the same compositional structure for both principled variation and arbitrary variation readings, all that is needed is to weaken the uniqueness requirement of the definite article.

This points to a broader theoretical conclusion. The classic Fregean analysis of definiteness, on which the definite article picks out the unique member of the NP extension, is not what is required for these constructions. What is required, we argue, is maximality in the sense of semantic strength, as proposed in Von Stechow et al. 2014. More specifically, for arbitrary variation readings, when the context doesn’t supply an external principle, reference to a function can instead be achieved through what we refer to as a ‘logical’ route, where choosing the right function involves looking at entailment relations between different functions. The idea is thus to weaken the Fregean uniqueness condition with reference to a notion of maximality so that the definite article picks out the maximally informative function with respect to the predicate of functions determined by the NP. For principled variation readings, moving to a strength-based definite makes no difference

since, in those cases, the intersection of the NP extension and C leads to a singleton set—meaning this lone function would be the maximal one, trivially.

As we'll see, this view of definiteness correctly predicts the existence of arbitrary variation readings, and moreover, under certain assumptions (that the domain condition of the function is asserted), makes nuanced predictions about the internal composition of the relative clause that turn out to be correct. The lesson, then, is that uniqueness, in the Fregean sense, isn't what's really required in these cases. Instead, what's required is maximality in the sense of semantic strength.

The paper proceeds as follows. Section 2.2 discusses principled variation readings and outlines how reference to a function takes place in these cases. Section 2.3 develops our proposal for arbitrary variation readings: we first outline the challenge in this case, then introduce maximal informativity as a solution. In Section 2.4, we show that our proposed analysis makes numerous correct predictions concerning arbitrary functional DPs. Section 2.5 discusses a residual puzzle. Section 2.6 concludes.

2.2 A route to the principled variation reading

We begin with principled variation readings, with the goals of illustrating the pragmatic route—contextual domain restriction—that enables reference to a function in this case. This will set the stage for showing that arbitrary variation readings involve a distinct route.

Target NP and DP denotation. There are several proposals on the market for analyzing functional definite DPs (cf. Jacobson 1994; Heim 2018). What they all have in common is that they all view the NP as a predicate of (ee) functions, so that the DP refers to an (ee) function. That is, to render (6) felicitous and true in worlds like w_2 , an external principle enables the functional definite DP to denote the "mother-of" function—which maps individuals to their respective mother—as shown in (11).

(11) Target DP denotation (principled variation reading):

$$\begin{aligned} & \llbracket [\text{DP the woman that each fourth grade boy invited}] \rrbracket = \\ & \lambda x_e: \exists! y [\text{MOTHER}(x)(y)]. \iota \text{MOTHER}(x) \end{aligned}$$

The idea is that, under the principled variation reading, the relativized DP, somehow, can end up referring to the "mother-of" function, leading to the denotation in (11). This is achieved by assuming that an external principle restricts the possible functions in the NP extension accordingly. Where these different analyses disagree is on the internal composition of the relative clause and the precise predicate of (ee) functions that the NP comes to denote. For the discussion at hand, the choice between different internal compositions (cf. Engdahl 1980; Jacobson 1994, 2018; Heim 2018, a.o.) is immaterial, since they all deliver the principled variation reading. For expositional purposes though, we will not get into the details of the internal composition at this stage. Instead, we opt for a different strategy: we will simply assume the NP denotation in (12) (for reasons that will become clear when we discuss arbitrary variation readings) and return to the internal composition of the relative clause in Section 2.4. The motivation for this expositional strategy is that we want to simply put on the table the target NP denotation that is needed for a unified analysis of principled and arbitrary variation readings. Then, in Section 2.4, we will work backwards to see what particular internal composition can deliver this target NP denotation. With this in mind, consider (12).

(12) Target NP denotation:

$$\begin{aligned} & \llbracket [\text{NP woman that each fourth grade boy invited}] \rrbracket \\ & = \lambda f_{(ee)}. \forall x [4GB(x) \rightarrow [x \in \text{dom}(f) \wedge \text{WOMAN}(f(x)) \wedge \text{INVITE}(f(x))(x)]] \end{aligned}$$

It is important to emphasize that, while the NP denotation in (12) differs slightly from those proposed in previous works, it still allows us to deliver the intended principled variation reading, as we will now show. At the same time, this denotation has the added benefit, as we will see in Section 2.3, that it also enables us to deliver the arbitrary variation reading. As a result, whatever compositional pathway delivers this denotation will be the same in both readings, meaning that a single internal composition for the relative clause underlies both readings—which is a central claim of the paper.

What (ee) functions are in the set characterized by the predicate denotation in (12)? To answer

this question, we can return to the facts in w_2 . First, every fourth grade boy invited his mother, meaning that the "mother-of" function is in this set. Recall also that, in w_2 , Billy invited both his mother and Ms Hakimi. This means that the function which maps every other fourth grade boy to his mother, and Billy to Ms Hakimi is also in this set. In other words, the functions that (12) applies to can map Billy to either his mother, or to Ms Hakimi, meaning that the set characterized by (12) contains either sort of function, as illustrated in (13):²

$$(13) \quad \text{Set characterized by NP denotation, in world } w_2: \left\{ \begin{array}{l} [\lambda x_e : \exists! y [\text{MOTHER}(x)(y)]. \iota \text{MOTHER}(x)], \\ [\lambda x_e : \exists! y [\text{MOTHER}(x)(y)]. \iota \text{MOTHER}(x)]^{\text{Ms. Hakimi/Billy}}, \\ \dots \end{array} \right\}$$

As a result, there are multiple (ee) functions in the set characterized by the predicate denotation in (12).

Deriving reference to a function. At the next stage of the derivation though, the definite article combines with this predicate of functions, of type ((ee)t). To accommodate this, Jacobson assumes a polymorphic definite article, as in (14), where $\alpha \in \{e, (ee), \dots\}$.

$$(14) \quad \text{Polymorphic definite article:} \\ \llbracket \text{the} \rrbracket = \lambda f_{(\alpha t)} : \exists! x \in D_\alpha [f(x)]. \iota \{x : f(x)\}$$

The issue though is that, given (13), the uniqueness condition imposed by (14) for the definite article is not actually met in w_2 . However, Jacobson suggests that a predicate like (12) can hold of a unique ‘external’ function in context; that is, a function that comes about by looking beyond the content of the sentence itself. This is illustrated in (15).

$$(15) \quad \text{A conceivable set of external functions in } w_2:$$

²The notation ‘Ms. Hakimi/Billy’ is meant to signal that, under this function, Billy gets mapped to Ms. Hakimi instead of his mother.

$$\left\{ \begin{array}{l} \lambda_{x_e} : \exists!y[\text{MOTHER}(x)(y)]. \iota\text{MOTHER}(x), \\ \lambda_{x_e} : \exists!y[\text{SISTER}(x)(y)]. \iota\text{SISTER}(x), \\ \lambda_{x_e} : \exists!y[\text{AUNT}(x)(y)]. \iota\text{AUNT}(x) \end{array} \right\}$$

Returning to the set of functions in (13), given the facts as in w_2 , the function mapping each boy to their mother and Billy to Ms Hakimi will be ruled out, since this is not such an external function. The entry in (16) makes tacit domain restriction explicit through a variable C (used here in both object and meta language for simplicity).

(16) Polymorphic (Fregean) definite article with domain restriction:

$$\llbracket \text{the}_C \rrbracket = \lambda f_{(\alpha t)} : \exists!x \in D_\alpha [C(x) \wedge f(x)]. \iota \{x : C(x) \wedge f(x)\}$$

Appealing to the revised entry for our definite article in (16), we would intersect the two sets in w_2 , (13) and (15), which results in a singleton set containing the "mother-of" function. As a result, we end up with the DP denotation in (17), since the "mother-of" function is the only external function that is also in (13).

(17) DP denotation in w_2 , assuming C as in (15):

$$\lambda_{x_e} : \exists!y[\text{MOTHER}(x)(y)]. \iota\text{MOTHER}(x)$$

Thus, when it comes to the principled variation reading, to satisfy the uniqueness condition imposed by the definite article, there is a heavy burden placed on tacit domain restriction. The definite article presupposes that there is a unique external function and we rely on the context to pick out said function. This gives us one way in which the definite can refer to an (ee) function. Once the definite article picks out the correct (ee) function, the sentence in (19) becomes a straightforward identity statement—assuming, as Jacobson 1994 does, that the copula is polymorphic, as in (18), where $\alpha \in \{e, (ee), \dots\}$, and the post copular expression also denotes the (ee) function in (17).³

(18) Polymorphic copula:

³Jacobson assumes that the pronoun *his* in *his mother* is a bound variable, deriving a function that takes an individual, x , as input and returns x 's mother as output—just as in (17).

$$\llbracket \text{is} \rrbracket = \lambda f_{\alpha}. \lambda g_{\alpha}. f = g$$

(19) The woman that each fourth grade boy invited is his mother.

As observed in Jacobson 1994, this line of analysis makes a welcome prediction: when multiple salient external functions are in the NP extension in a given world, uniqueness can no longer be satisfied and, as a result, the sentence should be infelicitous. This prediction is borne out.

- (20) a. *[Al, Bill and Chris are the fourth grade boys. All three of them invited their respective mother and aunt.]*
 b. #The woman that each fourth grade boy invited is his mother.

Given the facts in (20-a), the sentence in (20-b) is infelicitous under the principled variation reading. This is because both the "mother-of" function and the "aunt-of" function are salient principles, given that they are lexically encoded in language as 'natural functions'. But they are also made salient by the description of the facts in (20-a). Crucially, given these facts, both functions are in the NP extension (as they are both functions which map each fourth grade boy to a woman he invited). Thus, when we intersect the NP extension with C, a set of external functions as in (15), we end up with a doubleton set containing both the "mother-of" and the "aunt-of" functions, and so the definite article's uniqueness requirement is not satisfied.

In sum, the strategy outlined in this section is available precisely when salience and context conspire in the following way. In virtue of being lexicalized, we can assume that natural functions are always salient principles and, as a result, members of C. Thus, something like the "mother-of" function is always part of C and it is the facts in a given world that ensure this external principle is also a member of the NP extension in that world. Crucially, though, for the principled variation reading, we assume that the external principles in C are precisely the natural functions—C restricts the domain by 'purging' arbitrary (ee) functions from consideration. As a consequence, the definite article picks out the unique external function in both the set of functions given by the NP extension and the set of functions given by C. In worlds like w_2 , then, the referent of the definite article is

the "mother-of" function.⁴

In Section 2.3, we turn to arbitrary variation readings, where salience (i.e., the functions in C) does not help us. In these cases, we assume that C still contains all natural functions, in virtue of them being lexicalized. However, in this case, C will contain not *only* the natural functions, but all possible (ee) functions. Moreover, arbitrary variation readings involve worlds where no external principle (or natural function) is included in the NP extension in the relevant possible world, such as w_3 . What this means is that, in worlds like w_3 , intersecting the NP extension with C will not change anything, it will simply return the NP extension. To overcome this, we will argue that enabling these readings requires more than a different pragmatic strategy, it requires revisiting the semantics of the definite article itself.

2.3 A route to the arbitrary variation reading

2.3.1 A first attempt at the arbitrary variation reading: the challenge

Target NP and DP denotation. As previewed in Section 2.1, under the arbitrary variation reading, we assume that the relativized DP also denotes a type (ee) function. In this case though, there is no external principle underlying the variation, we simply want the domain of the function to include the fourth grade boys and to map each fourth grade boy to a woman he invited. This is shown in (21).

(21) Target DP denotation in w_3 :

$$\begin{aligned} & \llbracket [\text{DP the woman that each 4th grade boy invited}] \rrbracket = \\ & \iota f_{(ee)} [\forall x [4GB(x) \rightarrow [x \in \text{dom}(f) \wedge \text{WOMAN}(f(x)) \wedge \text{INVITE}(f(x))(x)]]] \end{aligned}$$

⁴Under this view, the predicted referent of the definite article crucially depends on the members of C. To illustrate the work that C is doing in these cases, consider the following. In our running example, the referent of the definite article turns out to be the "mother-of" function. But depending on what functions are in C, the precise content of the "mother-of" function could vary. For instance, as we've been assuming, C could contain the function that is defined for *any individual* with a mother and maps individuals to their respective mother—the 'unrestricted' "mother-of" function. Alternatively, C could contain the function that is defined for *just the fourth grade boys* with a mother and maps these individuals to their respective mother—the 'restricted' "mother-of" function. Depending on which of these "mother-of" functions are in C, the referent of the definite article will be different: it could be either the unrestricted or the restricted "mother-of" function.

To arrive at (21), then, we need for the NP that the definite article combines with to denote an appropriate predicate of (ee) functions. One possible candidate NP denotation is provided in (22)—the same NP denotation we assumed in (12).

(22) Target NP denotation:

$$\begin{aligned} & \llbracket [\text{NP woman that each fourth grade boy invited}] \rrbracket \\ & = \lambda f_{(ee)}. \forall x[4GB(x) \rightarrow [x \in \text{dom}(f) \wedge \text{WOMAN}(f(x)) \wedge \text{INVITE}(f(x))(x)]] \end{aligned}$$

The denotation in (22) imposes two requirements on each function in the set: (i) that every fourth grade boy is in the function's domain, and (ii) that each boy is mapped to a woman he invited. As noted in Section 2.2, for now, we simply assume that some compositional pathway can deliver the denotation in (22), as it has certain desirable properties that will prove crucial to our analysis of arbitrary variation readings.

Deriving reference to a function. An issue arises in trying to get from (22) to (21), though. The problem is that the predicate in (22) holds of many distinct functions, not just the intended one. Consider the three functions in (23), which differ only in the respective size of their domains (i.e., they are ordered by inclusion from smallest to largest).

(23) A conceivable set of (ee) functions:

$$\left\{ \begin{array}{l} \iota f_{(ee)}[\forall x[4GB(x) \rightarrow [x \in \text{dom}(f) \wedge \text{WOMAN}(f(x)) \wedge \text{INVITE}(f(x))(x)]]], \\ \iota f_{(ee)}[\forall x[4G\text{-STU.}(x) \rightarrow [x \in \text{dom}(f) \wedge \text{WOMAN}(f(x)) \wedge \text{INVITE}(f(x))(x)]]], \\ \iota f_{(ee)}[\forall x[\text{STU.}(x) \rightarrow [x \in \text{dom}(f) \wedge \text{WOMAN}(f(x)) \wedge \text{INVITE}(f(x))(x)]]], \\ \dots \end{array} \right\}$$

All three of the functions in (23) satisfy (22): each maps every fourth grade boy to a woman he invited, while the latter two functions additionally map non-fourth grade boy individuals (all fourth grade students, all students) to someone. In general, with functions viewed as sets of ordered pairs, any function that is a superset of the target denotation in (21) will satisfy (22). Since the NP extension has more than one member, the definite article in (14) cannot successfully refer to a

function.

We take this underdetermination to be a central challenge for the arbitrary variation reading. For the principled variation reading, uniqueness is achieved through contextual domain restriction: the unique function is the intersection of C , given by external principles like the "mother-of" function, and the NP extension in the world of evaluation (thus, it need not be context alone that delivers a singleton set). But arbitrary variation readings arise in worlds where intersecting the NP extension with C does not deliver a singleton set, since, in the world of evaluation, no external principle is present in the NP extension (due to the relevant facts in the world of evaluation). A different pathway is therefore required, one that picks out the intended *minimal* function from the set in (23): the function whose domain ranges exactly over the NP restrictor of the embedded universal, the fourth grade boys.

Thus, our goal is to deliver the arbitrary variation reading while preserving the proposed composition of principled variation readings. To achieve this, our strategy will be to instead revisit the contribution of the definite article in (14) and weaken the uniqueness requirement. That is, the intuition that we pursue builds on two ideas: (i) that the first function in (23) is the unique maximal member of the NP extension and (ii) by 'maximal', we mean maximal informativity. This means that the first function in (23) is the maximally informative one over the fourth grade boys; the other functions contain superfluous information. Fortunately, in the literature on definiteness, there exists an account that has proposed a way of turning this intuition into a concrete proposal.

2.3.2 Capturing the minimality requirement

2.3.2.1 Maximal informativity

In Section 2.2, we assumed a classic Fregean analysis of the definite article, under which it picks out the unique member of the NP extension. One way of weakening this uniqueness requirement is to instead assume that the definite article references maximality. For instance, one possibility is to replace our Fregean definite article with the one in (24), following Sharvy (1980).

$$(24) \quad \llbracket \text{the} \rrbracket = \lambda P_{(et)} : \exists ! x \in D_e [\text{MAX}_{\sqsubseteq}(P(x))]. \iota \text{MAX}_{\sqsubseteq}(\{x : P(x)\})$$

Under this proposal, the definite article no longer selects the unique member of the NP extension, but rather the unique maximal element, where maximality concerns the mereological ordering of individuals (i.e., the ‘part-of’ relation): a maximal individual is one that is not part of any other individual.

Let us see how this applies to basic cases involving plural definites, such as *the fourth grade boys*. In a world where the fourth grade boys are a, b, c, then the plurality a+b+c includes all sub-pluralities (e.g., a+b, b+c, a+c, a, etc.) as a part. Hence, a+b+c is maximal in terms of the part-of relation and is thus the unique maximal member of the NP extension in this world. As a result, it serves as the referent in this world.

While this works for pluralities of individuals—since they are related by the part-of relation—it doesn’t work for the (ee) functions that we are concerned with. Our (ee) functions are not related by the part-of relation. Even though the outputs of the functions are individuals, we can’t reach into each ordered pair and compare the mereological structure of these individuals. How else could we give content to ‘MAX’ in (24)? We need a different notion of maximality, one that is based off a different kind of ordering that can apply to the (ee) functions we are concerned with.

To this end, a different ordering that has been appealed to in the literature involves what we may refer to as a kind of ‘logical’ ordering, where the definite article selects the semantically strongest or most informative element satisfying the NP extension (Von Fintel et al., 2014). Thus, while Sharvy (1980) proposes a definite article involving part-based maximality, Von Fintel et al. (2014) proposes a definite article involving strength-based maximality. We can illustrate the motivation for a strength-based definite article by first considering (25).

- (25) a. **Facts in w_6 :** Mary has been eating too many nuts lately. As a result, her mom told her that she cannot have more than 10 nuts.
- b. The number of nuts Mary is allowed to eat is 10.
- c. NP extension in w_6 :
- $\llbracket \text{number of nuts Mary is allowed to eat} \rrbracket = \{0, \dots, 10\}$

Given the facts in w_6 , the NP extension can include all numbers from 0 to 10 (but not numbers above 10), as in (25-c). In this case, though, the definite article picks out the *maximal* number from the interval, as in (25-b). But now consider (26).

- (26) a. **Facts in w_7 :** The dish requires at least 10 nuts. Any number of nuts above 10 will work, but the dish cannot be made with less than 10 nuts.
- b. The number of nuts sufficient to make the dish is 10.
- c. NP extension in w_7 :
- $$\llbracket \text{number of nuts sufficient to make the dish} \rrbracket = \{10, \dots\}$$

Given the facts in w_7 , the NP extension can include all numbers above 10, as in (26-c). In this case, though, the definite article picks out the *minimal* number from the interval, as in (26-b). Thus, in some cases, the definite article picks out the maximal number, while in other cases, it picks out the minimal number. What Von Fintel et al. (2014); Alonso-Ovalle and Schwarz (2026) propose is that, in each case, the definite article picks out the maximally informative proposition. For example, given the facts in (25-a), if Mary is allowed to eat 10 nuts, then she is also allowed to eat 9 nuts, and so the proposition where the definite refers to 10 entails the proposition where the definite refers to 9 (and so on for all numbers of nuts below 10). Similar logic shows that (26-b) is the maximally informative proposition given the facts in w_7 , except now 10 nuts entails any larger number of nuts. It is examples like these that motivate the move to strength-based maximality.

Following Von Fintel et al. (2014); Alonso-Ovalle and Schwarz (2026), a maximality operator based on semantic strength is provided in (27), where P is an intensional property and X is the extension of P in a given world w (i.e., all the members satisfying P in w). In that world w , MAX looks at the propositions we get by applying P to all members of its extension in that world, $P(w)$, and selects members of $P(w)$ that yield the strongest propositions (i.e., there is no other member of X that entails it).

- (27) Strength-based maximality:
- $$\text{MAX}_P(X) = \{x \in X: \neg \exists y \in X[\{w : P(w)(y)\} \subset \{w : P(w)(x)\}]\}$$

For the relativized DPs we have been concerned with, P would correspond to the property expressed by the NP (i.e., the NP's intension) while X would be the extension of the NP in a given world w . In (28), we have a revised entry for the definite article that incorporates maximal informativity:

(28) Strength-based definite:

$$\llbracket \text{the} \rrbracket^w = \lambda P_{(s(\alpha t))}: \exists! x \in D_\alpha [\text{MAX}_P(P(w)(x))]. \iota \text{MAX}_P(\{x: P(w)(x)\})$$

The entry for the definite article in (28) is completely general insofar as it is assumed to be polymorphic: P has type $(s(\alpha t))$ and x has type α , where $\alpha \in \{e, (ee), d, \dots\}$.

Let us now return to see how strength-based maximality applies to basic cases involving plural definites, such as *the fourth grade boys*. Looking again at a world where the boys are a, b, c , then $\text{boys}(a+b+c)$ strictly entails $\text{boys}(a+b)$, $\text{boys}(b+c)$, etc. Hence $\text{boys}(a+b+c)$ is maximal in terms of semantic strength and $a+b+c$ is the unique maximal member of the NP extension in this world. As a result, it serves as the referent in this world: the most informative element is simply the largest element by the part-whole ordering and so, in these cases, the predictions of strength-based maximality and part-based maximality coincide.

Looking at our definition of strength-based maximality in (27), if we take α to be type (ee) , we get an instantiation of strength-based maximality as it applies to our (ee) functions:

(29) Strength-based maximality instantiated for (ee) functions:

$$\text{MAX}_P(F) = \{f \in F: \neg \exists g \in F[\{w : P(w)(g)\} \subset \{w : P(w)(f)\}]\}$$

where f is type (ee) ; g is type (ee) ; P is type $(s((ee)t))$

Having revised our assumptions concerning the definite article, we will now turn to the predicate of functions that it combines with. We will show that, provided we have the right NP denotation, as in (22), these NPs have certain ‘logical’ properties that have desirable implications when combined with the semantics of definiteness just discussed. Before moving on, though, it is worth flagging the broader significance of the move we are making. The proposal we develop in the remainder of the paper is that the definite article in functional definite DPs—both arbitrary and

principled—references strength-based maximality, in the sense of Von Stechow et al. (2014). Moving to a strength-based definite for principled variation readings makes no difference since, in those cases, the intersection of the NP and C leads to a singleton set. Therefore, this lone function would be the semantically strongest one, trivially. This single change in the semantics of the definite article will, we argue, account not only for the existence of arbitrary variation readings, but for a range of further empirical generalizations about functional definite DPs with relative clauses containing quantifiers. The remainder of Section 2.3 spells out how strength-based maximality interacts with the predicate of functions denoted by the NP to deliver the arbitrary variation reading. Section 2.4 develops the further empirical consequences of the proposal.

2.3.2.2 Monotonicity properties of functional NPs

Before turning to the NP denotation, in (30), we outline the relevant notion of subethood between two functions. Viewing functions as sets of ordered pairs, (30) simply states what the subset relation between functions amounts to, in terms of inputs and outputs: if g is a subset of h , then the domain of h includes the domain of g , and h agrees with g on all members of the domain of g .

(30) Subset relation between functions:

for all $g, h \in D_{(ee)}$

$$g \subseteq h \Leftrightarrow \text{dom}(g) \subseteq \text{dom}(h) \wedge \forall x[x \in \text{dom}(g) \rightarrow h(x) = g(x)]$$

“dom(h) includes dom(g) and h agrees with g on dom(g)”

With this in mind, when the relative clause contains a universal, as in (31), the resulting NP denotation is ‘upward’ monotone. Concretely, this means that, all the worlds where the NP denotation maps a given function f to true are worlds where it maps arbitrary supersets of f to true as well—but not arbitrary subsets of f . For instance, if a function g maps every boy to a woman he invited and h is a superset of g , then it will also be the case that h maps every boy to a woman he invited, since h will simply ‘inherit’ this from g . Conversely, if a function h maps every boy to a woman he invited and g is a subset of h , then it need not be the case that g also maps every boy to a woman he invited, since we don’t know what information concerning h is discarded when we move to g .

(31) [NP woman that each boy invited]

(32) **Each case: upward monotonicity**

for all $g, h \in D_{(ee)}$

- a. if $g \subset h$ then $\{w : \llbracket (31) \rrbracket(w)(g)\} \subseteq \{w : \llbracket (31) \rrbracket(w)(h)\}$
“g maps every boy to a woman he invited $\Rightarrow$
h maps every boy to a woman he invited”
- b. if $g \subset h$ then $\{w : \llbracket (31) \rrbracket(w)(h)\} \not\subseteq \{w : \llbracket (31) \rrbracket(w)(g)\}$
“h maps every boy to a woman he invited $\nRightarrow$
g maps every boy to a woman he invited”

Recall, in the previous section, we saw that the target intension for (31) is (33), which conveys that every boy is mapped by f to a woman he invited.

(33) Assumed intension for (31):

$$\lambda w. \lambda f_{ee}. \forall x[\text{BOY}_w(x) \rightarrow x \in \text{dom}(f) \wedge \text{WOMAN}_w(f(x)) \wedge \text{INVITE}_w(f(x))(x)]$$

“f maps every boy to a woman he invited”

Given (30) and (32), when the embedded quantifier is a universal, it follows that the strongest function is the one with a minimal domain; i.e., whose domain corresponds exactly to the NP restrictor of the universal. This is because the minimal function that maps every fourth grade boy to a woman he invited entails any larger function that also maps every fourth grade boy to a woman he invited. Once a function’s domain includes all the fourth grade boys (and maps each of them to a woman he invited), then an arbitrary larger function is also guaranteed to do so. However, if a function’s domain includes all the fourth grade boys (and maps each of them to a woman he invited), then an arbitrary smaller function is *not* guaranteed to do so—because we cannot be sure that all the fourth grade boys are still included in the smaller domain. Therefore, the smaller domain yields a stronger proposition.

This is a desirable property because of the way it interacts with a strength-based definite article. As we saw in the previous section, for (34), the strength-based definite article in (28) picks the

strongest functions from the set of (ee) functions in (33). Given the upward monotonicity of (33), these will be functions with a minimal domain that includes nothing but boys.

(34) [DP the [NP woman that each boy invited]]

We can further clarify this point by returning to the set of functions in (23) and illustrate how a strength-based definite article selects the intended function. Assume that the fourth grade boys are Al, Bill and Chris and the fourth grade students are the fourth grade boys plus Mary. The first two functions in (23) would then correspond to (35-a) and (35-b), respectively.

(35) a. Function over fourth grade boys:

$f_1 = [Al \rightarrow \text{woman}_1, \text{Bill} \rightarrow \text{woman}_2, \text{Chris} \rightarrow \text{woman}_3]$

b. Function over fourth grade students:

$f_2 = [Al \rightarrow \text{woman}_1, \text{Bill} \rightarrow \text{woman}_2, \text{Chris} \rightarrow \text{woman}_3, \text{Mary} \rightarrow \text{woman}_4]$

Given our definition of function subethood in (30), we can see that f_1 is a subset of f_2 (similarly, f_1 is also a subset of f_3). According to the definition of strength-based maximality in (29), and taking ‘P’ to be the intension in (33), we can now consider f_1 and f_2 as sets of ordered pairs, given by the extension of (33). We then check the strength of the proposition we get once we feed each function to the intension of (33). In this case, the definite article would pick f_1 , as it is the semantically stronger function with respect to (33): f_1 entails f_2 (but not vice versa). Therefore, the strength-based definite article correctly predicts that arbitrary variation readings are possible when the relative clause contains a universal quantifier and, furthermore, enables reference to the intended function by selecting the unique maximally informative one (where the unique function is the minimal one with the domain corresponding to the embedded NP restrictor).

An important point to note, though, is that there will only be a unique maximal function of this sort if, and only if, every boy invited exactly one woman. Therefore, this line of approach makes a prediction: to get a *unique* minimal function, each boy must have invited exactly one woman (unlike principled variation readings: recall, in w_2 , Billy invited two women). A strength-based definite article rules out potential functions that extend the domain beyond the minimal one,

but determining a unique function can also fail if any of the boys in the domain invite multiple women. This is because there would then be multiple possible functions over a minimal domain and each of these functions would qualify as maximally informative (as they would all be logically independent of each other), preventing successful reference. Stated differently: if the definite article presupposes that there is a unique maximal function, then each fourth grade boy must invite a unique woman as a unique function over a minimal domain implies that each element in the domain is mapped to a single output. We can now illustrate that this prediction is borne out.

As previewed in the introduction, the world w_3 establishes that each fourth grade boy invited exactly one woman. If we modify the facts so that one of the fourth grade boys invited multiple women, the arbitrary variation reading becomes unavailable: the sentence in (37) can no longer be used felicitously in $w_{3'}$.

(36) **Facts in $w_{3'}$:** The fourth grade boys are Al, Bill and Chris. Al invited Carol **and Susan**; Bill invited Diane; Chris invited Mary. No fourth grade boys invited any other women.

(37) The woman that each fourth grade boy invited is the woman that each of them admires.

Hence, under the assumptions that (i) the definite article references semantic strength and (ii) the definite article selects a unique maximal function, the constraint that each boy invited exactly one woman falls out.

2.4 Further benefits of our analysis

In Section 2.3, we argued that arbitrary variation readings can be derived by weakening the Fregean uniqueness condition on the definite article by invoking a strength-based maximality filter. The motivation for this move was, in the first instance, to deliver the arbitrary variation reading in worlds like w_3 , where contextual domain restriction is unavailable. In this section, we show that this move has substantially more empirical content than this. In particular, strength-based maximality predicts that arbitrary variation readings exhibit certain sensitivities, each of which is empirically attested. First, we predict that the availability of arbitrary variation readings is sensitive to the

quantifier inside the relative clause (Section 2.4.1). Second, we predict that arbitrary variation readings become unavailable in cases of domain expansion; i.e., when the external composition requires going outside the function’s minimal domain (Section 2.4.2). Finally, we predict that arbitrary variation readings are also available when the head noun of the relative clause is plural (Section 2.4.3). These results, we argue, provide independent support for the central theoretical claim of the paper: that the definite article in functional definite DPs references strength-based maximality.

2.4.1 Quantifier sensitivity

2.4.1.1 The prediction

What happens when we replace the universal inside the relative clause with a different quantifier? For instance, what does our analysis predict for the NP in (38), when the relative clause contains a negative quantifier? To answer this question, we can propose a denotation for (38) that parallels the denotation that we had for (31). This is done in (39).

(38) $[_{NP}$ woman that no boy invited]

(39) Assumed intension for (38):

$$\lambda w. \lambda f_{ee}. \neg \exists x [\text{BOY}_w(x) \wedge x \in \text{dom}(f) \wedge \text{WOMAN}_w(f(x)) \wedge \text{INVITE}_w(f(x))(x)]$$

“f maps no boy to a woman he invited”

When the relative clause contains a negative quantifier, we get the reverse property: the NP denotation is now ‘downward’ monotone. This means that, all the worlds where the NP denotation maps a given function f to true are worlds where it maps arbitrary subsets of f to true as well—but not arbitrary supersets of f . In this case, if a function h maps no boy to a woman he invited and g is a subset of h , then it will also be the case that g maps no boy to a woman he invited, since g will again ‘inherit’ this from h . Conversely, if a function g maps no boy to a woman he invited and h is a superset of g , then it need not be the case that h also maps no boy to a woman he invited, since we don’t know anything about the additional ordered pairs that are in h .

(40) **No case: downward monotonicity**

for all $g, h \in D_{(ee)}$

a. if $g \subset h$ then $\{w : \llbracket (38) \rrbracket(w)(h)\} \subseteq \{w : \llbracket (38) \rrbracket(w)(g)\}$

“h maps no boy to a woman he invited $\Rightarrow$

g maps no boy to a woman he invited”

b. if $g \subset h$ then $\{w : \llbracket (38) \rrbracket(w)(g)\} \not\subseteq \{w : \llbracket (38) \rrbracket(w)(h)\}$

“g maps no boy to a woman he invited $\nRightarrow$

h maps no boy to a woman he invited”

For (41), the strength-based definite article in (28) picks the strongest functions from the set of (ee) functions in (39). Given the downward monotonicity of (39), there will not be a *unique* maximal function. This is because, for any non-boy x , (39) is silent on who x is mapped to. Hence, regardless of the facts in the world, there will be multiple maximal functions in the NP extension—all those functions that include non-boys in their domain—yielding reference failure. In effect, as long as D includes any non-boys, there cannot be a unique maximally informative function. Therefore, we predict reference failure, ruling out arbitrary variation readings with negative quantifiers.

(41) [_{DP} the [_{NP} woman that no boy invited]]

This is, in fact, a welcome prediction as Sharvit (1999) observes that arbitrary variation readings are possible only with embedded universal quantifiers.⁵ In contrast, principled variation readings are possible when the relative clause contains not just universal quantifiers, but a range of other quantifiers (i.e., negative quantifiers, proportional quantifiers, etc.) as predicted by the analysis of principled variation readings presented in Section 2.2, since reference is enabled through contextual domain restriction (cf. Jacobson 1994; Sharvit 1999). Therefore, given our assumed NP denotations, our analysis makes numerous desirable predictions.

Finally, it is worth pointing out that our analysis does not only rule out relativized DPs with negative quantifiers. The same reasoning extends to other quantifiers as well—for example, pro-

⁵Note that a similar issue arises with pair-list readings of questions (Groenendijk and Stokhof, 1984; Chierchia, 1993; Krifka, 2001; Fox, 2012; Xiang, 2023).

portional quantifiers. Consider *most*, with the NP intension in (42):

$$(42) \quad \llbracket \text{woman that most 4th grade boys invited} \rrbracket \\ = \lambda w. \lambda f_{(ee)}. \text{MOST}(x)[\text{BOY}_w(x) \wedge x \in \text{dom}(f) \wedge \text{WOMAN}_w(f(x)) \wedge \text{INVITE}_w(f(x))(x)]$$

This NP denotation requires that more than half of the boys be in the domain of f , and that each of those boys be mapped to a woman he invited. The key question is whether this predicate of functions determines a unique maximally informative function that the strength-based definite article can pick out, and it does not. Even if every boy invited exactly one woman, there are many distinct ways to choose a majority subset of the boys, and the predicate is satisfied by any function whose domain is one of these majority subsets. None of these functions is more informative than the others—they are all equally maximal as their domains are just different majority subsets of the boys (i.e., different minimal domains)—and so they are all logically independent and there is no unique strongest function for the definite article to pick out. The result is reference failure, and an arbitrary variation reading is correctly predicted to be unavailable.

This case is instructive because it shows that the work being done by our analysis is more general than the monotonicity discussion in the previous section might suggest. With negative quantifiers, arbitrary variation readings are blocked because the NP denotation is downward monotone. With *most* (and other quantifiers), it is not necessarily about the monotonicity of the NP denotation, but arbitrary variation readings are blocked for a related reason: there is no unique maximal function that satisfies the predicate either. In both cases, what rules out the arbitrary variation reading is the absence of a unique maximally informative function—though the specific reason for that absence differs.

More generally, only universal quantifiers guarantee a unique minimal domain—namely, the one corresponding exactly to the embedded quantifier’s NP restrictor (Pafel, 1999; Fox, 2012; Xiang, 2023)—and, in turn, a unique maximal function. Arbitrary variation readings are therefore predicted to be available only when the relative clause contains a universal quantifier. We note that the analysis in Sharvit 1999 also captures this restriction on arbitrary variation readings, albeit

through different means. Under that analysis, Sharvit appeals to a semantic operator that relies on extracting the minimal witness set of the quantifier inside the relative clause to generate an arbitrary variation reading. Given that only universal quantifiers have minimal witness sets, the quantifier restriction also falls out under this analysis. Effectively, an analysis that invokes a strength-based definite article can be viewed as an alternative way of cashing out Sharvit 1999’s insight that we need to extract the minimal witness set from the quantifier, one that has independent motivation.

2.4.1.2 A note on the internal composition of the NP

We now have an explanation of the quantifier restriction on arbitrary variation readings: the monotonicity properties of the NP denotations interact with a strength-based definite article in way that delivers reference with universal quantifiers but not negative ones. However, this explanation of the quantifier restriction crucially depended on our assumed NP denotations in (44) and (46). Can these denotations actually be grounded in a credible internal composition? In other words, do analyses of functional relative clauses from the literature deliver these assumed denotations? We now show that two prominent proposals of the internal composition of the relative clause, Heim 2018 and Jacobson 1994, 2018, do not deliver the target NP denotations. Moreover, we will show that, to get the target denotations, then, the domain condition of the function must enter into the NP denotation as asserted content within the scope of the quantifier, what we refer to as the ‘asserted domain condition’. As a consequence, this leads to a proposed constraint on how the internal composition of functional relative clauses should proceed.

(43) $[_{NP} \text{ woman that each boy invited}]$

(44) Assumed intension for (43):

$$\lambda w. \lambda f_{ee}. \forall x[\text{BOY}_w(x) \rightarrow x \in \text{dom}(f) \wedge \text{WOMAN}_w(f(x)) \wedge \text{INVITE}_w(f(x))(x)]$$

“f maps every boy to a woman he invited”

(45) $[_{NP} \text{ woman that no boy invited}]$

(46) Assumed intension for (45):

$$\lambda w. \lambda f_{ee}. \neg \exists x [\text{BOY}_w(x) \wedge x \in \text{dom}(f) \wedge \text{WOMAN}_w(f(x)) \wedge \text{INVITE}_w(f(x))(x)]$$

“f maps no boy to a woman he invited”

Heim’s denotations The analysis from Heim 2018 delivers the following denotations for each of the above NPs. For the discussion at hand, the crucial assumption that Heim makes is that the complex trace, $f(x)$, is defined only when the value of x is in the domain of f (and is mapped by f to a woman). That is, the complex trace triggers a presupposition that x is in the domain of f . As a result, when a quantifier, like *each boy*, binds the variable x , this presupposition projects universally and we end up with the *presupposition* that every boy is in the domain of the function.

Turning to (47), the crucial features of this denotation is that it (i) presupposes that f maps every boy to a woman and (ii) asserts that f maps every boy to someone he invited. Putting these two things together, (47), as a whole, conveys that f maps every boy to a woman he invited (see the appendix for more details concerning the internal composition and how exactly these denotations are derived).

(47) Heim’s denotation for (43):

$$\lambda w. \lambda f_{ee}. \forall x [\text{BOY}_w(x) \rightarrow x \in \text{dom}(f) \wedge \text{WOMAN}_w(f(x))]. \forall x [\text{BOY}_w(x) \rightarrow \text{INVITE}_w(f(x))(x)]$$

presupposes: “f maps every boy to a woman”

asserts: “f maps every boy to someone he invited”

total: “f maps every boy to a woman he invited”

Similarly, the crucial features of the denotation in (48) is that it (i) presupposes that f maps every boy to a woman and (ii) asserts that f maps no boy to someone he invited. This feature in (i) is a consequence of the assumption, as Heim assumes for (48), that negative quantifiers also project their presuppositions universally (Chemla, 2009; Sudo et al., 2012; Sudo, 2014). Putting these two things together, (48), as a whole, conveys that f maps every boy to a woman he didn’t invite.

(48) Heim’s denotation for (45):

$\lambda w. \lambda f_{ee}: \forall \mathbf{x}[\mathbf{BOY}_w(\mathbf{x}) \rightarrow \mathbf{x} \in \mathbf{dom}(\mathbf{f}) \wedge \mathbf{WOMAN}_w(\mathbf{f}(\mathbf{x}))]. \neg \exists \mathbf{x}[\mathbf{BOY}_w(\mathbf{x}) \wedge \mathbf{INVITE}_w(\mathbf{f}(\mathbf{x}))(\mathbf{x})]$

presupposes: “f maps every boy to a woman”

asserts: “f maps no boy to someone he invited”

total: “f maps every boy to a woman he didn’t invite”

The problem that arises with this analysis is that, in both cases, it is presupposed that every boy is in the domain of the function (given that both quantifiers project their presuppositions universally), as shown by the bolded portion in each denotation. This means that, even in the *no* case, the NP denotation is upward monotone—just like in the *each* case: if a function *g* is in (48) (meaning it maps every boy to a woman he didn’t invite) and *h* is a superset of *g*, then *h* will also be in (48) since it will also map every boy to a woman he didn’t invite. As a result, we overgenerate by wrongly predicting that an arbitrary variation reading should also be possible with negative quantifiers.⁶

Jacobson’s denotations The analysis from Jacobson 1994, 2018 delivers the following denotations for each of the above NPs. The crucial assumption for the following discussion is that Jacobson assumes a classic head external analysis of the relative clause where the head noun of the relative clause, *woman*, gets interpreted in its surface position. For it to combine with the relative clause, then, we must assume that it can be type-shifted from a predicate of individuals, of type (et), to a predicate of functions, of type ((e)t), and denote the set of woman-valued functions.

Turning to (49), this denotation asserts that *f* maps every element of its domain (whatever it may be) to a woman and also that *f* maps every boy to someone he invited (again, see the appendix for more details concerning the internal composition and how these denotations are derived).

(49) Jacobsonian denotation for (43):

⁶In a sense, this is to be expected, and Heim views this as a virtue of her analysis of functional questions/relative clauses since she only considers principled variation questions/relative clauses and not arbitrary variation ones. That is, Heim’s analysis is meant to generate functional readings with both universal and negative quantifiers alike.

$$\lambda w. \lambda f_{ee}. \quad \forall x[x \in \mathbf{dom(f)} \rightarrow \mathbf{WOMAN}_w(\mathbf{f(x)})] \wedge \forall x[\text{BOY}_w(x) \rightarrow \text{INVITE}_w(f(x))(x)]$$

“f maps every element of its domain to a woman &
f maps every boy to someone he invited”

Similarly, the denotation in (50) asserts that f maps every element of its domain (whatever it may be) to a woman and also that f maps no boy to someone he invited.

(50) Jacobsonian denotation for (45):

$$\lambda w. \lambda f_{ee}. \quad \forall x[x \in \mathbf{dom(f)} \rightarrow \mathbf{WOMAN}_w(\mathbf{f(x)})] \wedge \neg \exists x[\text{BOY}_w(x) \wedge \text{INVITE}_w(f(x))(x)]$$

“f maps every element of its domain to a woman &
f maps no boy to someone he invited”

The problem that arises with this analysis is that, because of the strong condition that every element of the function’s domain, whatever it may be, be mapped to a woman (the bolded portion in each denotation), these denotations cannot be upward monotone. To see why, consider the minimal function whose domain ranges just over the boys. Assume that the denotation in (49) maps this function to true. Even if this is the case, this does not guarantee that an arbitrary extension of this minimal function is also mapped to true; simply because nothing about the minimal function being mapped to true ensures that the non-boy members of the larger domain are mapped to a woman—and the denotation in (49) requires that every element of the function’s domain must be mapped to a woman. Therefore, the denotation in (49) is not upward monotone. As a result, when it comes to the arbitrary variation reading, the *each* case cannot be captured and we undergenerate by wrongly predicting that arbitrary variation readings are not possible in this case.

Scope sites for negation. We now make one final remark concerning the internal composition of the relative clause. As discussed above, it has been observed that arbitrary variation readings are not available with negative quantifiers. We now build upon this observation and show that this restriction isn’t just about having negation in the relative clause, but rather about the different scope sites for negation that the internal composition provides. That is, if negation scopes high—as it does with negative quantifiers—arbitrary variation readings are unavailable; alternatively, if

negation scopes low, then arbitrary variation readings are possible. To illustrate this point, consider the facts in w_3'' in (51), with the two possible target sentences in (52).

(51) **Facts in w_3'' :** The fourth grade boys are Al, Bill and Chris. Al didn't invite Carol; Bill didn't invite Diane; Chris didn't invite Mary. All the fourth grade boys invited everyone else.

- (52) a. The woman that no fourth grade boy invited is the woman that each fourth grade boy admires. (*✗arbitrary variation*)
- b. The woman that each fourth grade boy didn't invite is the woman that each fourth grade boy admires. (*✓arbitrary variation*)

The observation is that an arbitrary variation reading is unavailable for (52-a), as expected, but it is available for (52-b). As a result, only (52-b) is felicitous and true in worlds like w_3'' . This is somewhat surprising since one might expect the two sentences to be semantically equivalent, given that *No fourth grade boy invited John* is logically equivalent to *Each fourth grade boy didn't invite John*. However, comparing the two target denotations in (54) and (56), we can isolate the relevant difference (for (56), we are assuming the sort of target NP denotation assumed earlier for the *each* case). In (54), negation sits high in the structure, next to the existential quantifier. In contrast, in (56), negation sits lower in the structure, on the VP and, crucially, below the universal quantifier.

(53) [_{NP} woman that no boy invited]

(54) Assumed intension for (53):

$$\lambda w. \lambda f_{ee}. \neg \exists x [\text{BOY}_w(x) \wedge x \in \text{dom}(f) \wedge \text{WOMAN}_w(f(x)) \wedge \text{INVITE}_w(f(x))(x)]$$

"f maps no boy to a woman he invited"

(55) [_{NP} woman that each boy didn't invite]

(56) Assumed intension for (55):

$$\lambda w. \lambda f_{ee}. \forall x [\text{BOY}_w(x) \rightarrow x \in \text{dom}(f) \wedge \text{WOMAN}_w(f(x)) \wedge \neg \text{INVITE}_w(f(x))(x)]$$

"f maps every boy to a woman he didn't invite"

As a result, this empirical contrast between (56-a) and (56-b) is captured under our analysis, in virtue of (56-b) containing a universal quantifier that scopes above negation. According to the denotation for (55) in (56), this case makes the same prediction as the *each* case, meaning the *each not* case empirically patterns with the *each* case, and not the *no* case. Following the same reasoning as above, if a function g maps every boy to a woman he didn't invite and g is a subset of h , then it will also be the case that h maps every boy to a woman he didn't invite.

(57) ***Each not* case: upward monotonicity**

for all $g, h \in D_{(ee)}$

- a. if $g \subset h$ then $\{w : \llbracket (55) \rrbracket (w)(g) \} \subseteq \{w : \llbracket (55) \rrbracket (w)(h) \}$
" g maps every boy to a woman he didn't invite $\Rightarrow$
 h maps every boy to a woman he didn't invite "
- b. if $g \subset h$ then $\{w : \llbracket (55) \rrbracket (w)(h) \} \not\subseteq \{w : \llbracket (55) \rrbracket (w)(g) \}$
" h maps every boy to a woman he didn't invite $\nRightarrow$
 g maps every boy to a woman he didn't invite "

(58) $[_{DP} \text{ the } [_{NP} \text{ woman that each boy didn't invite }]]$

Thus, the internal composition of the relative clause provides different scope sites for negation. Depending on where negation scopes, this has consequences for the monotonicity properties of the NP denotation: when negation scopes high, we get a downward monotone NP; when negation scopes low, we get an upward monotone NP.

2.4.1.3 Taking stock

What we have seen is that the strength-based maximality proposal makes concrete predictions about the internal composition of the relative clause: to derive the quantifier sensitivity discussed above, the NP must have the assumed denotations we have been working with. These predictions are non-trivial since existing analyses of functional relative clauses (i.e., Heim's and Jacobson's) do not deliver them. In some sense, though, Heim's denotations are closer to the target denotations

than Jacobson’s are. The tension is that, to capture the intended monotonicity properties, *woman* must be interpreted in the scope of the quantifier—to avoid the problem with Jacobson’s denotations. At the same time, the presupposition posited by Heim must be prevented from projecting in the general case. To get the target denotations, then, the purported triggered presupposition must enter into the NP denotation as asserted content within the scope of the quantifier, what we refer to as the ‘asserted domain condition’. Our proposal therefore imposes a constraint on how the internal composition of functional relative clauses should proceed—a constraint that, to our knowledge, is not satisfied by any existing analysis in the literature.

One route to satisfying this constraint is to assume that there is a requirement, for some unknown reason, that the scope of the quantifier contains an ‘accommodation’ operator (cf. Beaver and Krahmer 2001): an operator that turns presupposed content into asserted meaning (by turning conditions for truth-valuedness into conditions for truth). This line of thought could involve the insertion of a silent operator, the so-called ‘floating-A’ operator of Beaver and Krahmer (2001), which converts presuppositions into entailments. Supplementing the above Heimian analysis with this kind of operator would lead to an LF as in (59), which would lead to the denotation in (60). In (59), following Heim (2018), we assume a head-raising analysis of the relative clause so that the head noun, *woman*, is interpreted in the scope of the quantifier (see the Appendix for more details concerning this head-raising structure and how to interpret the complex trace, which invokes trace conversion, as proposed in Fox 2002). In addition, the A operator attaches right below the universal DP.

$$(59) \quad [_{CP} \text{ wh } \lambda_2 [[\text{each boy}] \lambda_1 A [t_{1,e} \text{ invited } [\text{THE woman IDENT } [f_{2,(ee)} \text{ pro}_{1,e}]]]]]$$

$$(60) \quad \lambda w. \lambda f_{ee}. \forall x [\text{BOY}_w(x) \rightarrow [x \in \text{dom}(f) \wedge \text{WOMAN}_w(f(x))]] \wedge \forall x [\text{BOY}_w(x) \rightarrow \text{INVITE}_w(f(x))(x)]$$

“f maps every boy to a woman and f maps every boy to someone he invited”

In (60), the presupposition triggered by the complex trace (which gets projected universally) is converted to asserted content by A and shows up as the first conjunct. Crucially, although written differently, (60) is equivalent to the target NP denotation in (22).

Another route to satisfying this constraint would be to assume that the domain condition is not triggered as a presupposition in the first place but arises locally as an asserted conjunct.⁷

In sum, the strength-based maximality proposal places a constraint on the internal composition of functional relative clauses: we have argued that our assumed denotations are useful in accounting for arbitrary variation readings and so we believe there should be some compositional pathway to these denotations. While no existing analysis straightforwardly delivers these denotations, we have suggested that a Heimian analysis supplemented with an accommodation operator points to one route by which the constraint could be satisfied. We take this to be a non-trivial contribution of the proposal: it tells us something about how the internal composition of functional relative clauses should be constrained, which is a question that has not been settled in the existing literature.

2.4.2 Domain expansion blocks arbitrary variation

We now turn to a consequence concerning how the specifics of the internal composition interact with the external composition. When it comes to the internal composition of arbitrary functional DPs (with a universal quantifier in the relative clause), we saw that the definite article picks out the *minimal* function (i.e., the function whose domain is exactly the embedded quantifier’s NP restrictor). Our analysis then predicts that these kinds of functional DPs should be unavailable whenever the semantic composition, for some reason, *requires* the function’s domain to extend beyond the embedded quantifier’s NP restrictor. In this section, we show that this prediction is borne out.

Assessing this prediction involves looking at cases where a distributive matrix subject binds into the relativized DP—henceforth, ‘dependency readings’. When the matrix and embedded quan-

⁷Pursuing this line of thought could involve treating the complex trace as an operator, as suggested in (i). In (i), Op takes an (ee) function and an individual argument, x , and asserts that x is in the domain of the function. Thus, we can pass the complex trace—consisting of a function variable and a pronoun—to Op, one at a time. Once the pronoun gets bound by the universal quantifier, we end up with the desired result: an *asserted* conjunct that says that every boy is in the domain of the function, as in (i-c). We leave a more detailed discussion of this line of thought to future work.

- (i) a. $\llbracket \text{Op} \rrbracket^g = \lambda f_{(ee)} \lambda x_e \lambda P_{(et)}. x \in \text{dom}(f) \wedge P(f(x))$
- b. $\llbracket \llbracket [\text{Op } f] \text{ pro}_{1,e} \rrbracket \rrbracket = \lambda P_{(et)}. g(1) \in \text{dom}(f) \wedge P(f(g(1)))$
- c. $\llbracket [\text{each boy}] \lambda_1 \llbracket \text{woman } \llbracket [\text{Op } f] \text{ pro}_{1,e} \rrbracket \rrbracket \rrbracket = \forall x [\text{BOY}(x) \rightarrow [x \in \text{dom}(f) \wedge \text{WOMAN}(f(x))]]$

tifiers have matching domains (e.g., both quantify over fourth grade boys), arbitrary functional DPs are possible. But when their domains are different (e.g., the matrix universal quantifies over third grade boys, while the embedded universal quantifies over fourth grade boys), only principled functional DPs are possible. This asymmetry follows directly from our analysis.

2.4.2.1 Functional DPs in argument positions

Consider the sentence in (61), where the relevant functional DP, *the woman that each of them invited last year*, is in object position.

- (61) (This year) Every fourth grade boy invited the (same) woman that each of them invited last year.

Intended: for each fourth grade boy x , x invited the woman that x invited last year.

Before looking at cases of domain matching and expansion, we need to understand how functional DPs can saturate argument positions, like the complement of *invite*. The compositional challenge is that the functional DP in object position is of type (ee) , as shown in (62), and the predicate, *invite*, expects an argument of type e . Thus, somehow, we need a way to turn something of type (ee) into something of type e .

- (62) Sample functional DP denotation:

$$\begin{aligned} \llbracket \text{the woman that each of them invited last year} \rrbracket = \\ \lambda x_e: 4GB(x). \iota \{y: WOMAN(y) \wedge INVITE-L-Y(y)(x)\} \end{aligned}$$

One way to resolve this tension is to propose a compositional structure like (63). According to this compositional structure, the functional DP gets saturated, to feed the matrix predication, by having the complex definite description first combine with an argument, x , before combining with the matrix predicate, *invite*. To achieve this, we can assume that the definite DP combines with a covert pronoun (Heim, 2018).⁸ This gives us the right kind of argument for the matrix predicate.

⁸Alternatively, one can adopt a convention in for evaluating an individual predicate as it is applied to a function, as done in Hulsey and Sauerland 2006. This would involve a meaning for the definite article as in, (i-a), and the rule in (i-b). As far as we can tell, nothing we say here hinges on the assumption that a covert pronoun combines with the relativized DP.

(63) External LF for function in argument position for (61):

$$\begin{aligned} & \llbracket \text{invited} [\text{pro}_{1,e} [\text{DP the woman that each of them invited last year}]] \rrbracket = \\ & \llbracket \text{invited} \rrbracket (\lambda z_e: 4GB(z). \iota \{y: \text{WOMAN}(y) \wedge \text{INVITE-L-Y}(y)(z)\}(x)) = \\ & \llbracket \text{invited} \rrbracket (\iota \{y: \text{WOMAN}(y) \wedge \text{INVITE-L-Y}(y)(x)\}) \end{aligned}$$

In what follows, then, we will assume that functional DPs saturate argument positions by first combining with a covert pronoun to derive something of type e .

2.4.2.2 Dependency reading with matching domains

Before examining cases of domain expansion, we first present cases where the matrix and embedded quantifiers have matching domains. To illustrate, we again consider the sentence in (61), repeated in (64), which contains two universal DPs: one inside the relative clause and one in the matrix subject position. Both universal DPs quantify over the same domain of individuals: the set of fourth grade boys.

(64) (This year) Every fourth grade boy invited the (same) woman that each of them invited last year.

Intended: for each fourth grade boy x , x invited the woman that x invited last year.

Suppose the facts are as follows:

(65) **Facts in w_4 :** The fourth grade boys are Al, Bill and Chris. Last year, Al invited Mary; Bill invited Carol; Chris invited Sue. This year, Al invited Mary; Bill invited Carol; Chris invited Sue. No fourth grade boy invited anyone else (last year or this year).

In world w_4 , (64) is intuitively true. Furthermore, the mapping from boys to women is arbitrary—there's no external function that the relativized DP can refer to, meaning we have an arbitrary

-
- (i) a. $\llbracket \text{the} \rrbracket (P_{((ee)t)}) = f$, where f is the unique function such that
 i. $\text{dom}(f) = \min\{\text{dom}(g) \mid g \in \text{dom}(P)\}$
 ii. $P(f) = 1$
 b. $P_{(et)}(f_{(ee)}) = 1$ iff $\forall x \in \text{dom}(f) : P(f(x)) = 1$

functional DP, not a principled one.⁹

This means that (64) has a reading where the matrix universal seemingly binds into the relativized DP to convey the following:

$$(66) \quad \forall x[4GB(x) \rightarrow INVITE(\iota\{y: WOMAN(y) \wedge INVITE-L-Y(y)(x)\})(x)]$$

This point can be clarified by considering the parallel between (64) and (67): it seems that the intended reading of (64) is equivalent to (67) in a bound variable reading.

$$(67) \quad \text{Every fourth grade boy invited the woman that he invited last year.}$$

That is, given our analysis of relativized DPs (and assuming that each fourth grade boy invited exactly one woman), the relativized DP ends up denoting the (minimal) function in (68).

$$(68) \quad \text{DP denotation in } w_4:$$

$$\begin{aligned} & \llbracket \text{the woman that each of them invited last year} \rrbracket = \\ & \lambda x_e: 4GB(x). \iota\{y: WOMAN(y) \wedge INVITE-L-Y(y)(x)\} \end{aligned}$$

One way to derive the target meaning in (66) is to propose a compositional structure like (69). The structure in (69) builds on the one in (63) by having the definite DP combine with a covert pronoun, to derive an *e* type object. The target sentence meaning in (66) then falls out under the assumption that the universal subject binds this pronoun.

$$(69) \quad \text{External LF and composition for (64):}$$

$$\begin{aligned} & \llbracket [\text{every fourth grade boy}] \lambda_1 [t_{1,e} \text{ invited } [\text{pro}_{1,e} [\text{DP the woman that each of them invited} \\ & \text{last year}]]] \rrbracket = \\ & \llbracket \text{every 4gb} \rrbracket (\lambda x_e. \llbracket \text{invited} \rrbracket (\lambda z_e: 4GB(z). \iota\{y: WOMAN(y) \wedge INVITE-L-Y(y)(z)\}(x))(x)) = \end{aligned}$$

⁹Further evidence that (64) involves an arbitrary functional DP comes from replacing the universal DP in the relative clause with a negative DP, as in (i). Given that arbitrary functional DPs are not possible with negative DPs, the object DP in (i) should only license a principled variation reading. This is borne out: we can no longer have an arbitrary mapping from one fourth grade boy to Mary, another fourth grade boy to Carol, and another fourth grade boy to Sue, rather, we need an external principle, like mapping each boy to his mother, to enable the variation.

(i) (This year) Every fourth grade boy invited the woman that none of them invited last year.

$$\llbracket \text{every 4gb} \rrbracket (\lambda x_e. \llbracket \text{invited} \rrbracket (\iota \{y: \text{WOMAN}(y) \wedge \text{INVITE-L-Y}(y)(x)\})(x))$$

Crucially, the higher quantifier's domain is the fourth grade boys, therefore, the pronoun it binds also ranges over fourth grade boys. Luckily each possible value of the pronoun is in the domain of the function, hence composition succeeds and we derive the intended interpretation.

2.4.2.3 Domain Expansion: mismatched quantifiers

Now consider what happens when the matrix and embedded quantifiers have different domains, as in (70).¹⁰

- (70) Every third grade boy invited the woman that each fourth grade boy invited (namely, his mother).

In (70), the matrix universal subject quantifies over third grade boys, while the universal in the relative clause quantifies over fourth grade boys.¹¹ If our analysis is on the right track, under an arbitrary functional DP, this should create a problem for the composition since they require functions defined *exactly* over the embedded quantifier's domain. This is because, following our analysis of dependency readings from the previous section, the binding relationship between the matrix subject and the pronoun fed to the relativized DP requires the function to apply to third grade boys (since the matrix subject binds the pronoun). But third grade boys are not in the domain of the minimal function. The prediction, then, is that, since the composition requires the function to

¹⁰Both Heim 2018 and Jacobson 2018 discuss examples like (70); albeit for a different purpose.

¹¹We do note a class of examples where the quantifier domains are mismatched and yet an arbitrary variation reading seems to be available. Consider (i).

- (i) a. *[There are three boys (Al, Bill, Chris) and three girls (Diane, Erica, Francine). The boys and girls are placed into pairs for the party. Each girl invited a woman to the party: Diane invited Mary; Erica invited Carol; Francine invited Sue. Unfortunately: Al dislikes Mary; Bill dislikes Carol; Chris dislikes Sue. None of the girls invited anyone else and none of the boys dislike anyone else.]*
 b. Every boy dislikes the woman that each girl invited.

The authors' intuitions are that the target sentence in (i-b) is felicitous and true, given the facts in (i-a), suggesting that an arbitrary variation reading is available. It is worth pointing out, though, that these kinds of examples seem to require a bijection between the two quantifier domains (meaning the notion of a 'minimal function' may somehow still be satisfied, perhaps by assuming some sort of function composition operator). Alternatively, it could be that domain of the function in this case is actually the pairs of boys and girls, and not just the girls. We leave a detailed discussion of these kinds of examples to future work.

apply to individuals outside its minimal domain, an arbitrary functional DP should be unavailable. Only a principled functional DP should be possible—since these functions can be defined over a larger domain via the external principle that is provided by the natural function.

This prediction is borne out. Sentence (70) is acceptable, but only with a principled functional DP; an arbitrary functional DP is unavailable. To clarify why this is the case in (70), we can look at how the external composition works in (71).

(71) External LF and composition for (70):

$$\begin{aligned} & \llbracket [\text{every third grade boy}] \lambda_1 [t_{1,e} \text{ invited } [\text{pro}_{1,e} [\text{DP the woman that each fourth grade boy} \\ & \text{invited}]]]] \rrbracket = \\ & \llbracket \text{every 3gb} \rrbracket (\lambda x_e. \llbracket \text{invited} \rrbracket (\lambda z_e: 4GB(z). \iota \{y: \text{WOMAN}(y) \wedge \text{INVITE}(y)(z)\}(x))(x)) = \\ & \llbracket \text{every 3gb} \rrbracket (\lambda x_e. \llbracket \text{invited} \rrbracket (\iota \{y: \text{WOMAN}(y) \wedge \text{INVITE}(y)(x)\})(x)) \end{aligned}$$

In (71), the external composition parallels what we saw above in (69), where the universal matrix subject binds the pronoun that gets fed into the relativized DP. In this case though, the pronoun that gets fed into the relativized DP has to range over third grade boys. If the domain of the (ee) function is just the set of fourth grade boys and we try to apply the (ee) function to this pronoun, the composition would fail: third grade boys are not in the function's domain. Instead, the relativized DP must refer to a function which is defined over a larger domain than the fourth grade boys. Given the binding relationship between the matrix subject and the covert pronoun, to interpret the structure in (71), the (ee) function must be defined for both third and fourth grade boys. Principled variation readings can succeed because they can involve functions defined over larger domains—there's no requirement that the function's domain be minimal. For instance, the "mother-of" function, repeated in (72), is defined for any individual with a mother—including both third and fourth grade boys. Since this function's domain includes third grade boys, then, it can successfully apply to the pronoun bound by the matrix universal.

(72) Possible DP denotation for (71):

$$\lambda x_e: \exists! y [\text{MOTHER}(x)(y)]. \iota \text{MOTHER}(x)$$

Thus, the "mother-of" function—with its large domain—is an acceptable denotation for the relativized DP. In contrast, for the DP in (69), we don't need to expand the domain at all; the minimal function in (68) suffices.

It is worth pointing out that we can also convey the same point concerning mismatched domains by considering analogous copular clauses. Consider the contrast between (73-a) and (73-b).

- (73) a. The woman that each fourth grade boy invited is the woman that each fourth grade boy admires.
- b. The woman that each fourth grade boy invited is the woman that each third grade girl admires.

As we saw in worlds like w_3 , (73-a) can be true with respect to an arbitrary mapping between boys and women. In contrast, when the pre-copular relativized DP and the post-copular relativized DP have different domains (i.e., fourth grade boys vs third grade girls), an arbitrary variation reading becomes unavailable. The sentence in (73-b) is only interpretable to the extent that we can accommodate a natural function, like *their favorite teacher*, that each relativized DP could then refer to.

In sum: when the semantic composition requires an (ee) function to apply to individuals outside the quantifier's NP restrictor, arbitrary functional DPs become unavailable. This happens in dependency readings where the matrix quantifier and the one in the relative clause have mismatched domains. Minimal functions are defined over the smallest possible domain—the quantifier's NP restrictor. When composition requires applying the function to something outside this domain, composition fails and principled functional DPs (with larger domains) are forced. Empirically, this asymmetry between matched and mismatched domains provides independent empirical support for our claim that arbitrary functional DPs with universal quantifiers involve minimal functions while principled variation DPs involve natural functions with potentially larger domains. It also directly reflects the difference between the two kinds of functional DPs: minimal functions are precisely

those that do not require domain expansion, while natural functions impose no such restriction.¹²

2.4.3 Plural relativized DPs: multiple smallest functions

So far we have only focused on relativized DPs where the head noun of the relative clause is singular. When we move from singular to plural relativized DPs, as in (27), the fact that these cases are possible is predicted by our analysis—an observation which, as far as we know, has yet to be pointed out in the literature.

(74) $[_{DP}$ the women that each fourth grade boy invited]

Consider (27) in a scenario where Andy, Bill and Chris are the fourth grade boys. Andy invited Mary, Sue and Jess, Bill invited Carol and Diane, and Chris invited Erica and Fiona. In this scenario, we want the relativized DP in (27) to denote a type (ee) function mapping each fourth grade boy to all the women he invited, as in (75).

(75) Intended function:

$$f = \{ \langle \text{Andy}, \text{Mary+Sue+Jess} \rangle, \langle \text{Bill}, \text{Carol+Diane} \rangle, \langle \text{Chris}, \text{Erica+Fiona} \rangle \}$$

One consequence of our analysis is that it extends to these cases without any further assumptions: appealing to a definite article which references strength-based maximality makes the right predictions in these cases as well. To see why, we first note that the denotation of the NP that combines with the definite article would be:

(76) NP denotation:

$$\llbracket [_{NP} \text{ women } [_{CP} \text{ wh } \lambda_2 [\text{that } [\text{each fourth grade boy } \lambda_1 [t_{1,e} \text{ invited } [f_{2,(ee)} \text{ pro}_{1,e}]]]]]] \rrbracket$$

¹²Strictly speaking, this isn't true. For the dependency reading with an arbitrary functional DP, the domains of both universals don't need to match exactly—as long as the subject universal quantifies over a subset of the embedded universal DP. To illustrate, consider the example in (i).

- (i) Every student except John revised the paper that each of them wrote.
- (ii) Every boy except Herb was happy about the woman that each of them invited.

The intended reading in (i) conveys that every student, including John, wrote a paper and, furthermore, that John is the only student that didn't revise their paper.

$$= \lambda w. \lambda f_{(ee)}. \forall x[4GB_w(x) \rightarrow [x \in \text{dom}(f) \wedge \text{WOMEN}_w(f(x)) \wedge \text{INVITE}_w(f(x))(x)]]$$

The denotation in (76) imposes two requirements on each function in the set: (i) that every fourth grade boy is in the function's domain, and (ii) that each boy is mapped to a some *plurality* of women he invited. Unlike the singular case, though, there is no single minimal function here: many functions map each fourth grade boy to *some* plurality of women he invited and these functions can share the same minimal domain, differing only in the plurality of women each boy gets mapped to. This is illustrated by in (77), where the only difference between the functions concerns who Andy invited.

- (77) a. $f_1 = \{\langle \text{Andy}, \text{Mary+Sue} \rangle, \langle \text{Bill}, \text{Carol+Diane} \rangle, \langle \text{Chris}, \text{Erica+Fiona} \rangle\}$
 b. $f_2 = \{\langle \text{Andy}, \text{Mary+Sue+**Jess**} \rangle, \langle \text{Bill}, \text{Carol+Diane} \rangle, \langle \text{Chris}, \text{Erica+Fiona} \rangle\}$

Understanding why f_2 is the maximally informative function among the two functions in (77) parallels the ordinary plural case discussed in Section 2.3.2.1. There, we saw that a strength-based definite article selects the most informative element in the NP extension under the part-whole ordering on individuals: the most informative element simply is the maximal individual under the part-whole ordering. The same principle applies here: the entailment ordering on (ee) functions not only ensures a minimal domain, but it also ensures a maximal range.

To see this, recall that under our strength-based definite article, the selected function must be one that is not entailed by any other function in the NP extension—it must be the semantically strongest one. Now, since *invite* is distributive, $\text{invite}(m+s+j)(a)$ strictly entails $\text{invite}(m+s)(a)$. Hence, $\text{invite}(m+s+j)(a)$ is maximal in terms of semantic strength and $m+s+j$ is the maximally informative element—with respect to who Andy invited. This means that f_2 asymmetrically entails f_1 and is more informative than f_1 : whenever f_2 satisfies the NP predicate, f_1 is guaranteed to as well, but not vice versa. Therefore, f_1 is ruled out by the strength-based definite article as there exists a stronger function in the NP extension that entails it. It follows then that f_2 is selected as the unique maximally informative function, delivering the intended denotation in (75).

Crucially, this result follows directly from the independently motivated decision to weaken

the Fregean uniqueness condition with reference to a notion of strength-based maximality. No additional assumptions are needed to handle plural relativized DPs: with universal quantifiers, the same mechanism that delivers a unique minimal function in the singular case delivers a unique maximal function in the plural case, simply because semantic strength is sensitive to the outputs of the function and not just its domain.¹³

2.5 Lingering issue: projection readings

In this section, we make one final observation concerning principled and arbitrary functional DPs. We already saw that a distributive matrix subject can bind the covert pronoun that we assumed gets fed into the relativized DP. As a consequence, if the individuals quantified over by the distributive matrix subject did not match the domain of the function, we got cases of domain expansion, in turn ruling out arbitrary functional DPs. Let us now look at the flip side of domain expansion: cases where, if you stay within the domain for the possible binders of the pronoun, an arbitrary functional DP is expected to be available. Interestingly, this prediction is not borne out. For this reason, we hypothesize that independent pragmatic considerations can render arbitrary functional DPs unavailable.

We begin by observing that (79) is intuitively true given the facts in world w_5 , in (78).

(78) **Facts in w_5 :** Every fourth grade boy invited his own mother. Herb is one of the fourth grade boys.

(79) Herb invited [the woman that each fourth grade boy invited].

Just as we saw for w_2 in (5), in (78), the relative clause doesn't restrict the NP *woman* to a singleton set of individuals. Rather, the sentence permits variation of women by fourth grade boy

¹³For non-distributive predicates, as in (i), the same problem doesn't arise. Assume there are three TAs, and each TA arranged for three students to form a group. If the function maps TA_1 to a group of two students (instead of 3), this function will no longer satisfy the predicate of functions since it will no longer be true that the TA got the two students to form a group. As a result, the only function that ends up satisfying the predicate of functions is the maximally informative one.

(i) Mary wrote down the students that each TA got to form a group.

since different boys invited different women. Again, this perceived variation is principled though: for each fourth grade boy, the woman he invited is his mother. Thus, given the facts in w_5 , the target sentence in (79) conveys that, just as every other fourth grade boy invite his own mother, Herb also invited his own mother.

How can we derive the reading illustrated in (78)? Since the facts in w_5 imply that Herb is in the domain of this function, it can be taken to apply to Herb in the composition. This means that Herb can serve as input to the function, and the output is Herb's mother. This leads to a compositional structure like (80), where, again, the complex definite description first combines with an argument, *Herb*, before combining with the matrix predicate, *invite*.

(80) Target sentence denotation:

$$\begin{aligned} & \llbracket \text{Herb invited } [_{DP} \text{ the woman that each fourth grade boy invited}] \rrbracket = \\ & \llbracket \text{invite} \rrbracket (\llbracket [_{DP} \text{ the woman that each fourth grade boy invited}] \rrbracket (\text{Herb})) (\text{Herb}) \Leftrightarrow \\ & \llbracket \text{invite} \rrbracket (tMOTHER(\text{Herb})) (\text{Herb}) \end{aligned}$$

According to this compositional structure, the DP denotation is applied to the subject denotation, which is Herb in this case. In this case then, we can have the definite DP combine with a covert pronoun that is anaphoric to the subject, as illustrated through the LF in (81) and the denotation in (82).

(81) External LF:

$$\text{Herb } \lambda_2 [t_2 \text{ invited } [pro_{2,e} [_{DP} \text{ the woman that each fourth grade boy invited}]]]$$

$$\begin{aligned} (82) \quad & \llbracket \text{Herb } \lambda_2 [t_2 \text{ invited } [pro_{2,e} [_{DP} \text{ the woman that each fourth grade boy invited}]]] \rrbracket = \\ & \llbracket \text{invite} \rrbracket (\llbracket [_{DP} \text{ the woman that every fourth grade boy invited}] \rrbracket (\text{Herb})) (\text{Herb}) \Leftrightarrow \\ & \llbracket \text{invite} \rrbracket (tMOTHER(\text{Herb})) (\text{Herb}) \end{aligned}$$

In what follows, we refer to the reading illustrated in (78) as the ‘projection reading’ since the reading involves selecting one of the particular outputs of the functional DP.

We now observe that this kind of reading is not available with an arbitrary functional DP. To

illustrate, consider the world w_5 in (83).

- (83) **Facts in w_5 :** The fourth grade boys are Billy, Herb and Jim. Billy invited Mary; Herb invited Carol; Jim invited Sue. None of the fourth grade boys invited anyone else.

Given the facts in w_5 , the target sentence in (79) cannot be understood to convey that Herb invited Carol.¹⁴

Why is this so? While we cannot offer a definitive explanation for this contrast, to conclude this section, we offer some preliminary remarks concerning this restriction. In particular, given the dependency readings discussed above, it doesn't seem to be the case that projection readings with arbitrary functional DPs are ruled out as a constraint on possible LFs (i.e., a grammatical constraint). In fact, this is confirmed by considering sentences like (85).

- (85) Only Al revised the paper that each of them wrote in the research seminar.

The observation is that (85) has a reading that can be paraphrased as: nobody but Al revised the paper that he or she wrote in the research seminar. To derive this reading compositionally, we could posit the LF in (86), where the relativized DP gets fed a covert pronoun that is bound by the matrix subject.

- (86) External LF:

Only [Al λ_2 [t₂ revised [pro_{2,e} [the paper that each of them wrote]]]]

The availability of this reading for (85) (with an arbitrary functional DP) and its corresponding LF in (86) confirms that the projection reading is not outright grammatically blocked with arbitrary functional DPs. If it were, no sentence with that configuration should be grammatical, yet (85) is acceptable under exactly that configuration. This suggests that the unavailability of the projection reading in worlds like w_5 reflects a pragmatic constraint rather than a grammatical one. This

¹⁴We also observe that the definite DP denotation needn't be applied to the subject denotation. For instance, consider (84), where now the definite DP denotation can be applied to the direct object denotation, Herb. That is, (84) can be understood to convey that John assigned Herb to Herb's mother.

- (84) John assigned Herb to the woman that every boy wanted to be assigned to (namely, his mother).

conclusion sits comfortably with the general picture developed throughout this paper: what distinguishes principled and arbitrary functional DPs is not an ambiguity at LF concerning the way the relative clause is parsed, but a difference in the way reference to a function is achieved. While we do not claim this to be a fully worked-out analysis, the projection asymmetry, under this view, is simply an instance of pragmatic regulation—one that we leave for future work to fully explain.¹⁵

2.6 Conclusion

The starting point of this paper was a familiar observation: definite DPs with relative clauses containing quantifiers can give rise to multiple readings. What is less clear in the literature is how the principled variation and arbitrary variation readings are related, and whether they require different compositional structures for the relative clause itself. In this paper, we have argued that they do not. Both readings involve a functional parse of the relative clause that delivers a predicate of (ee) functions, and the ambiguity between the two readings is not rooted in how the relative clause is internally composed. What distinguishes the two readings is the route by which the definite article achieves reference to a function.

¹⁵Relatedly, one possible line of approach is to argue that the projection reading with an arbitrary functional DP is blocked since the speaker could have used the simpler sentence in (i) to convey the same thing.

- (i) Herb invited Carol.

Naturally, this raises the question of why the projection reading with a principled functional DP is not blocked. Let's consider again the sentence in (ii).

- (ii) Herb invited the woman that each fourth grade boy invited (namely, his mother).

(ii) conveys that Herb invited his own mother. In addition, it conveys that each fourth grade boy also invited his own mother. That is, even though the sentence asserts something about Herb and who *he* invited, the use of the relative clause still contributes further information. Consequently, the simpler sentence in (iii) doesn't convey as much information as (ii).

- (iii) Herb invited his mother.

In contrast, with an arbitrary functional DP, the fourth grade boys all invited different women, so the relative clause will not contribute any further information and (79) conveys the same meaning as (i), albeit, in a convoluted way. In other words, with an arbitrary functional DP, the projection reading doesn't convey any extra information when we add the relative clause. Given that (i) is much simpler, it is preferred. In contrast, with a principled functional DP, (ii) conveys more information than (iii)—since it also tells us who each fourth grade boy other than Herb invited. Perhaps, then, the extra information conveyed by the relative clause is enough to license the projection reading in this case. This would suggest that a kind of *pragmatic* blocking effect—i.e., a preference grounded in informativeness—could be at play in the projection reading.

For principled variation readings, this mechanism is pragmatic in nature, as it involves contextual domain restriction: the unique function is the intersection of C , given by external principles like the "mother-of" function, and the NP extension in the world of evaluation. For arbitrary variation readings, no such natural function is available and, as a result, domain restriction does not help in these cases. We argued that reference to a function can nonetheless be achieved, but through a different route. The key move is to weaken the Fregean uniqueness requirement with reference to strength-based maximality, following Von Stechow et al. 2014. Under this view, the definite article picks out the maximally informative function with respect to the predicate of functions determined by the NP it combines with. Given the upward monotonicity of NPs with universal quantifiers in the relative clause, this turns out to be the minimal function whose domain corresponds exactly to the NP restrictor of the embedded universal.

We have argued that this move has genuine utility: it is not merely a technical revision to the semantics of the definite article, but one that makes correct and non-trivial predictions. For instance, under the assumption that the domain condition of the function enters the NP denotation as asserted content, rather than presupposed content, the proposal makes nuanced predictions about the internal composition of the relative clause that turn out to be correct. It correctly predicts that arbitrary variation readings are restricted to relativized DPs with universal quantifiers in the relative clause. It correctly predicts that arbitrary variation readings become unavailable when the semantic composition requires the function's domain to extend beyond the quantifier's NP restrictor, as in dependency readings with mismatched quantifier domains. And it extends naturally to plural relativized DPs, where the same mechanism selects the correct function without any additional assumptions.

The broader lesson concerns the theory of definiteness itself. Strength-based maximality has typically been invoked as a refinement of the Fregean analysis, motivated by cases where uniqueness is too strong a requirement (cf. plurals and degree DPs). The findings of this paper suggest a converging perspective. We have shown that, in the domain of functional definite DPs as well, strength-based maximality is not merely a way of relaxing uniqueness when it fails, but is the

mechanism actively responsible for delivering reference to a function in the arbitrary variation case. The Fregean uniqueness requirement and strength-based maximality requirement happen to coincide in many cases, which is why the choice between them has seemed like a matter of theoretical preference. The functional domain provides a case where they come apart in meaningful ways—and that when they do, it is strength-based maximality, not Fregean uniqueness, that makes the right predictions. This suggests that strength-based maximality deserves a more central place in the theory of definiteness than it has been given so far: not simply as a solution for problem cases, but perhaps as the right general analysis of what the definite article contributes.

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A.1 Heim’s analysis of principled variation readings

Heim 2018’s analysis of functional relative clauses builds on a head-raising approach to relative clauses combined with Fox 2002’s mechanism of trace conversion. To make this appendix self-contained, we first explain the two pieces of machinery the analysis depends on—head-raising and trace conversion—and then show how they interact to deliver the predicate of functions in (92).

Head-raising. On a head-raising analysis of relative clauses, the head noun does not originate in its surface position external to the relative clause. Rather, it originates inside the relative clause, in the position of the gap, and is moved out. The motivation for this kind of analysis comes from a range of reconstruction effects suggesting that the head noun behaves, for interpretive purposes, as if it were inside the relative clause (cf. Heim 2018 and references therein). What matters for present purposes is that, under such an analysis, the head noun is available for interpretation inside the relative clause. In Heim’s analysis of principled variation readings, this is exactly where it is interpreted.

Trace conversion. When a quantificational expression moves, it leaves behind a trace. The question is what the trace denotes. Under a copy theory of movement (cf. Chomsky 1993; Bošković et al. 2007), the trace is a full copy of the moved expression, but this copy cannot be interpreted as-is: a quantificational expression in argument position would yield a type mismatch. Trace conversion (Fox, 2002) is the mechanism that converts a quantificational trace into something interpretable. It does so in two steps: first, it inserts an identity predicate of the form $\lambda y. y = x$, where x is a variable bound by the moved phrase (‘Variable Insertion’); second, it replaces the quantificational determiner with the definite determiner THE (‘Determiner Replacement’). The result, [THE NP [$\lambda y. y = x$]], is a definite description that, by virtue of its uniqueness presupposition and the inserted identity predicate, denotes the unique individual identical to x . That is, it just denotes x , while preserving information about the descriptive content of the original NP. The conversion process is summarized in (87).

(87) Trace Conversion (cf. Fox 2002):

a. Variable Insertion:

(Det) $\text{Pred} \rightarrow (\text{Det}) [\text{Pred } \lambda y . y \ x]$

b. Determiner Replacement:

(Det) $[\text{Pred } \lambda y . y \ x] \rightarrow \text{the } [\text{Pred } \lambda y . y \ x]$

Since trace conversion occurs at LF, it has no phonological consequences—which is why we will write THE in small caps in our LFs. It is the original quantificational determiner that is pronounced. Following Heim, we break the inserted predicate into two parts: a bound index x and the polymorphic function IDENT, defined in (88). IDENT takes an entity and returns the property of being identical to that entity, which is exactly the predicate that Variable Insertion introduces. The result, [THE NP IDENT t], is the converted trace.

(88) $\llbracket \text{IDENT} \rrbracket = \lambda x_e . \lambda y_e . x = y$

The principled variation reading. For principled variation readings, Heim’s key move is to make the index in the converted trace not a simple individual variable but the output of a function variable applied to an individual variable (cf. Engdahl 1980; Chierchia 1993). The intuition is the following. On the principled variation reading, the relativized DP picks out a function from individuals to individuals. To get a function variable into the composition, we need a position for it. The position Heim exploits is the index inside the converted trace: rather than this index being a plain individual variable x , it is the complex expression $f(x)$, where f is a function variable and x is an individual variable. The function variable f will be bound from outside the relative clause by the *wh* trace, and the individual variable x —represented syntactically as a pronoun, *pro*—will be bound inside the relative clause by the embedded quantifier. The LF representing the internal composition of the relative clause is illustrated in (89).

(89) LF for predicate:

$[_{CP} \text{ wh } \lambda_2 [\text{that } [\text{each fourth grade boy } \lambda_1 [t_{1,e} \text{ invited } [\text{THE woman IDENT } [f_{2,(ee)} \text{ pro}_{1,e}]]]]]$

To see how this composes, we begin with the converted trace. Because THE triggers a presupposition that its restrictor be uniquely satisfied—and given that IDENT restricts the set of women to those identical to $f(\text{pro}_{1,e})$ —the trace [THE woman IDENT [f $\text{pro}_{1,e}$]] is defined only when the value of $\text{pro}_{1,e}$ is in the domain of f and is mapped by f to a woman. Where it is defined, it denotes $f(\text{pro}_{1,e})$. When we then abstract over the individual variable $\text{pro}_{1,e}$ to feed the embedded quantifier, the result is a partial function: it is defined only for individuals satisfying the presupposition just described, as shown in (90). (Following standard assumptions, presuppositions associated with a clause translate into restrictions on the domain of the abstracted function under lambda-abstraction.)

$$\begin{aligned}
 (90) \quad & \llbracket \lambda_1 [t_{1,e} \text{ invited } [\text{THE woman IDENT } [f_{2,(ee)} \text{ pro}_{1,e}]]] \rrbracket^g \\
 & = \lambda x: x \in \text{dom}(g(f)) \wedge \text{WOMAN}(g(f)(x)). \text{ INVITE}(g(f)(x))(x) \\
 & \approx \text{the set of individuals who invited the woman that } f \text{ maps them to}
 \end{aligned}$$

When the embedded universal *each fourth grade boy* combines with this partial predicate and binds the variable x , the presupposition projects universally. That is, the presupposition in the scope of the universal quantifier—that x is in the domain of f —is inherited as a universal presupposition on every possible value of x . The result is that every fourth grade boy must satisfy the conditions for the predicate to be defined: every fourth grade boy must be in the domain of f and mapped to a woman. This yields the partial proposition in (91).

$$\begin{aligned}
 (91) \quad & \llbracket [\text{each fourth grade boy}] \lambda_1 [t_{1,e} \text{ invited } [\text{THE woman IDENT } [f_{2,(ee)} \text{ pro}_{1,e}]]] \rrbracket^g \\
 & = \forall y [\text{G4B}(y) \rightarrow \text{INVITE}(g(f)(y))(y)] \\
 & \text{Defined only if } \forall x [\text{G4B}(x) \rightarrow [x \in \text{dom}(g(f)) \wedge \text{WOMAN}(g(f)(x))]]; \text{ otherwise undefined}
 \end{aligned}$$

Finally, abstracting over the function variable f , we arrive at the denotation of the NP: a predicate of functions that (i) presupposes that every fourth grade boy is in the domain of f and is mapped to a woman, and (ii) asserts that every fourth grade boy is mapped to someone he invited.

$$(92) \quad \text{Predicate denotation:}$$

$$\begin{aligned} & \llbracket [\text{CP wh } \lambda_2 [\text{that } [\text{each fourth grade boy } \lambda_1 [\text{t}_{1,e} \text{ invited } [\text{THE woman IDENT } [f_{2,(ee)} \\ & \text{pro}_{1,e}]]]]]] \rrbracket = \\ & \lambda f_{(ee)}: \forall y[G4B(y) \rightarrow [y \in \text{dom}(f) \wedge \text{WOMAN}(f(y))]]. \forall y[G4B(y) \rightarrow \text{INVITE}(f(y))(y)] \end{aligned}$$

The crucial feature of this denotation, for our purposes, is that the domain condition on f —namely, that every fourth grade boy be in the domain of f and mapped to a woman—enters as presupposed content that projects universally from the scope of the embedded quantifier. This is the property that the main text takes issue with: as discussed in Section 2.4.1, the strength-based maximality proposal requires that this condition enter the NP denotation as asserted content, not as a presupposition.

B.2 Jacobson’s analysis of principled variation readings

Jacobson 1994’s analysis of functional relative clauses is developed within a variable-free framework, in which binding is mediated by a type-shifting operation on the verb (the z -rule) rather than by lambda-abstraction over individual variables. We will not import the full variable-free machinery here; for present purposes, what matters is the predicate of functions that the analysis assigns to the relative clause, and the assumptions that yield it. We highlight five.

First, common nouns are polymorphic. The noun *woman* has its usual (et) meaning—a predicate of individuals—but can also be interpreted as a predicate of functions, of type $((ee)t)$, denoting the set of woman-valued functions, as in (93-a). Second, when a predicate of functions combines with another predicate of functions—as the head noun and the relative clause must, under this analysis—they combine intersectively. Third, the definite article is polymorphic: it applies to predicates of individuals as well as predicates of functions, returning the unique element of the relevant set. Fourth, in the post-copular DP *his mother*, the pronoun *his* denotes the identity function on individuals, with the result that *his mother* denotes the “mother-of” function—exactly the kind of object that a standard bound-variable analysis would deliver for *his mother* relative to a binding subject, but obtained here through type composition rather than variable binding. Fifth, the copula is polymorphic and, in this configuration, expresses identity between two (ee) functions.

With these assumptions in place, the relative clause itself denotes a predicate of functions, characterizing the set of functions that map every boy to someone he invited. Jacobson adds a further restriction: the function must be a "natural" function, encoded by a predicate 'NAT', which restricts the set to functions that are 'natural', in some sense which Jacobson is admittedly non-committal about. This is what rules out arbitrary mappings as values for the relativized DP, and, when considering arbitrary functional DPs, can be left out (under our presentation of functional relative clauses in Section 2.2, domain restriction did the work of NAT). The denotation of the relative clause is given in (93-b).

- (93) a. $\llbracket \text{woman} \rrbracket = \lambda f_{(ee)}. \forall x \in \text{dom}(f)[\text{WOMAN}(f(x))]$
 b. $\llbracket \text{that each boy invited} \rrbracket = \lambda f_{(ee)}. \text{NAT}(f) \wedge \forall x[\text{BOY}(x) \rightarrow \text{INVITE}(f(x))(x)]$

The head noun and the relative clause both denote predicates of functions, so they combine intersectively, yielding (94).

- (94) $\llbracket \text{woman that each boy invited} \rrbracket =$
 $\lambda f_{(ee)}. \forall x \in \text{dom}(f)[\text{WOMAN}(f(x))] \wedge \text{NAT}(f) \wedge \forall x[\text{BOY}(x) \rightarrow \text{INVITE}(f(x))(x)]$

The polymorphic definite article then applies to this predicate of functions, returning the unique natural function whose domain consists of women-valued mappings and which maps every boy to someone he invited.

- (95) $\llbracket \text{the woman that each boy invited} \rrbracket =$
 $\iota h[\forall x \in \text{dom}(h)[\text{WOMAN}(h(x))] \wedge \text{NAT}(h) \wedge \forall x[\text{BOY}(x) \rightarrow \text{INVITE}(h(x))(x)]]$

Finally, given that *his mother* denotes the "mother-of" function and the copula expresses identity between (ee) functions, the full copular sentence asserts that the function picked out by the pre-copular DP is identical to the "mother-of" function.

- (96) $\llbracket \text{the woman that each boy invited is his mother} \rrbracket =$
 $\iota h[\forall x \in \text{dom}(h)[\text{WOMAN}(h(x))] \wedge \text{NAT}(h) \wedge \forall x[\text{BOY}(x) \rightarrow \text{INVITE}(h(x))(x)]] =$

$$\lambda_{x\iota y}[\text{MOTHER-OF}(x)(y)]$$

The crucial feature of this denotation, for our purposes, is the contribution of the head noun: because *woman* is interpreted as a predicate of functions outside the scope of the embedded universal, the resulting NP denotation imposes a strong condition—that *every* element of the function’s domain must be mapped to a woman. This is the property that the main text takes issue with: as discussed in Section 2.4.1, the strength-based maximality proposal requires an NP denotation that constrains only the values of the function from the embedded quantifier’s restrictor, not its full domain. Jacobson’s analysis, by interpreting the head noun above the embedded quantifier, places the womanhood condition outside the quantifier’s scope and thereby blocks the upward monotonicity that our proposal requires.

Part II

Finite complement clauses as scope islands

Preface to Chapter 3

Chapters 1 and 2 have been concerned with a single syntactic environment: relative clauses. Together, these two chapters establish that variation of the referent of the definite in relative clauses is better understood as a reflex of functional interpretations than as the result of non-local scope-shifting. Chapter 3 shifts the empirical focus to a different syntactic environment: finite complement clauses. The variation readings that have been discussed in relation to finite complement clauses—sentences where the universal DP appears to take matrix scope over the singular indefinite subject, have been among the most influential examples in the literature arguing for non-local scope (cf. Farkas and Giannakidou 1996; Barker 2022; Hoeks et al. 2022; Wang 2024, a.o.). Understanding whether these cases, too, can be explained by a pseudo-scope mechanism is therefore central to the dissertation’s overall argument.

Finite complement clauses differ from relative clauses insofar as they are not islands for overt extraction. While a *wh*-phrase cannot move out of a relative clause without violating the Complex NP Constraint, *wh*-movement out of a finite complement clause is freely permitted in English. This means that, while both syntactic environments are considered to be scope islands, their theoretical status differs: on accounts that derive scope-island behavior from extraction-island behavior, relative clauses would be expected to resist non-local scope, while finite complement clauses would not. That finite complement clauses nevertheless appear to constrain scope-taking is therefore not straightforwardly derivable from locality constraints on movement per se, suggesting that whatever limits scope-taking in this environment cannot be stated in purely structural terms. It is against this backdrop that the predicate sensitivity that we will discuss in Chapter 3 becomes particularly significant: the relevant constraint on variation readings is not uniform across all finite complement

clauses, but instead tracks properties of the embedding predicate itself. This empirical observation also suggests that the clause-boundedness hypothesis is insufficient: why should the availability of variation readings be predicate sensitive if the operative constraint on scope-taking is the presence of a clause boundary?

The discussion of finite complement clauses in Chapter 3 connects to the overall goals of the dissertation in a transparent way. The empirical investigation rounds out the dissertation's central argument. In relative clauses, Chapters 1 and 2 showed that variation is better explained by functional interpretations than by non-local scope-shifting. In finite complement clauses, Chapter 3 shows that variation is also better explained by appealing to the inferential profile of the embedding predicate than by non-local scope-shifting. In both environments, then, the data that has been adduced as arguments for exceptional scope does not hold up to further scrutiny: what looks like a universal DP taking matrix scope turns out to be the signature of a different compositional process entirely. The theoretical conclusions point toward the dissertation's broader contribution. The pseudo-scope mechanisms identified in the three chapters are distinct—functional interpretations vs. cumulative inferences—and yet they converge on a common theme: explaining variation without scope-shifting. Understanding what unifies them, and whether that unity reflects a deeper principle governing how variation is licensed across grammatical environments, is among the most significant open questions that this dissertation leaves for future work.

Exceptional wide scope and cumulation

Finite complement clauses are often assumed to be scope islands for quantifier raising (QR) of universal quantifiers. However, as observed by Farkas and Giannakidou 1996; Barker 2022; Hoeks et al. 2022, a.o., there are apparent counterexamples to this claim, where a universal DP appears to scope out of a finite complement clause to take scope over a singular indefinite in the matrix clause, henceforth ‘variation readings’. Barker 2022 argues that the availability of variation readings is predicate sensitive, and attributes this predicate sensitivity to non-local QR. In this paper, based on experimental results, we challenge Barker’s conclusion. First, we demonstrate that variation readings are also possible when we replace the embedded universal quantifier with a negative indefinite, and non-local QR is insufficient to explain these cases. Second, we argue instead for the following empirical generalization: embedding predicates license variation readings if they also license a form of cumulativeness in relational pred-

ication. Building on this generalization, we argue that the predicate sensitivity of both phenomena is rooted in the inferential profile of the embedding predicate (Pasternak, 2018; Harada, 2023).

3.1 Introduction

Finite complement clauses are often assumed to be scope islands for quantifier raising (QR) of universal quantifiers (Chomsky, 1975; May, 1977; Fox and Sauerland, 1996). However, as observed by Farkas and Giannakidou 1996; Barker 2022; Hoeks et al. 2022, a.o., there are apparent counterexamples to this claim, where a universal DP appears to scope out of a finite complement clause to take scope over a singular indefinite in the matrix clause, henceforth ‘variation readings’. At the same time, Barker 2022 (a.o.) has observed that the availability of such variation readings is predicate sensitive: there are also examples where the universal DP seems unable to take matrix scope from an embedded position, and this seems to depend on the embedding predicate. To illustrate, consider the contrast in (1).

(1) Variation readings with universal quantifiers:

- a. A student made sure that every invited speaker had a ride. ✓*variation*
(Farkas and Giannakidou, 1996, 37)
- b. A student claimed that every professor was wrong. ✗*variation*
(cf. Barker 2022)

On the one hand, in addition to a reading about a particular student, (1-a) licenses a weaker reading where the student may vary by speaker (i.e., a variation reading). On the other hand, in contrast to (1-a), (1-b) does not license a variation reading—the sentence only receives a reading about a particular student who made a claim about all the professors.¹

Barker 2022 interprets the contrast in (1) as evidence that universal DPs can, in principle, undergo QR out of finite complement clauses, but the availability of this kind of non-local QR depends on the embedding predicate (for other proposals that finite clauses do not impose locality constraints on QR, for lexical or processing reasons, see: Anderson 2004; Syrett 2015; Wurmbrand 2018; Hoeks et al. 2022; Wang 2024).

¹At the very least, the variation reading is much more difficult in (1-b) than it is in (1-a). The claim that we defend is not that the variation reading is impossible in (1-b), but that there is a relative contrast when it is compared to (1-a).

In this paper, supported by experimental evidence, we challenge this proposal. First, we observe that variation readings are still available when we replace the universal quantifier with a negative indefinite, and they seem to have the same sensitivity to the embedding predicate as variation readings with universal quantifiers, as illustrated in (2).

(2) Variation readings with negative quantifiers:

- a. A student organizer made sure that no speaker was late for their talk ✓*variation*
- b. A student claimed that no speaker was right. ✗*variation*

The contrast in (2-a) and (2-b) seems quite parallel to the case of variation readings with universal quantifiers: the former permits variation of student organizers by speakers, when the embedding predicate is *make sure*, while the latter does not permit variation, when the embedding predicate is *claim*. The challenge is that non-local QR cannot derive variation readings with negative quantifiers, meaning variation is not an argument for QR in this case. This calls the non-local QR proposal into question: it shows that there must be an alternative path to variation readings with negative quantifiers. If there is such an alternative path, then it seems plausible that such a path might also exist for variation readings with universal quantifiers.

Second, we will present evidence that the predicate sensitivity observed in variation readings (with universal or negative quantifiers) parallels the predicate sensitivity observed in instances of what we will refer to as ‘non-local cumulation’ (we will also sometimes refer to this simply as ‘cumulation’). To illustrate what we mean by ‘non-local cumulation’ and how it is predicate sensitive, consider (3).

(3) Cumulation with universal quantifiers:

- a. *[There are four speakers. Ann made sure that speaker 1 and speaker 2 had a ride; but not speaker 3 and speaker 4. Bea made sure that speaker 3 and speaker 4 had a ride; but not speaker 1 and speaker 2.]*
Ann and Bea made sure that every speaker had a ride.
- b. *[There are four speakers. Ann claimed that speaker 1 and speaker 2 were wrong; but*

not speaker 3 and speaker 4. Bea claimed that speaker 3 and speaker 4 were wrong; but not speaker 1 and speaker 2.]

Ann and Bea claimed that every speaker was wrong.

The target sentence in (3-a) can be understood as true in the provided context under a reading that is weaker than the distributive reading one might expect. In contrast, in (3-b), the target sentence is not perceived as true in the given context. Therefore, given the same cumulative scenario in each example, the target sentence in (3-a) is felicitous while the target sentence in (3-b) feels degraded.

Thus, our central finding in this paper is that we observe the same predicate sensitivity in (1), (2) and (3), leading to the following generalization: an embedding predicate licenses variation readings if it licenses cumulation with plural subjects. As a consequence, the proper compositional path to variation should share a key component with the compositional path to cumulation. The correlation with cumulation offers a possible explanation for why variation readings are predicate sensitive: the literature offers an analysis that captures the predicate sensitivity of non-local cumulation, without invoking QR. At the heart of this analysis is the inferential profile of the embedding predicate: the way the lexical meaning of the predicate interfaces with a particular context to license what we will refer to as ‘cumulative inferences’ (Krifka, 1992; Kratzer, 2002; Pasternak, 2018; Harada, 2022a,b, 2023). Given this, our findings suggest that an account is needed for variation readings that also references the inferential profile of the embedding predicate. The most simple-minded reaction to our findings, then, is to directly extend the account in Harada 2023 to variation readings, reducing variation to a form of cumulation. What we learn, though, is that this simple-minded elaboration of Harada 2023’s approach, as suggested by the experimental results, falls short. Therefore, a fully adequate development of the approach is left for future work, leaving us with an unresolved puzzle and an open question concerning how to best capture the precise truth conditions of variation readings.

To support the argument that the same mechanism is involved in (1), (2) and (3), this paper presents results from four acceptability rating experiments. First, in Section 3.2, we present evidence from two rating experiments establishing that variation readings are indeed predicate sensi-

tive, and moreover the predicate sensitivity appears parallel with universal quantifiers and negative indefinites. Then, in Section 3.3, we present evidence from two more rating experiments establishing that non-local cumulation with plural subjects is also predicate sensitive, and this predicate sensitivity is also parallel to the variation readings with singular indefinite subjects.

3.2 Assessing the predicate sensitivity of variation readings

Previous literature has claimed that variation readings are available with universal quantifiers, providing potential evidence for the proposal that the universal DP undergoes QR out of the finite complement clause. We observed that similar variation readings are possible with negative indefinites as well and, furthermore, these readings seem to be subject to the same predicate sensitivity. If our claim is true, this is theoretically significant since analyzing variation readings through non-local QR is insufficient to explain variation readings with negative indefinites. The challenge is (i) that negative indefinites do not typically license inverse scope readings, (4), and (ii) that non-local QR only delivers the right truth conditions when the embedded quantifier is a universal. Even if the negative indefinite were to undergo QR to a position above the subject indefinite, as illustrated by the LF in (5-a), the resulting truth conditions correspond to an unattested reading. These truth conditions are provided in (5-b), where we can see that they do not correspond to a variation reading.

- (4) a. Two students read no books.
b. *Unavailable reading*: There are no books x such that two students didn't read x
- (5) A student organizer made sure that no speaker was late for their talk.
a. *LF*: $[\text{no speaker}] \lambda_1 [\text{a student organizer}_1 \text{ made sure that } {}_{\text{TP}}[\text{t}_1 \text{ was late}]]$
b. $\neg \exists y [\text{speaker}(y) \wedge \exists x [\text{student-volunteer}(x) \wedge \text{make-sure}(x)(\text{late}(y))]]$
'there's no speaker y s.t. there's a student organizer that made sure y was late.'

This means the sentence in (5) is a case that permits variation of the singular indefinite, but this variation cannot be due to QR. Therefore, whether variation readings are possible with negative

indefinites (and whether they are predicate sensitive in the same way) is important for adjudicating between different analyses of variation readings. It is clear that, if variation readings are indeed possible with negative indefinites, then a different route than QR is needed to derive at least these variation readings, because QR would derive the wrong truth conditions. If, moreover, the predicate sensitivity is the same for the two types of quantifiers, this would support the view that both types of variation readings obtain through the same non-QR route. Such a correlation, in which predicates are sensitive to both types of quantifiers, is unlikely to be a coincidence.

Before exploring our alternative route to variation, in this section, we put these claims to the test. We first demonstrate that variation readings with universal quantifiers and those with negative indefinites do, in fact, exhibit a similar predicate sensitivity. By ‘predicate sensitivity’, we mean that, with respect to variation readings, predicates differ systematically in their acceptability across both phenomena, yielding a similar relative ordering of predicates. To properly assess the parallel between both kinds of variation readings, we conducted two sentence rating experiments, each with 40 participants. Our goal with these experiments was to determine a scale of acceptability reflecting the predicate sensitivity for both kinds of variation readings, to see how similar they were.

One experiment looked at variation readings with embedded universal quantifiers, while the other experiment looked at variation readings with embedded negative indefinites. Given that the effect size was rather small and the judgments rather subtle, our experiments focused explicitly on contexts that encouraged variation readings, as in (6) and (7), to not introduce unnecessary noise.² That is, to facilitate the task and help participants better understand the relevant notion of ‘variation’, the instructions for each experiment introduced participants to variation readings in simple transitive sentences (i.e., ‘inverse scope readings’) by providing them with a varying

²This was determined through a series of pilot experiments where we tested different ways of getting at the predicate sensitivity by modifying both the instructions and the stimuli for the experiment. Part of the reason why we focused just on these kinds of ‘varying’ contexts is that, in previous versions of the experiments, we found that including positive and negative baselines shrank the effect even further and introduced unwanted noise into the experiment: effectively, many participants answered at ceiling for the positive baseline, floor for the negative baseline and the mid-point of the likert scale for the test stimuli. As a result, this obscured the predicate sensitivity effect, which was the main goal of our experiments.

context and a target sentence which can be understood as true in that context (the target sentence involved a singular indefinite subject and a universal DP object).³ Then, in test trials, participants were shown similar varying contexts and target sentences with singular indefinite subjects and universal quantifiers/negative indefinites embedded within finite complement clauses, as in (6), and were asked to rate if the sentence in bold sounded correct, given the varying context, on a 6-point Likert scale (where 1 corresponded to ‘clearly incorrect’ and 6 corresponded to ‘clearly correct’). Participants were recruited on Prolific., and we restricted participation to those who self-report to be native speakers of North American English, having grown up and currently live in the United States or Canada. We also restricted the experiment to participants with at least a 97% approval rating on Prolific. These experiments aimed to replicate the experiments in Palucci 2024 but with a different experimental design, different stimuli, more predicates and more participants.

- (6) VARYING INDEFINITE CONTEXT—UNIVERSAL: *[There were three homework problems to review. As teaching assistants, Ann, Bea and Carol were asked to review the homework problems. Ann made sure that the first problem was error-free. Bea made sure that the second problem was error-free. Carol made sure that the third problem was error-free.]*

A teaching assistant made sure that every problem was error-free.

- (7) VARYING INDEFINITE CONTEXT—NEGATIVE: *[There were three homework problems to review. As teaching assistants, Ann, Bea and Carol were asked to review the homework problems. Ann made sure that the first problem did not contain errors. Bea made sure that the second problem did not contain errors. Carol made sure that the third problem did not contain errors.]*

A teaching assistant made sure that no problem contained errors.

The experimental stimuli involved 36 embedding predicates. We wanted predicates that fell on a scale regarding the extent to which they licensed variation readings. Some of the embedding

³This means that, in the experiment concerning variation readings with negative quantifiers, participants learned about inverse scope readings with universal DP objects, and had to transfer this knowledge to the task involving negative DP objects, which may be a non-trivial thing to do and could have influenced the results of the experiment. We return to this point in Section 3.3.

predicates we chose were based off the discussion in Barker 2022, where it is proposed that finite complement clauses of attitude and communication verbs block variation readings. To this end, we hypothesized that a subset of the predicates we chose—the attitude and communication verbs—would pattern more like *claim* (in terms of not licensing variation readings with singular indefinite subjects or cumulation with plural subjects). The other predicates we chose were determined by initially relying on our own introspective judgments.⁴ We hypothesized that some of these predicates would pattern more like *make sure* (in terms of licensing variation readings with singular indefinite subjects and cumulation with plural subjects) while others would lie somewhere in the middle of the scale, since intuitions were less clear.

For all experiments, there was a total of 36 items—where each item involved a different embedding predicate—and every participant saw every item. The list of predicates is provided in (8).

- (8) **Embedding predicates:** *make sure, confirm, establish, prove, verify, determine, guarantee, corroborate, double-check, note down, ascertain, attest, certify, validate, enforce, demonstrate, check, affirmclaim, notice, confess, heard, believe, hope, fear, know, say, realize, recognize, think, speculate, whisper, exclaim, judge, hypothesize, observe*

All of the experiments reported in this paper followed the same methodology and experimental design, all that varied was the stimuli and the instructions. The results for our first two experiments are shown in Figure 3.1.

The first thing to note in Figure 3.1 is that the mean acceptability ratings for all the predicates lie in the middle of the scale, indicating that the predicate sensitivity of these variation readings (with both quantifiers) is a small effect and that the judgments are not always clear. We believe that this is because the task we are running is actually quite involved and the judgments we are probing are rather subtle.⁵ Furthermore, the error bars for the predicates all overlap, indicating that there

⁴As we will see throughout the paper, we use statistical models to validate these pre-theoretic intuitions.

⁵This is validated by our pilot experiments where we constantly found the acceptability ratings of the different predicates to lie in the middle of the scale. Even trained linguists that we consulted with found these judgments rather subtle and difficult, so it may not be all that surprising that the effect is small, due to inherent noise in the

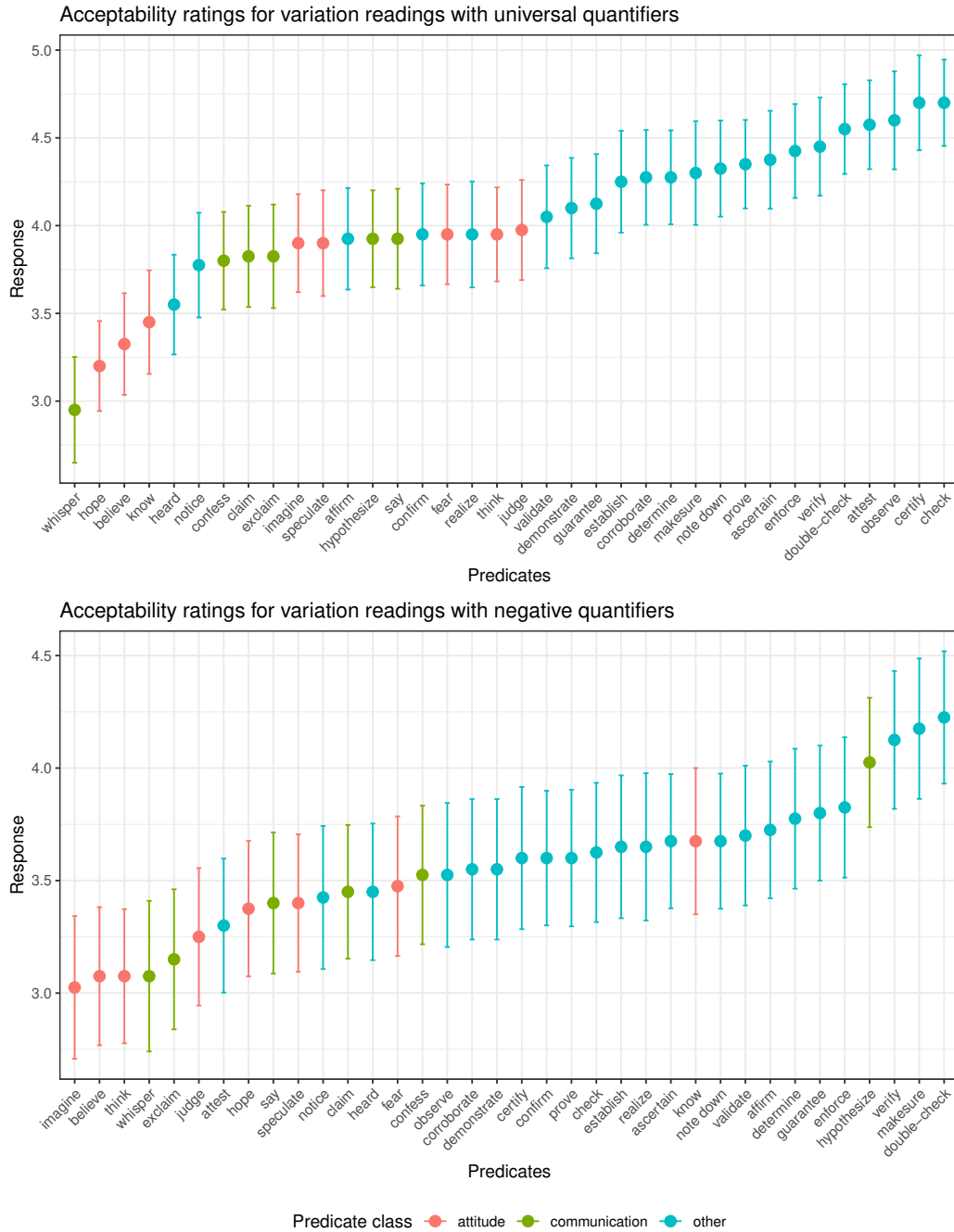


Figure 3.1: Top: Raw mean acceptability ratings for variation readings with universal quantifiers, for each predicate.

Bottom: Raw mean acceptability ratings for variation readings with negative quantifiers, for each predicate.

1 = ‘clearly incorrect’; 6 = ‘clearly correct’.

experimental task. It is also worth pointing out that it is quite common for ratings to cluster in the middle of the scale with acceptability rating tasks, especially when the judgments being probed are subtle.

could, in fact, be some variability in the precise ordering of predicates.

That being said, looking at Figure 3.1, a comparison of mean acceptability ratings across predicates for variation readings with universal quantifiers and variation readings with negative quantifiers reveals a similar pattern of predicate sensitivity. In both panels of Figure 3.1, the ratings of the different predicates show an increasing pattern of acceptability and the overall ordering of predicates is largely preserved, with the same clusters of predicates emerging. In particular, attitude and communication predicates occupy the lower end of the scale, in line with Barker 2022's hypothesis, whereas the other predicates tend to receive higher ratings. The fact that a similar pattern of predicate sensitivity is observed across both experiments at least validates to some extent the empirical methodology and the underlying linguistic intuitions. Furthermore, the fact that the predicates at both ends of the scale agree with linguistics intuitions is reassuring, serving as a kind of sanity check for the methodology.⁶ It is also worth pointing out that overall ratings were slightly lower for variation readings with negative quantifiers. However, we emphasize that our claim is not that the acceptability ratings for both kinds of variation readings should be the same, but rather that the two readings are correlated insofar as predicates that are rated better in one case tend to be rated better in the other.

Regarding why the ratings lie in the middle of the scale, there could be a few reasons. We hypothesize that one reason why the predicates on the upper end of the scale were not rated higher could be because there are also contextual factors that will inevitably influence what readings participants access. For example, these contextual factors might decide whether participants gravitate towards a single individual reading (i.e., a reading about a particular individual), versus a variation reading. Relatedly, if this individual reading is more readily available for the provided target sentence, it could simply be that participants have difficulties overcoming this reading, leading to the variation reading being dispreferred. For these reasons, we believe that the small effect size is still perfectly compatible with our hypothesis: even given the effect size, there is still a clear and

⁶We do acknowledge that it would be beneficial to find a methodology that is more fine-grained insofar as it better tracks the predicate sensitivity of these different phenomena. We view this as an important step for future research. At the same time, we believe that the experiments reported in this paper set the stage for further research, including further experimental validation.

observable predicate sensitivity in both kinds of variation reading.

We can more directly assess the parallel between the two phenomena by plotting the mean acceptability ratings for each predicate from the experiment looking at variation readings with universal quantifiers on the x-axis and the mean acceptability ratings for each predicate from the experiment looking at variation readings with negative quantifiers on the y-axis, as in Figure 3.2.

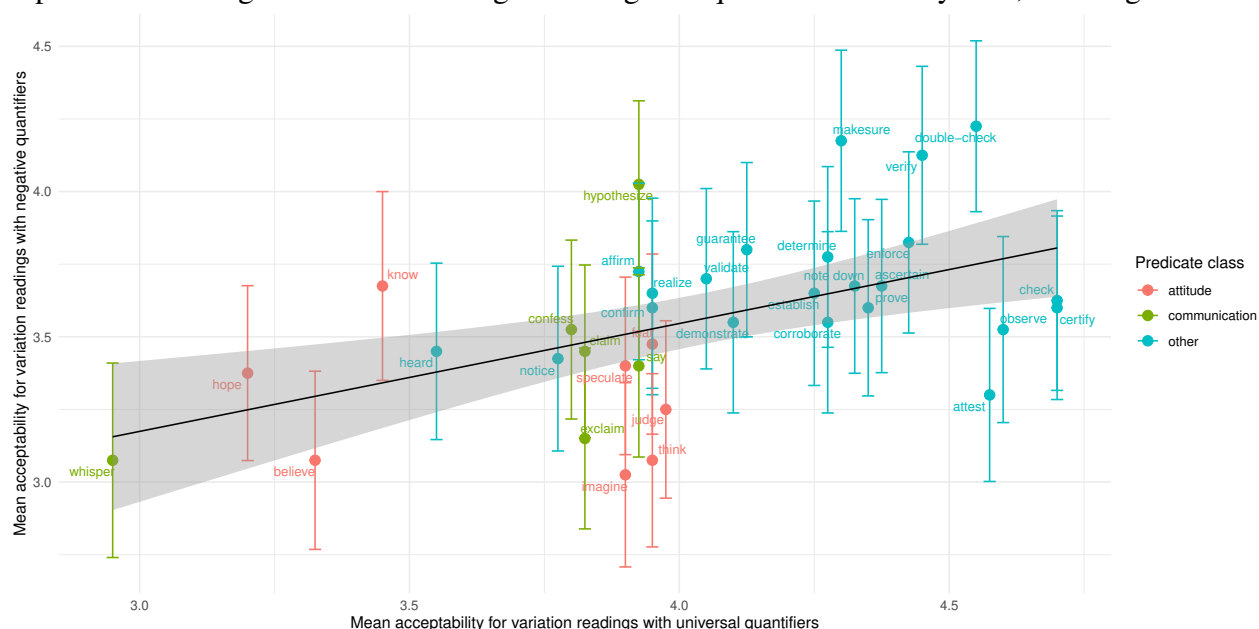


Figure 3.2: Mean acceptability ratings (for each predicate) of variation readings with embedded negative quantifiers plotted against mean acceptability ratings (for each predicate) of variation readings with embedded universal quantifiers.

Lower values indicate lower acceptability; higher values indicate higher acceptability.

The main observation in Figure 3.2 is that mean acceptability ratings in the two experiments exhibit a clear positive correlation, as illustrated by the regression line. What this means is that predicates that receive relatively high ratings for variation readings with universal quantifiers also tend to receive relatively high ratings for variation readings with negative quantifiers, and vice versa. Again, the parallel predicate sensitivity is further reflected in the distribution of predicate classes, with attitude and communication predicates clustering at the low end in both conditions, and the other predicates occupying the higher end. This parallel distribution of predicate classes suggests that both phenomena exhibit a similar predicate sensitivity.

Finally, if variation readings and negative variation readings share the same predicate sensi-

tivity, then acceptability in one case should predict acceptability in the other. This prediction is borne out by a cumulative link mixed model (using the ordinal package in R (Christensen, 2025)) with acceptability ratings for variation readings with negative quantifiers as the response and acceptability ratings for variation readings with universal quantifiers as the predictor, with random intercepts and by-participant random slopes (to control for participant-level variability in both intercepts and slopes).⁷ The model indicates a significant positive effect of acceptability ratings of variation readings with universal quantifiers on acceptability ratings of variation readings with negative quantifiers ($\beta = 0.64$, $SE = 0.18$, $z = 3.62$, $p < 0.001$).⁸ These results confirm that variation readings with universal quantifiers and variation readings with negative quantifiers exhibit similar predicate sensitivity, insofar as predicates that more readily license the former also tend to license the latter (even if there was increased variability in the latter case).

Therefore, we have strong evidence that these two kinds of variation readings are correlated, and exhibit a similar predicate sensitivity, meaning both readings may derive from a single source. Given that we need an alternative route to variation for the cases with negative quantifiers, in what follows, we explore what this other route might look like and how it could apply uniformly to variation readings with universal quantifiers and those with negative quantifiers. It is also worth pointing out that the parallel predicate sensitivity between these two kinds of variation readings shows that non-local QR is not a very viable alternative for analyzing the cases with universal quantifiers. Even if one were to analyze variation readings with universal quantifiers and variation readings with negative quantifiers separately—with the former receiving a QR parse—one would still need to explain where the predicate sensitivity of QR comes from, and why it is similar to the predicate sensitivity that we observe with negative quantifiers. This means we would need two distinct mechanisms that are predicate sensitive in the same way: the predicate sensitivity of non-local QR

⁷ $\text{Response} \sim \text{VariationIndex} + (1 \mid \text{participant}) + (0 + \text{VariationIndex} \mid \text{participant})$ where ‘Response’ is the raw acceptability ratings for variation readings with negative quantifiers and ‘VariationIndex’ corresponds to the standardized mean acceptability ratings for each predicate from the experiment looking at variation readings with universal quantifiers.

⁸Because the predictor was standardized in our model, β represents the effect of a 1 standard deviation (SD) increase in acceptability ratings of variation readings with universal quantifiers: a 1 SD increase was associated with a 1.89-fold increase in the odds of a higher response concerning variation readings with negative quantifiers, corresponding to a moderate increase in the probability of higher acceptability judgments.

would have to mirror the predicate sensitivity of this alternative mechanism. Moreover, proposing a predicate sensitive version of non-local QR is not a trivial task. The predicate sensitivity of variation readings would have to amount to predicate sensitivity of this covert movement: some embedding predicates allow the required covert movement of the embedded universal DP, while others do not. While logically possible, there is no expectation that there is such a thing since there is no independent evidence suggesting that predicates differ in this particular way (i.e., in their permeability to covert movement). As a result, this approach does not tell us which embedding predicates allow or disallow the universal DP to move across the finite clause boundary—since we have no insight into why there is a division between predicates that allow non-local QR and those that do not.

Thus, as a step towards determining what an alternative route to variation could be, we will instead draw inspiration from the literature on non-local cumulation. This is because, in this literature, there is already a viable analysis that captures the predicate sensitivity, which we will return to in Section 3.4. Before turning to that analysis, though, we present experimental results showing that, not only is non-local cumulation also predicate sensitive in the same way as variation readings, but it exhibits predicate sensitivity with universal quantifiers and negative quantifiers alike.

3.3 Assessing the predicate sensitivity of non-local cumulation

As previewed in Section 3.1, non-local cumulation is possible with embedded universal quantifiers, (9). This is paralleled by non-local cumulation with embedded negative quantifiers, as in (10).

(9) Cumulation with universal quantifiers:

- a. *[There are four speakers. Ann made sure that speaker 1 and speaker 2 had a ride; but not speaker 3 and speaker 4. Bea made sure that speaker 3 and speaker 4 had a ride; but not speaker 1 and speaker 2.]*

Ann and Bea made sure that every speaker had a ride.

- b. *[There are four speakers. Ann claimed that speaker 1 and speaker 2 were wrong; but not speaker 3 and speaker 4. Bea claimed that speaker 3 and speaker 4 were wrong;*

but not speaker 1 and speaker 2.]

Ann and Bea claimed that every speaker was wrong.

(10) Cumulation with negative quantifiers:

- a. *[There are four speakers. Ann made sure that speaker 1 and speaker 2 were not late; but not speaker 3 and speaker 4. Bea made sure that speaker 3 and speaker 4 were not late; but not speaker 1 and speaker 2.]*

Ann and Bea made sure that no speaker was late.

- b. *[There are four speakers. Ann claimed that speaker 1 and speaker 2 were not right; but not speaker 3 and speaker 4. Bea claimed that speaker 3 and speaker 4 were not right; but not speaker 1 and speaker 2.]*

#Ann and Bea claimed that no speaker was right.

In this section, we demonstrate that cumulation with universal quantifiers and cumulation with negative quantifiers exhibit a similar predicate sensitivity. We show this through two more rating experiments that assess the predicate sensitivity of non-local cumulation with universal quantifiers and negative quantifiers, respectively (just as we did with variation readings).

Again, given that the effect size was rather small and the judgments rather subtle, our experiment focused explicitly on the kinds of cumulative contexts in (11) and (12) that encouraged non-local cumulation. To facilitate the task and help participants with the relevant notion of ‘cumulation’, the instructions for both experiments introduced participants to cumulative readings in simple transitive sentences by providing them with a varying context (where two individuals, between them, completed a task) and a target sentence which can be understood as true in that context (the target sentence involved a conjoined subject and a universal DP object; concerning cumulative readings with every DPs in object position see Kratzer 2002; Champollion 2010; Haslinger and Schmitt 2018; Chatain 2022a,b).⁹ Then, in test trials, participants were shown context-sentence

⁹This means that, in the experiment concerning non-local cumulation with negative quantifiers, participants learned about cumulative readings with universal DP objects, and had to transfer this knowledge to the task involving negative DP objects. Again, this may not have been a trivial thing to do for participants.

pairs, like (11), and asked to rate if the sentence in bold sounded correct, given the varying context, on a 6-point Likert scale (where 1 corresponded to ‘clearly incorrect’ and 6 corresponded to ‘clearly correct’). The results are presented in Figure 3.3.

- (11) CUMULATION CONTEXT—UNIVERSAL: *[There are four speakers. Ann made sure that speaker 1 and speaker 2 had a ride; but not speaker 3 and speaker 4. Bea made sure that speaker 3 and speaker 4 had a ride; but not speaker 1 and speaker 2.]*

Ann and Bea made sure that every speaker had a ride.

- (12) CUMULATION CONTEXT—NEGATIVE: *[There are four speakers. Ann made sure that speaker 1 and speaker 2 were on time; but not speaker 3 and speaker 4. Bea made sure that speaker 3 and speaker 4 were on time; but not speaker 1 and speaker 2.]*

Ann and Bea made sure that no speaker was late.

Just as we observed with variation readings, the first thing to note in Figure 3.3 is that the mean acceptability ratings for all the predicates lie in the middle of the scale, indicating that the predicate sensitivity of non-local cumulation (with both quantifiers) is a subtle effect. Again, we believe that this is because the task we are running is actually quite difficult and the judgments we are probing are rather subtle. In this case too, the error bars for the predicates all overlap, indicating that there could, in fact, be quite some variability in the precise ordering of predicates.

That being said, looking at Figure 3.3, a comparison of mean acceptability ratings across predicates for cumulation with universal quantifiers and cumulation with negative quantifiers reveals a similar pattern of predicate sensitivity. In both panels of Figure 3.3, the ratings of the different predicates show an increasing pattern of acceptability and the overall ordering of predicates is largely preserved. Strikingly, as was the case with both kinds of variation readings, the same clusters of predicates emerge: attitude and communication predicates occupy the lower end of the scale, whereas the other predicates tend to receive higher ratings. Hence, if Barker 2022’s hypothesis is on the right track for variation readings, the same constraint seems to be operative for non-local cumulation. More generally, the overall ordering of predicates that was observed in the

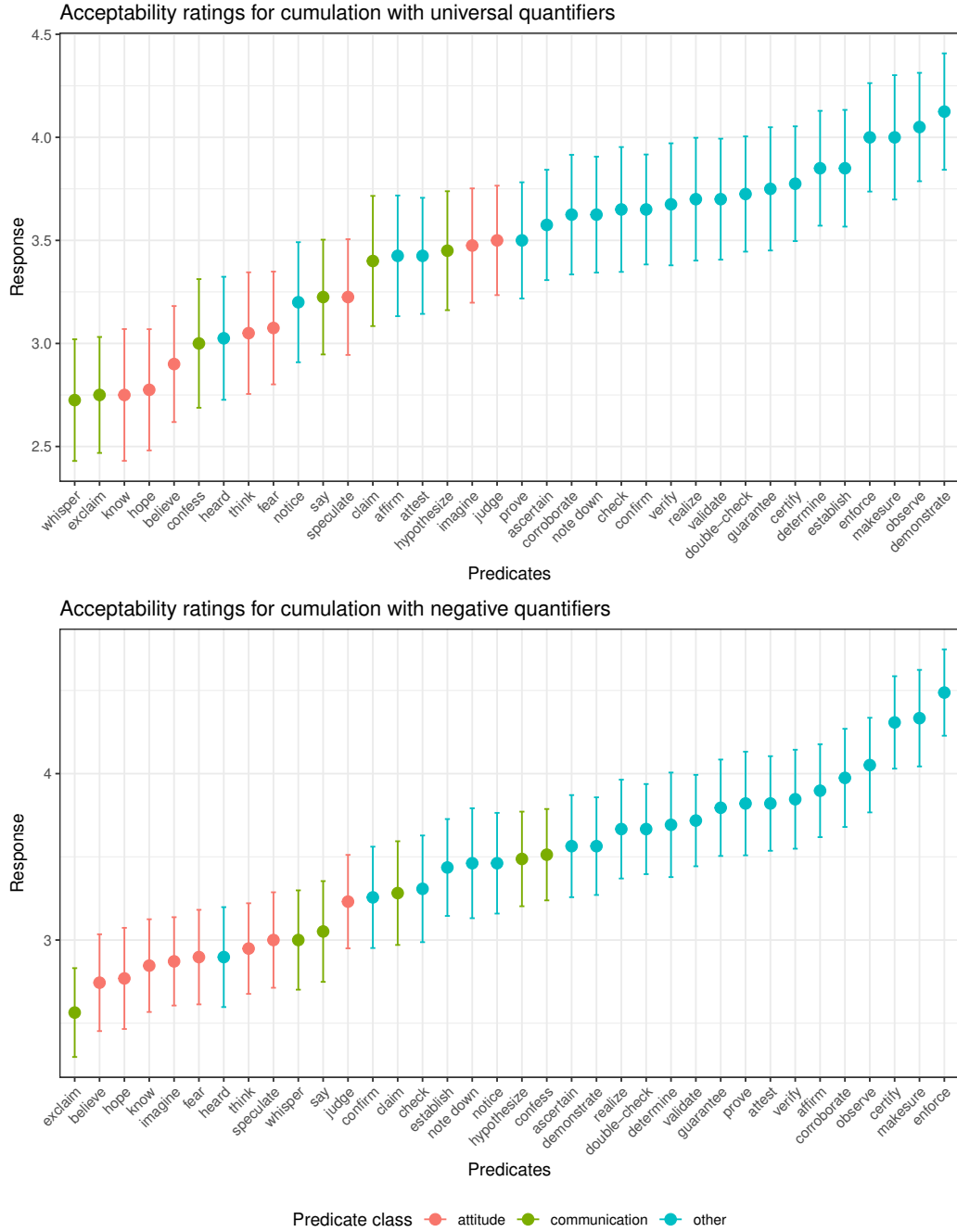


Figure 3.3: Top: Raw mean acceptability ratings for cumulation with universal quantifiers, for each predicate.

Bottom: Raw mean acceptability ratings for cumulation with negative quantifiers, for each predicate.

1 = ‘clearly incorrect’; 6 = ‘clearly correct’.

experiments on variation readings is preserved in these cases as well.

There are also several reasons why the ratings in these experiments lie in the middle of the scale. As discussed in Section 3.2, we hypothesize that one reason why the predicates on the upper end of the scale were not rated higher could be because, for the target sentences in this experiment too, there are also contextual factors that will inevitably influence what readings participants access. For example, these contextual factors might decide whether participants gravitate towards a distributive interpretation, versus a non-distributive one (or vice-versa). If the non-distributive reading is more readily available for the provided target sentence, it could simply be that participants have difficulties overcoming this reading, leading to the non-distributive reading being dispreferred. For these reasons, we believe that the small effect size observed in these experiments is still perfectly compatible with our hypothesis: once again, there is still a clear and observable predicate sensitivity in both kinds of non-local cumulation.

In fact, as shown in Figure 3.4, when we plot the mean acceptability ratings for each predicate from the experiment on cumulation with universal quantifiers on the x-axis and the mean acceptability ratings for each predicate from the experiment on cumulation with negative quantifiers on the y-axis, we get a better sense of how closely related these two phenomena are.

In Figure 3.4, mean acceptability ratings for cumulation with universal quantifiers and cumulation with negative quantifiers exhibit a clear positive correlation, as illustrated by the regression line. Thus, predicates that receive relatively high ratings for cumulation with universal quantifiers also tend to receive relatively high ratings for cumulation with negative quantifiers, and vice versa. Moreover, in this case, the regression line even closely approximates the identity line, $y=x$ (which would correspond to a perfect correlation between the two phenomena), indicating not only a positive relationship, but a near one-to-one correspondence in the relative acceptability across predicates for the two phenomena: on average, predicates received nearly the same mean acceptability rating in both cases of cumulation.

This visual pattern is confirmed by a cumulative link mixed model predicting acceptability ratings of cumulation with negative quantifiers from acceptability ratings of cumulation with universal

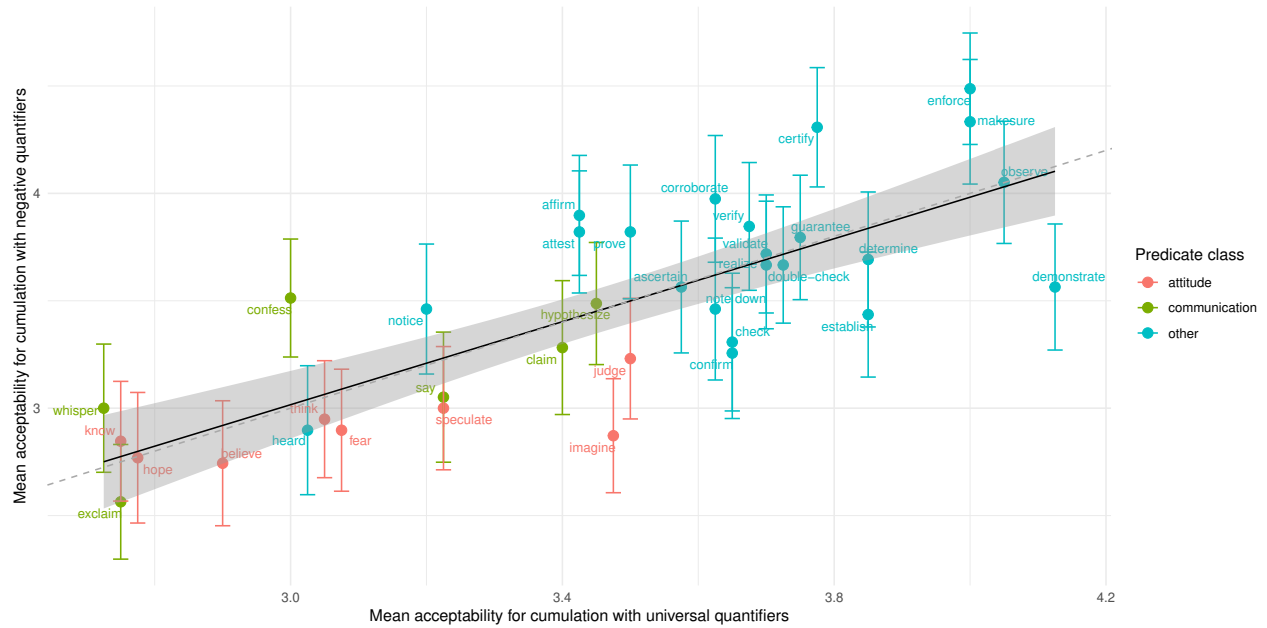


Figure 3.4: Mean acceptability ratings (for each predicate) of non-local cumulation with embedded negative quantifiers plotted against mean acceptability ratings (for each predicate) of non-local cumulation with embedded universal quantifiers.

Lower values indicate lower acceptability; higher values indicate higher acceptability.

quantifiers, with random intercepts and by-participant random slopes (to control for participant-level variability in both intercepts and slope).¹⁰ The model indicates a significant positive effect of acceptability ratings of cumulation with universal quantifiers on acceptability ratings of cumulation with negative quantifiers ($\beta = 1.11$, $SE = 0.23$, $z = 4.87$, $p < 0.001$).¹¹ Together, these results demonstrate that non-local cumulation exhibits similar predicate sensitivity—whether the embedded quantifier is a universal or a negative quantifier—paralleling the pattern observed for variation readings.

In sum, all four experiments reveal a clear similarity in predicate sensitivity across all four

¹⁰Response ~ CumulationIndex + (1 | participant) + (0 + CumulationIndex | participant) where ‘Response’ is the raw acceptability ratings for cumulation with negative quantifiers and ‘CumulationIndex’ corresponds to the standardized mean acceptability ratings for each predicate from the experiment looking at cumulation with universal quantifiers.

¹¹Since we standardized the predictor in our model, β represents the effect of a 1 SD increase in acceptability ratings of cumulation with universal quantifiers: a 1 SD increase was associated with an approximately threefold increase in the odds of a higher response concerning cumulation with negative quantifiers, corresponding to a substantial shift in the probability of higher acceptability ratings, indicating a strong and systematic relationship between the two phenomena.

phenomena we considered (variation readings with universal quantifiers, variation readings with negative quantifiers, non-local cumulation with universal quantifiers and non-local cumulation with negative quantifiers). We are led to hypothesize, then, that the predicate sensitivity observed across all four experiments is no coincidence, but a consequence of the following empirical hypothesis about the nature of the predicate generalization, where sentences with singular indefinites in the matrix subject position (cf. (1-a)) and those with plural subjects (cf. (3-a)) exhibit a parallel.

- (13) *Empirical hypothesis:* the embedding predicates which license variation readings with singular indefinites also license non-local cumulation with plural subjects.

The empirical claim in (13) can be understood as conveying that the availability of variation readings *correlates* with the availability of non-local cumulation. Therefore, a more direct test of this hypothesis is to look for a correlation between these two phenomena. More specifically, we can use the results from the experiment concerning cumulation with universal quantifiers as a predictor of variation readings with universal quantifiers (the prediction: higher acceptability rating for cumulation leads to higher acceptability rating for variation reading).¹²

First, we can visually observe the connection between variation readings with universal quantifiers and cumulation with universal quantifiers by looking at Figure 3.5, where we have mean acceptability ratings of cumulation with universal quantifiers (for each predicate) on the x-axis and mean acceptability ratings of variation readings with universal quantifiers (for each predicate) on the y-axis.

Once again, the main observation from Figure 3.5 is that mean acceptability ratings of variation readings with universal quantifiers and cumulation with universal quantifiers exhibit a clear positive correlation, as illustrated by the regression line. Thus, predicates that receive relatively high ratings for cumulation with universal quantifiers also tend to receive relatively high ratings for variation readings with universal quantifiers, and vice versa. We note that, in this case, the regression line is shifted upward along the y-axis, showing that, in general, variation with universal

¹²Even though we focus on the correlation between variation readings with universal quantifiers and non-local cumulation with universal quantifiers, we also tested other correlations, which were all qualitatively similar.

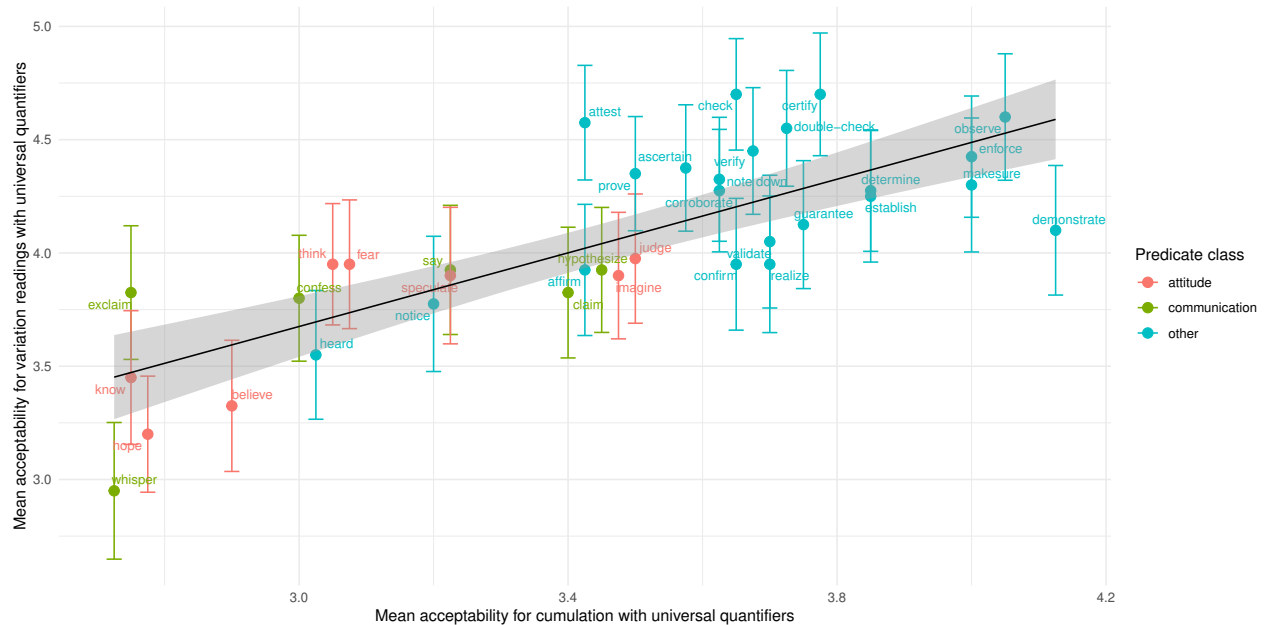


Figure 3.5: Mean acceptability ratings (for each predicate) of variation readings with embedded universal quantifiers plotted against mean acceptability ratings (for each predicate) of non-local cumulation with embedded universal quantifiers.

Lower values indicate lower acceptability; higher values indicate higher acceptability.

quantifiers was rated higher than cumulation with universal quantifiers. That being said, the slope of the regression line is still approximately 1, suggesting that *differences* in acceptability between predicates are preserved to a similar degree across both experiments. These observations support the hypothesis that predicates that license variation readings with singular indefinites tend to also license non-local cumulation with plural subjects. For completeness, we can also look at the plot in Figure 3.6, which now looks at the connection between variation readings with negative quantifiers and cumulation with negative quantifiers: we have mean acceptability ratings of cumulation with negative quantifiers (for each predicate) on the x-axis and mean acceptability ratings of variation readings with negative quantifiers (for each predicate) on the y-axis. While the correlation is a little weaker in this case (due to the variability in the acceptability ratings for variation readings with negative quantifiers), qualitatively, the results parallel what we observe in Figure 3.5.

We also fit a cumulative link mixed model predicting acceptability ratings for variation readings with universal quantifiers from acceptability ratings for cumulation with universal quantifiers, with

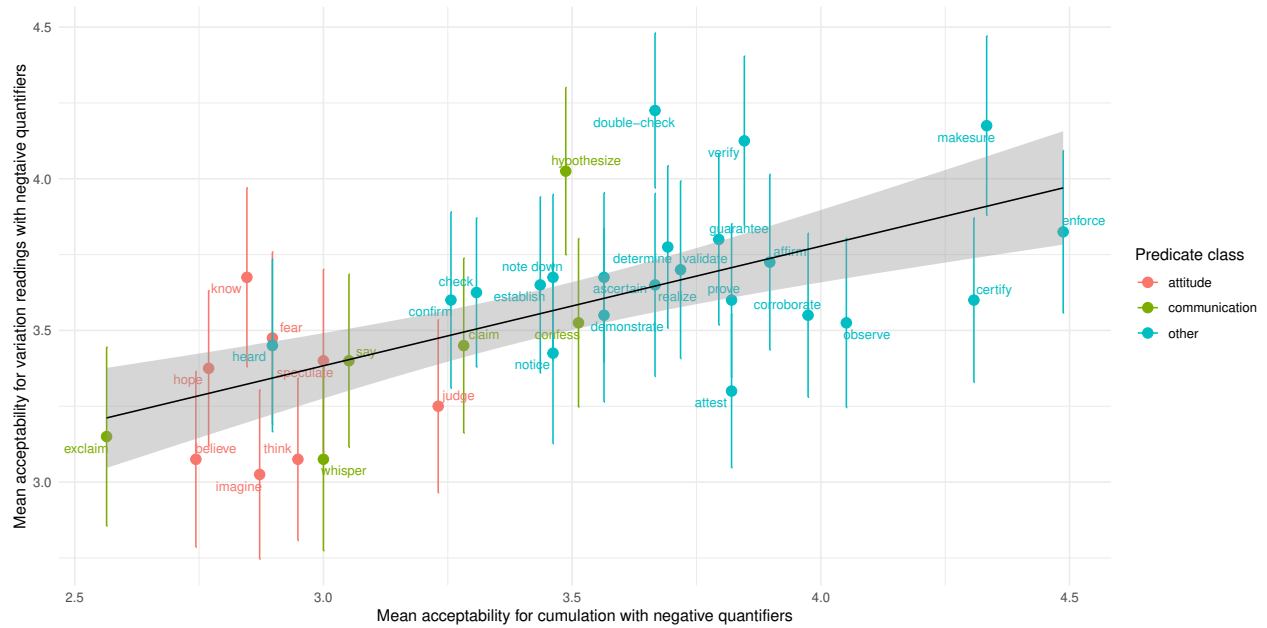


Figure 3.6: Mean acceptability ratings (for each predicate) of variation readings with embedded negative quantifiers plotted against mean acceptability ratings (for each predicate) of non-local cumulation with embedded negative quantifiers.

Lower values indicate lower acceptability; higher values indicate higher acceptability.

random intercepts and by-participant random slopes (to control for participant-level variability in both intercepts and slope).¹³ Acceptability ratings of cumulation with universal quantifiers were a significant positive predictor of acceptability ratings of variation readings with universal quantifiers ($\beta = 1.21$, $SE = 0.21$, $z = 5.68$, $p < 0.001$), confirming that predicates judged more acceptable for cumulation with universal quantifiers are also more likely to receive higher acceptability ratings for variation readings with universal quantifiers.¹⁴ These results indicate that variation readings with universal quantifiers and non-local cumulation with universal quantifiers are strongly correlated and exhibit similar predicate sensitivity, suggesting that they could involve a shared underlying

¹³Response ~ CumulationIndex + (1 | participant) + (0 + CumulationIndex | participant) where ‘Response’ is the raw acceptability ratings for variation readings with universal quantifiers and ‘CumulationIndex’ corresponds to the standardized mean acceptability ratings for each predicate from the experiment looking at cumulation with universal quantifiers.

¹⁴Again, because our predictor was standardized in our model, β represents the effect of a 1 SD increase in acceptability ratings of cumulation with universal quantifiers: a 1 SD increase was associated with a 3.37-fold increase in the odds of a higher response concerning variation readings with universal quantifiers, indicating a strong alignment between the two measures. The magnitude of this effect indicates a substantial shift in the probability of higher responses, reinforcing the conclusion that the two phenomena involve the same underlying predicate sensitivity.

mechanism.¹⁵

We note that the plot in Figure 3.5 only compares variation readings with universal quantifiers and cumulation with universal quantifiers. Recall, though, that we have shown that variation readings with universal quantifiers and variation readings with negative quantifiers exhibit a similar predicate sensitivity as well. This means that, if acceptability ratings for cumulation with universal quantifiers are a good predictor of variation readings with universal quantifiers, then they should also be a good predictor of variation readings with negative quantifiers. Figure 3.7 shows the connection between all three phenomena by taking Figure 3.2 and replacing the color-coding of different predicate classes (i.e., communication, attitude and other) with the acceptability ratings of cumulation with universal quantifiers. That is, on the x-axis, we have mean acceptability ratings for each predicate from the experiment on variation readings with universal quantifiers. On the y-axis, we have mean acceptability ratings for each predicate from the experiment on variation readings with negative quantifiers. The gradient color-coding (going from red to blue) indicates mean acceptability ratings for each predicate from the experiment on cumulation with universal quantifiers (red means lower acceptability for cumulation and blue means higher acceptability for cumulation).¹⁶

The main observation in Figure 3.7 is that mean acceptability ratings for both kinds of variation reading are positively correlated—as we noted when discussing Figure 3.2—and, crucially, this relationship correlates with acceptability ratings for cumulation with universal quantifiers. This is reflected in the plot by predicates with higher acceptability ratings for cumulation clustering in the upper-right region of the plot (as indicated by the fact that the majority of the predicates in the upper-right region are blue), whereas those with lower acceptability ratings for cumulation clustering in the lower-left (as indicated by the fact that the majority of the predicates in the lower-

¹⁵We also ran a cumulative link mixed model predicting acceptability ratings for variation readings with negative quantifiers from acceptability ratings for cumulation with negative quantifiers. The model indicated a significant positive effect ($\beta = 0.77$, $SE = 0.25$, $z = 3.11$, $p = 0.0018$), indicating that predicates that are more acceptable for cumulation with negative quantifiers are also more acceptable for variation readings with negative quantifiers.

¹⁶Given that the correlation between cumulation with universal quantifiers and cumulation with negative quantifiers was nearly perfect, we just used the former as a proxy for cumulation in general (Figure 3.7 already conveys a lot of information).

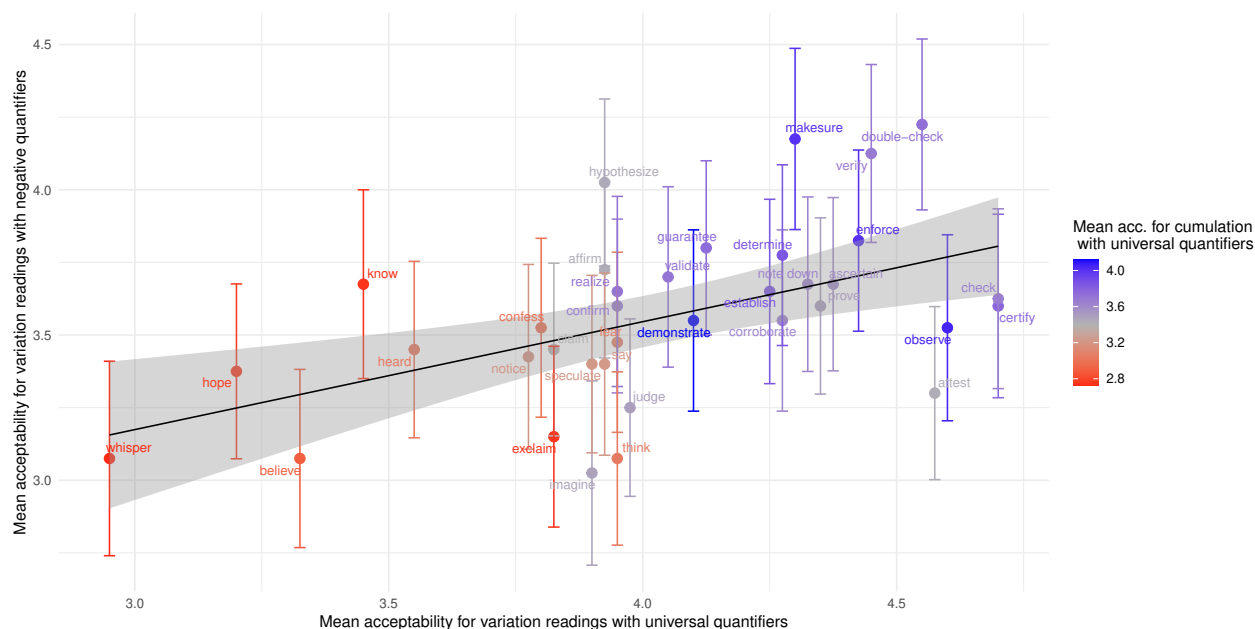


Figure 3.7: Mean acceptability ratings (for each predicate) of variation readings with embedded negative indefinites plotted against mean acceptability ratings (for each predicate) of variation readings with embedded universal quantifiers.

Lower values indicate lower acceptability; higher values indicate higher acceptability.

Gradience indicates the mean acceptability ratings (for each predicate) for cumulation with universal quantifiers. Red means lower acceptability for cumulation while blue means higher acceptability for cumulation.

left region are red). Together, these patterns indicate a consistent ordering of predicates across all three phenomena, suggesting that they all exhibit a similar predicate sensitivity.

Before moving on, we note that one possible reason why the correlation between variation readings with negative quantifiers and variation readings/cumulation with universal quantifiers is weaker (and more dispersed) could be because the experimental results are simply messier—due to the fact that variation readings with negative quantifiers are more variable and difficult than variation readings with universal quantifiers and cumulation with universal quantifiers. Alternatively, this may be because the task itself was more difficult in the experiment looking at variation readings with negative quantifiers: given that the instructions introduced participants to inverse scope readings with *universal* quantifiers, they had to transfer that knowledge to variation readings with *negative* quantifiers—which may be non-trivial for participants and could lead to increased

variability and confusion in their responses. In any case, even though the plot in Figure 3.7 shows some dispersion, we can still observe that both kinds of variation readings correlate fairly well with cumulation with universal quantifiers. Furthermore, the positive results obtained in these experiments provide support for the methodology that we used, namely, acceptability rating tasks. Our results illustrate that, even for an experiment this size, this methodology is in fact sensitive enough to detect relative contrasts in acceptability for subtle judgments, like variation readings and non-local cumulation.

In sum, the results from these experiments reveal a consistent predicate sensitivity across variation readings with universal quantifiers, variation readings with negative quantifiers, and non-local cumulation. At the predicate level, mean ratings are positively correlated across all four phenomena, indicating a shared ordering of predicates: predicates that are judged more acceptable in one case tend to be judged more acceptable in the others. This pattern is reinforced by mixed-effects modeling, which shows that acceptability in one case reliably predicts acceptability in another. Although the strength of these relationships varies—being stronger between variation readings and cumulation, and somewhat weaker for negative variation readings—the overall pattern is robust. These findings provide converging evidence for the above empirical hypothesis, repeated in (14), namely that predicates which license variation readings also license non-local cumulation (and, more broadly, variation readings with negative quantifiers).

- (14) *Cumulation-variation correspondence*: An embedding predicate licenses variation readings if it licenses cumulation with plural subjects.

Of course, the acceptability ratings across all four experiments did not parallel each other perfectly. This is to be expected given the difficulty in comparing results across experiments. We believe that the important takeaway is that the same empirical pattern can be observed in each experiment: the relative predicate sensitivity. We also believe that these experiments set the stage for further research, including further experimental validation.

More generally, the fact that all four phenomena exhibit closely aligned predicate sensitivity

suggests that they may be a consequence of the same mechanism operating on embedding predicates. Moreover, it seems that whatever is driving the predicate sensitivity of cumulation could also be driving the predicate sensitivity of variation. Pursuing this line of thought, in the next section, we present an analysis of cumulation where the predicate sensitivity stems from the inferential profile of the predicate itself, and then we try to leverage this account to explain the predicate sensitivity of variation.

3.4 The cumulative inference approach to non-local cumulation

One strategy to explain the predicate sensitivity of non-local cumulation draws on inferential analyses of cumulativity (Kratzer, 2002; Harada, 2022b). As we will see, the main advantage of this strategy is that it is inherently predicate-sensitive, since the mechanism of how to derive the relevant reading relies on the lexical/inferential properties of the predicates and its pragmatic interaction with context and world knowledge.¹⁷

Underlying this analysis is the claim put forth in Krifka 1992 and Kratzer 2002 in which certain relations between individuals and events can be cumulated, as in (15).¹⁸

$$(15) \quad \textit{Cumulative schema: } \forall e \forall e' \forall x \forall y [[R(x)(e) \wedge R(y)(e')] \rightarrow R(x+y)(e+e')]$$

¹⁷Our choice to focus on a particular analysis of non-local cumulation is by no means an exhaustive comparison of proposals. Our decision to focus on this analysis, though, is motivated by the parallel between non-local cumulation and variation readings. Furthermore, non-local cumulation, alone, is not the central focus of the paper, and so, our main concern is to simply outline an analysis of non-local cumulation that offers potential extensions to variation readings. In particular, one prominent approach of non-local cumulation that we do not discuss is the plural projection account put forth in Schmitt 2019, 2020. This is a very appealing approach to analyzing non-local cumulation, with many empirical benefits. However we do not consider it in detail for a couple of reasons. First, one of our general aims is to explore the hypothesis that connects cumulation to variation. As such, we choose one representative account of cumulation in which to illustrate this: the cumulative inference analysis. Second, and related to the first, the reason we choose this account is because we believe it provides a clearer explanation of the observed predicate sensitivity (as we understand it, plural projection is not a predicate sensitive mechanism). That being said, if one were able to put forth a predicate sensitive version of plural projection, then, in principle, one could replace the cumulative inference analysis with plural projection and our main argument would not be compromised—that cumulation can be leveraged to also explain variation.

¹⁸In what follows, we assume a standard semantics of plurals, where a conjunction like *Ann and Bea* denotes a sum of entities, $a+b$. Similarly, we assume that events can be singular or plural, construed as mereological sums (Bach, 1986). So the plural event comprising of e_1 and e_2 denotes a sum of events, e_1+e_2 .

In the antecedent in (15), the relation R holds between x and e , and also between y and e' . Due to the properties of the relation, we infer that the relation R also holds of the plurality, $x+y$, and the plural event $e+e'$. In what follows, for simplicity, we assume that embedding predicates like *make sure* are relations between individuals and propositions, as in (16). We could also enrich our semantics to take an event argument as well, but we leave this out for simplicity since they are not essential for the analysis. What is crucial for the analysis is that cumulation takes place between the matrix plurality and some other plurality coming from the embedded clause (i.e., a plurality of individuals or propositions)—but that does not need to be a plurality of events.¹⁹

$$(16) \quad \llbracket \text{make sure} \rrbracket = \lambda p_{\langle s,t \rangle} \lambda x_e. \text{make-sure}(x)(p)$$

The central observation underlying this analysis can be traced back to analyses of plurals more generally, where it has been claimed that the interpretation of plural subjects can depend on the meaning of the sentence predicate (Kroch, 1974; Scha, 1984; Dowty et al., 1987; Winter, 2000; De Vries, 2014; Champollion, 2017). For a basic motivation, it is known that plural predication is observed to be predicate sensitive. Consider for instance the following contrast:

- (17) a. John touched the books.
 b. John stole the books.

The sentence in (17-a) can be understood as true if John touched a single book. Alternatively, the sentence in (17-b) requires that John stole all the books. Therefore, even basic examples like (17) show us that different predicates can license different inferences—due to their lexical content.

More closely related to the cases at hand, it has been observed that sentences containing plural subjects and an embedded proposition can receive non-distributive interpretations in certain contexts (Pasternak, 2018). Building on Pasternak’s observation, Harada 2022b goes on to propose that these non-distributive interpretations between a plural subject and an embedded proposition is predicate dependent: weak (non-distributive) sentence meanings that arise when an embedding

¹⁹Notation: we take v to be the type of eventualities and s to be the type of worlds. Though since we will be focused on extensional contexts, we will mostly ignore the world index for the propositional argument.

predicate relates a plurality of individuals to an embedded proposition is enabled by the meaning of the embedding predicate.²⁰

Building on Harada 2022b, our proposal is that predicates like *make sure*, due to certain inferential properties tied to their lexical semantics, have a similar cumulating property as outlined in (15), and this gives rise to weak truth conditions.^{21,22} This means that, similar to (15), separate propositional arguments of *make sure* can be cumulated into a conjunction of (atomic) propositions. Analogously, the relevant individuals responsible for rendering each (atomic) proposition true are also cumulated together into a plurality of individuals. This is shown in (18), which illustrates the relevant inferential property as it applies to *make sure*—henceforth, a ‘cumulative inference’.²³

$$(18) \quad \frac{\forall x \forall y \forall p \forall q [[\text{make-sure}(x)(p) \wedge \text{make-sure}(y)(q)]]}{\rightarrow \text{make-sure}(x+y)(p \wedge q)}$$

²⁰Drawing on observations put forth in Harada 2022b, the empirical claim is that a cumulative reading is more readily available for the target sentence in (i), in contrast to the target sentence in (ii).

- (i) a. *[Chef 1 made sure the first recipe was flawless, but didn’t do anything for the second recipe. Chef 2 made sure that the second recipe was flawless, but didn’t do anything for the first recipe.]*
b. The two chefs made sure that the two recipes were flawless.
- (ii) a. *[Chef 1 claimed the first recipe was flawless, but didn’t say anything about the second recipe. Chef 2 claimed that the second recipe was flawless, but didn’t say anything about the first recipe.]*
b. #The two chefs claimed that the two recipes were flawless.

While we share this judgment concerning the relative contrast between (i) and (ii), an important step for future research would be to validate this contrast experimentally—especially since it has been contested in the literature (Haslinger and Schmitt, 2018; Schmitt, 2020). This would provide further evidence for the parallel between cumulation and variation. However, in this paper, we are primarily focused on non-local cumulation where the embedded plurality comes from a universal DP. For these cases, as we saw, we have experimental evidence that cumulation is predicate sensitive.

²¹In some sense, the claim we are making is a generalization of (15) since we propose that the propositional arguments of the embedding predicate get cumulated, along with the subject arguments.

²²Which properties of their lexical semantics enable this is a question which we leave for future research. We also note that this claim about embedding predicates is somewhat akin to imposing a meaning postulate on the predicate. However, we are reluctant to call this a meaning postulate because there are reasons to believe that it is not entirely a matter of lexical semantics but also influenced by pragmatics/context. Unfortunately, due to space, we cannot properly address these concerns here but we do offer some speculative remarks in the conclusion.

²³Rather than conjoining the propositional arguments, one can also make use of a cross-categorical sum operator, to derive a sum/plurality of propositions (Haslinger and Schmitt, 2018; Schmitt, 2019). For simplicity, in what follows, we will conjoin propositions rather than deal with pluralities of propositions. As far as we can tell, nothing hinges on this assumption.

One way to understand (18) is as follows: if the context renders the two conjuncts in the antecedent true, then this is sufficient to conclude that the consequent is also true. In a sense then, these inferential properties of the predicate provide sufficient conditions for the target sentence to be understood as true—since, as we will see, the target sentence will be contextually equivalent (or be entailed by) the proposition in the consequent of the cumulative inference. In contrast, we assume that a predicate like *claim* does not as readily license this kind of inference, due to it lacking the relevant inferential properties. This means that, for the predicate *claim*, something analogous to (18) would not be easily accessible.

With these assumptions in place, we can return to the above example, repeated in (19). The intended interpretation is captured as a result of the inferential property of *make sure* provided in (18). The intended interpretation of (19-b), given the context in (19-a), is provided in (19-c).

- (19) a. *[There are four speakers. Ann made sure that speaker 1 and speaker 2 had a ride; but not speaker 3 and speaker 4. Bea made sure that speaker 3 and speaker 4 had a ride; but not speaker 1 and speaker 2.]*
- b. Ann and Bea made sure that every speaker had a ride.
- c. $\llbracket (19-b) \rrbracket = 1$ if $[\text{make-sure}(a+b)(\forall x[\text{speaker}(x) \rightarrow \text{ride}(x)])]$
‘Each member of the plurality, Ann and Bea, did some making sure, resulting in all the speakers having a ride’

The first step towards deriving this interpretation involves the cumulative inference in (20), which is an instantiation of the schema in (18).

- (20) *Cumulative inference:*
- $$\text{make-sure}(a)(\text{ride}(s_1+s_2)) \wedge \text{make-sure}(b)(\text{ride}(s_3+s_4))$$
- $$\rightarrow \text{make-sure}(a+b)(\text{ride}(s_1+s_2) \wedge \text{ride}(s_3+s_4))$$

In (20), the first conjunct corresponds to Ann’s contribution in (19-a): Ann made sure that speaker 1 and speaker 2 had a ride. The second conjunct corresponds to Bea’s contribution in

(19-a): Bea made sure that speaker 3 and speaker 4 had a ride. Again, due to *make sure*'s inferential properties, we can combine both contributions and conclude that the consequent in (20) is true: Ann and Bea, between them, made sure that the four speakers had a ride. That is, cumulation combines $ride(s_1+s_2)$ and $ride(s_3+s_4)$ into the conjoined proposition, $ride(s_1+s_2) \wedge ride(s_3+s_4)$. The second step towards deriving the interpretation in (19-c) involves what we refer to as a ‘universal inference’, illustrated in (21): assuming there are only four speakers, speakers 1 and 2 having a ride and speakers 3 and 4 having a ride contextually entails that every speaker had a ride.

(21) *Universal inference:*

$$[ride(s_1+s_2) \wedge ride(s_3+s_4)] \rightarrow \forall x[speaker(x) \rightarrow ride(x)]$$

Therefore, combining these two contributions then amounts to an interpretation where Ann and Bea, between them, made sure that every speaker had a ride—which is the desired cumulative interpretation. In addition, the requisite underlying mechanism appealed to, to derive the intended truth conditions, is sensitive to the embedding predicate: cumulative inferences depend on certain properties of the predicate.

A couple of remarks are in order concerning the cumulative inference approach. First, we note that while this approach provides an account of the predicate sensitivity, it does not really tell us *why* we observe cumulation with some predicates but not with others. However, it does at least tell us that the source of the predicate sensitivity stems from the lexical semantics of the embedding predicate: the underlying mechanism—cumulative inferences—directly interfaces with the lexical semantics of the embedding predicate in a way that could be leveraged to explain the predicate sensitivity. Under this view, non-local cumulation crucially depends on certain inferential properties that differ from predicate to predicate. We do not yet have a complete theory of exactly what those properties are or how they interact with context, but we take the claim that predicates have different inferential profiles due to their lexical semantics as rather uncontroversial.

Related to this point, the gradient nature of our experimental results raises the question of whether predicate sensitivity is inherent or epiphenomenal—reducible, for instance, to constraints

on which QUDs make cumulative construals available (Haslinger, 2021; Poortman, 2014).²⁴ In fact, the small effect sizes observed in our experiments could reflect the fact that the inferential profile of the embedding predicate is not fully encoded in its lexical meaning, but emerges from the combination of lexical meaning in conjunction with linguistic context and context of use. Under this line of thought, then, we agree that pragmatics plays a role, but we nonetheless want an analysis in which the embedding predicate is referenced directly: whatever the pragmatic contribution turns out to be, the lexical semantics of the predicate clearly mediates how it interfaces with context and the QUD. The cumulative inference approach naturally accommodates this: since cumulative inferences arise from the interaction of lexical content with world knowledge, and world knowledge varies across individuals and predicate-context pairs, gradient acceptability is expected.

Second, as we now demonstrate, an approach relying on the inferential profile of the embedding predicate naturally extends to negative quantifiers as well. An explanation of examples concerning cumulation with negative quantifiers essentially comes for free, given the way we have formulated the cumulative inference analysis. All that needs to be said is that, instead of the cumulative and universal inferences that we had above, we have the corresponding ones in (22)—but these inferences are still made possible by the exact same inferential properties of *make sure*.

(22) Ann and Bea made sure that no speaker was late.

a. *Cumulative inference:*

$$\begin{aligned} &\text{make-sure}(a)(\text{on-time}(s_1+s_2)) \wedge \text{make-sure}(b)(\text{on-time}(s_3+s_4)) \\ &\rightarrow \text{make-sure}(a+b)(\text{on-time}(s_1+s_2) \wedge \text{on-time}(s_3+s_4)) \end{aligned}$$

b. *Negative universal inference:*

$$[\neg \text{late}(s_1+s_2) \wedge \neg \text{late}(s_3+s_4)] \rightarrow \neg \exists x[\text{speaker}(x) \wedge \text{late}(x)]$$

The observation is that the inferential profile of this example parallels that of examples concerning cumulation with universal quantifiers and, as such, we would expect a cumulative inference to be licensed—which is what we see. Crucially, the same inferential properties of *make sure* that license

²⁴We thank an anonymous reviewer for raising this point. Haslinger 2021 in particular proposes that the QUD drives the availability of cumulative readings.

cumulation with universal quantifiers also license cumulation with negative quantifiers.²⁵ In a sense then, cumulation with negative quantifiers provides independent support for the cumulative inference analysis as these examples show that predicates differ in the exact way predicted by the cumulative inference analysis. As a result, these examples strongly suggest that what underlies the predicate sensitivity of non-local cumulation is the inferential profile of the embedding predicate.

Therefore, Harada 2023’s account of non-local cumulation provides a compelling reason to believe that the predicate sensitivity of non-local cumulation is driven by the inferential profile of the embedding predicate. Combining this account with the experimental findings reported earlier, we have strong evidence that variation also references the embedding predicate’s inferential profile. The most obvious reaction to this result, then, is that variation arises from cumulation, which is what we explore in the next section.

3.5 Variation from cumulation

Having outlined the cumulative inference approach to cumulation, we now explore the hypothesis that variation readings involve the same underlying inferential mechanism (i.e., cumulative inferences), and this is why we observe the same predicate sensitivity. Our goal in the remainder of the paper, then, is to explore how the cumulative inference approach to non-local cumulation can be extended to handle variation readings, providing an initial discussion of the consequences for variation readings. In doing so, we point out what modifications need to be made to the cumulative inference analysis to handle variation readings and the costs that this kind of analysis incurs. That is, in Section 3.5.2, we outline some prominent challenges that arise in naively assuming that variation readings are just like cases of cumulation with a plural subject, and also involve cumulative inferences with underlyingly plural matrix subjects.

Our motivation for exploring this particular analysis is that this inferential mechanism is in-

²⁵It is worth pointing out that, in contrast, any approach to non-local cumulation that involves two pluralities would need to stipulate that negative indefinites also denote pluralities of individuals—something which, as far as we know, has not been proposed in the literature (unlike with universal DPs). Furthermore, even with such a stipulation, it would need to explain why negative indefinites pattern with universals in licensing cumulation but differ from them in other respects. The cumulative inference approach requires no such stipulations: the same inferential properties of *make sure* naturally extend to both cases.

herently predicate sensitive and applies uniformly to negative and universal quantifiers, meaning it has the potential to handle both variation readings with universal quantifiers and variation readings with negative quantifiers—which we view as an important desideratum.

3.5.1 A simple-minded extension of the cumulative inference approach

As we have seen, the cumulation we are concerned with involves a plurality in the matrix subject position. If variation readings also overtly involved such a plurality, then they could be analyzed just like the example in (19). But since they do not contain overtly plural subjects, this raises the following question: where does the subject plurality come from with a singular indefinite? A key element of the proposal put forth in this section is that variation comes from cumulation, implying that singular indefinites can participate in cumulation, and hence, they can somehow contribute a plurality. As a consequence, contrary to appearances, variation readings actually do involve a hidden plurality in the matrix subject position, which is supplied by the singular indefinite. The challenge is that we do not have any independent evidence that this is possible. We outline what this kind of analysis looks like by focusing on the cumulative scenario in (23-a).

- (23) a. *[There are three speakers. The first student, Ann, made sure speaker 1 got to campus; the second student, Bill, made sure speaker 2 got to campus; the third student, Carl, made sure speaker 3 got to campus.]*
- b. A student made sure that every invited speaker got to campus.

To maintain the parallel with cumulation with plural subjects in (19-c), the target sentence meaning for (23-b) should be something like (24).

- (24) Target sentence meaning for (23-b):
- $\llbracket (23-b) \rrbracket = 1$ if $\text{make-sure}(a+b+c)(\forall x[\text{speaker}(x) \rightarrow \text{campus}(x)])$
- ‘Ann (student 1), Bill (student 2) and Carl (student 3) each did some making sure resulting in all the speakers getting to campus’*

To arrive at this target sentence meaning, we need a way of retrieving the plurality of individuals, $a+b+c$, from the singular indefinite. Effectively, what we need to say is that, due to the inferential properties of the embedding predicate, cumulating the individual contributions—just as we did with overtly plural subjects—allows us to interpret the target sentence as if a plural subject were present. The idea would be that this plurality comes from collecting the different witnesses of the indefinite (Ann, Bill and Carl) into a plural subject. To illustrate, consider the cumulative inference in (25): if the context renders the three conjuncts in the antecedent true (where each conjunct corresponds to an individual's contribution), then this is sufficient to conclude that the consequent is also true. But the consequent of the cumulative inference delivers a proposition containing a plurality in the subject position. If the meaning for the target sentence in (23-b) is the result of the cumulative inference in (25) then the singular indefinite needs to be interpreted as if it were a plural subject. This means that, contrary to appearances, we must assume that the inferential properties of the embedding predicate enable us to access a plurality in the matrix subject position.

(25) *Cumulative inference:*

$$\begin{aligned} & [\text{make-sure}(a)(\text{campus}(s_1))] \wedge \\ & [\text{make-sure}(b)(\text{campus}(s_2))] \wedge \\ & [\text{make-sure}(c)(\text{campus}(s_3))] \\ & \rightarrow [\text{make-sure}(a+b+c)(\text{campus}(s_1) \wedge \text{campus}(s_2) \wedge \text{campus}(s_3))] \end{aligned}$$

As laid out in Section 3.4, the cumulative inference also conjoins the three embedded propositions to get the conjunction, $\text{campus}(s_1) \wedge \text{campus}(s_2) \wedge \text{campus}(s_3)$. The second step towards deriving the interpretation in (24) involves the universal inference in (26): assuming there are only three speakers, speakers 1, 2 and 3 getting to campus contextually entails that every speaker got to campus.

(26) *Universal inference:*

$$[\text{campus}(s_1) \wedge \text{campus}(s_2) \wedge \text{campus}(s_3)] \rightarrow \forall x[\text{speaker}(x) \rightarrow \text{campus}(x)]$$

Combining these three contributions then amounts to an interpretation where Ann, Bill and Carl,

between them, made sure that every speaker got to campus. This interpretation captures the perceived variation since different students were responsible for different speakers getting to campus. Therefore, whereas cumulation with plural subjects involves an overtly plural subject, to extend the analysis to variation readings, we must posit that cumulation with singular indefinites involves a hidden plurality in the matrix subject position, where the plurality is derived from the different witnesses of the indefinite.

3.5.2 Challenges for the simple-minded extension

The proposed analysis so far enables variation with singular indefinite subjects. However, positing a hidden plurality alone does not resolve our issues: the proposed analysis derives too weak a meaning, as the variation it permits is too unconstrained and does not align with speaker intuitions. Assuming we can derive a plurality in the matrix subject position, this would suggest that the singular indefinite is really interpreted like a plural (in)definite. As a result, the meaning of (1-a), repeated in (27-a), would be equivalent to (27-b), where the singular indefinite, *a student*, is replaced by the plural definite, *the students*.

- (27) a. A student made sure that every speaker had a ride.
b. The students made sure that every speaker had a ride.

But how can we even derive a plurality in the matrix subject position? One compositional route that enables us to retrieve a plurality from a singular indefinite involves looking at the set of individuals supplied by the indefinite's restrictor. Compositionally, this can be achieved by type-shifting the singular indefinite to a predicate, using Partee's **be** operator, as defined in (28).

- (28) a. $\llbracket \text{be} \rrbracket = \lambda \mathcal{P}_{\langle e,t \rangle, t} \lambda y. \mathcal{P}(\lambda z. z = y)$
b. $\text{be}(\llbracket a \text{ student} \rrbracket) = \text{be}(\lambda P. \exists x [\text{student}(x) \wedge P(x)])$
 $= \lambda x. \text{student}(x)$
 $= \llbracket \text{student} \rrbracket$

The predicative meaning of the singular indefinite can then be used to retrieve the relevant set of

students. This solution is technically straightforward, but it does lead to an immediate challenge: singular indefinites do not regularly behave like plural arguments. In fact, as we now show, the two sentences in (27) are true in different scenarios and therefore have different meanings. What we observe is that the variation reading is stronger than ordinary cumulation with plural subjects in precisely the way that an LF with non-local scope would be. As a result, this compositional move would need to be appropriately constrained to somehow derive a stronger meaning (as would the claim that singular indefinites can furnish a plurality in general).²⁶

First, for the variation reading, the propositional conjuncts that participate in cumulation—each propositional argument of an instance of making sure—need to be ‘projected’ from the universal. In our running example, we need one conjunct per speaker; i.e., $\text{campus}(s_1) \wedge \text{campus}(s_2) \wedge \dots$. Second, the matrix subject must be related to each of these conjuncts in a ‘pointwise’ manner. In our running example, this means that we need a single student per conjunct: there is a ‘singular restriction’ due to the singular indefinite. These two points can be clarified by looking at the truth conditions that result from a non-local QR parse: for each speaker x , a student made sure that x got to campus. These truth conditions capture that the conjunction must be by speaker and also that the matrix indefinite must be related to each conjunct in a pointwise manner.

We can further emphasize these points by considering the example in (29).

- (29) a. *[There are three invited speakers. To get from the airport to campus, one has to take*

²⁶Another possibility, as suggested by a reviewer, is that the subdomain alternatives of indefinites are grammatically accessible and this is one way of accessing the witnesses of the indefinite in order to retrieve a plurality (Bar-Lev, 2021). While this is an appealing solution, it would face similar challenges as Partee’s **be** operator. In particular, either move does not even seem to be general enough to capture the full range of distribution of variation readings. This is because variation readings are also possible when the matrix subject is a singular definite, as illustrated in (i).

- (i) a. *[There were three colloquium events this year, each with a different invited speaker. For each colloquium, the speaker was assigned a designated student volunteer to help them get to campus. For the first event, the student volunteer, Ann, made sure the speaker got to campus. For the second event, the student volunteer, Bill, made sure the speaker got to campus. Finally, for the third event, the student volunteer, Carl, made sure the speaker got to campus.]*
- b. The student volunteer made sure that every invited speaker got to campus.

In this case, cumulation cannot take place through the plurality of individuals supplied by the definite’s restrictor—or the subdomain alternatives of the definite—since the definite determiner presupposes that its restrictor is a singleton. Therefore, while there are different compositional pathways to retrieving a plurality for cumulation, they do not actually appear to be very promising avenues.

a shuttle bus, then get a ride. Ann, a student, made sure every speaker had a shuttle ticket and Bill, another student, made sure every speaker had a ride to campus.]

- b. The students made sure that every invited speaker got to campus.
- c. #A student made sure that every invited speaker got to campus.

In (29-a), the conjuncts that participate in cumulation are: *every speaker had a shuttle ticket* and *every speaker had a ride to campus*. This conjunction of propositions contextually entails the proposition that *every speaker got to campus*. This means that, in (29-a), the two students, between them, made sure that every speaker got to campus. As a result, the sentence with a plural definite subject, (29-b), is felicitous and true in this context. If the singular indefinite subject is simply interpreted as a plural definite, we would expect (29-c) to also be felicitous and true in this context, contrary to what is observed. This contrast in (29) shows us that cumulation with singular indefinites is more restricted than cumulation with plural subjects.

We can highlight the tension in (29) by considering again our inferential schema for *make sure*, repeated in (30),

$$(30) \quad \forall x \forall y \forall p \forall q [[\text{make-sure}(x)(p) \wedge \text{make-sure}(y)(q)] \\ \rightarrow \text{make-sure}(x+y)(p \wedge q)]$$

In (29-a), the conjunct *p* corresponds to the proposition *every speaker had a shuttle ticket* and the conjunct *q* corresponds to the proposition *every speaker had a ride to campus*, as in (31).

$$(31) \quad \text{make-sure}(a)(\forall x[\text{speaker}(x) \rightarrow \text{bus}(x)]) \wedge \\ \text{make-sure}(b)(\forall x[\text{speaker}(x) \rightarrow \text{ride}(x)]) \\ \rightarrow \text{make-sure}(a+b)(\forall x[\text{speaker}(x) \rightarrow \text{campus}(x)])$$

This conjunction of propositions is at odds with the variation reading. This is because we want each conjunct to project from the universal. That is, what we need for the variation reading is to restrict the universal quantification over *p* and *q* in (30): they must be members of the set of

propositions ‘projected’ by the universal quantifier, as illustrated in (32) where, now, each conjunct that enters into cumulation (i.e., *campus(s_i)*) is projected from the universal DP in the sense that we get one conjunct per speaker: *campus(s₁)* $\wedge$ *campus(s₂)* $\wedge$

- (32) *make-sure(a)(campus(s₁))* $\wedge$
 make-sure(b)(campus(s₂)) $\wedge$
 make-sure(c)(campus(s₃))
 $\rightarrow$ *make-sure(a+b+c)($\forall x$ [speaker(x) $\rightarrow$ campus(x)])*

Given that the variation reading is stronger than ordinary cumulation with plural subjects in precisely the way that an LF with non-local scope is, does this suggest that a non-local QR analysis may be on the right track after all—since the problem is avoided by QR? We do not believe this to be the case since negative variation readings are also constrained in the same way as variation readings. That is, even with negative indefinites, the variation reading is stronger than ordinary cumulation with plural subjects insofar as it requires the propositional conjuncts to project from the negative indefinite DP, (33).

- (33) a. *[There are three invited speakers. To ensure that none of the speakers were late, they needed to take a shuttle bus and then get a ride. Ann, a student, made sure every speaker had a shuttle ticket and Bill, another student, made sure every speaker had a ride to campus.]*
- b. The students made sure that no invited speaker was late.
- c. #A student made sure that no invited speaker was late.

The fact that we see the same constraints on variation readings with universal quantifiers and variation readings with negative quantifiers reinforces that whatever the underlying mechanism is, it should apply to both kinds of variation readings—something QR does *not* do.

Relatedly, the context in which (29-c) is used involves two students getting each speaker to campus, not one, and this conflicts with the use of the singular indefinite. This tension does not

arise with plural (in)definite subjects. To see this, consider the example in (34). Given the cumulative scenario in (34-a), the target sentence in (34-b) is felicitous, even though Carl and Dan were jointly responsible for the third speaker. In contrast, the target sentence in (34-c) is not felicitous, due to the fact that Carl and Dan were jointly responsible for the third speaker.

- (34) a. *[Ann, Bea, Carl and Dan are students. There are three invited speakers. Ann made sure that speaker 1 got to campus. Bea made sure that speaker 2 got to campus. Carl and Dan made sure that speaker 3 got to campus.]*
- b. The students made sure that every invited speaker got to campus.
- c. #A student made sure that every invited speaker got to campus.

Therefore, the use of the singular indefinite imposes a number effect which requires the singular indefinite to be related to the conjunction of propositions (projected from the universal) in a point-wise manner. What we observe then is that cumulation with plural subjects is available in a range of cumulative scenarios, leading to a weaker meaning. Sentences with singular indefinite subjects involve a more restricted form of cumulation, as evidenced by the infelicity of the target sentence in (29-c). For variation readings, not only would we need to somehow retrieve a plurality from a singular indefinite for cumulation to take place, but this derived plurality should lead to a different meaning than sentences with overtly plural subjects. In other words, we want to derive a plural subject in a way that also suitably restricts the meaning of the sentence.²⁷

- (35) **Challenge:** replacing a plural subject with a singular indefinite subject leads to a stronger

²⁷ Another example illustrating the difference between plural indefinite and singular indefinite subjects is provided in (i), where the universal quantifier is replaced with a singular definite (Harada, 2022b, Ch.2).

- (i) a. *[The ramen recipe is comprised of two parts: the noodle recipe and the broth recipe. Chef 1 proved that the noodle recipe is flawless. Chef 2 proved that the broth recipe is flawless.]*
- b. Two chefs proved that the ramen recipe is flawless.
- c. #A chef proved that the ramen recipe is flawless.

In (i), even though the embedded proposition contains a singular definite, *the ramen recipe*, the context establishes that this recipe is divided into two parts. As a result, cumulation is possible with the sentence containing a plural indefinite subject while the sentence with a singular indefinite subject is not felicitous in the provided context.

meaning for the sentence as a whole.

Since we do not have a principled way of deriving this stronger meaning with singular indefinite subjects, we would be forced to simply stipulate that the use of the singular indefinite is only compatible with a cumulative scenario like (23-a). Ideally, we would like to tie this restricted form of cumulation to the use of the singular indefinite, and the fact that, in the scenario in (29-a), two students were required to get each speaker to campus—which is at odds with the use of the singular indefinite. In short, we propose that the singular indefinite restricts the possible ways of dividing the content of the embedded proposition by requiring that the cumulative mapping between students and propositions be injective: each instance of making sure must involve a single student and a single speaker getting to campus.

It is worth noting that there is a partial solution to the projection problem observed with variation readings. The idea is that, with a singular indefinite matrix subject, the relevant plurality is parasitic on a set of propositions retrieved from the complement clause rather than from context. One way to capture this is to have the part structure of the universal DP ‘project’ to the propositional level (Schmitt, 2019): each individual in the domain of the universal pointwise combines with the complement denotation, yielding a proposition for each speaker x (i.e., x getting to campus). The required conjunction of propositions is then retrieved from the universal’s part structure, from which a plurality in the matrix subject position can be constructed.

This solves the projection problem but only partially. It does not ensure that the cumulative mapping between the subject plurality and the projected propositions is injective, and it faces a further difficulty: extending the same projection mechanism to sentences with negative quantifiers is non-trivial, since negative DPs do not standardly supply a plurality of individuals.

A further challenge for the cumulative inference analysis arises with plural existential subjects. Consider the sentence in (36-a).²⁸

- (36) a. Two students made sure that every professor had a ride.

²⁸We thank a reviewer for bringing this point to our attention.

- b. *Cumulative reading*: There is a plurality of students that cumulatively made sure that every professor had a ride.
- c. *Distributive reading*: For every professor, there are two students that (cumulatively) made sure this professor had a ride.

The cumulative reading, paraphrased in (36-b), is straightforwardly captured by the cumulative inference analysis. The variation reading, paraphrased in (36-c), however, is not: the two readings require different subject pluralities—a single fixed plurality in (36-b), but a potentially different plurality for each professor in (36-c). It is unclear how the cumulative inference approach can derive both from a unified mechanism. A non-local QR analysis handles this contrast more cleanly: the universal can tuck in under the plural existential—delivering the cumulative reading via a ****-operator (Krifka, 1986; Beck and Sauerland, 2000)²⁹—or take matrix scope over it—delivering the variation reading. One might therefore take (36) as evidence for non-local QR. However, again, we believe the parallel example in (37) undermines this conclusion.

- (37)
- a. Two students made sure that no professor was late to the conference.
 - b. *Cumulative reading*: There is a plurality of students that cumulatively made sure that no professor was late.
 - c. *Distributive reading*: For every professor, there are two students that (cumulatively) made sure this professor wasn't late.

If the observations concerning (37) are correct, the same two readings arise when we replace the universal with a negative quantifier. Since negative DPs cannot straightforwardly take distributive scope the way universal DPs can, deriving (37-c) via QR is non-trivial (in fact, it is also unclear how to derive the cumulative reading in (37-b) under a QR approach). The apparent advantage of the non-local QR account thus dissolves: it faces the same explanatory gap as the cumulative inference approach when the universal is replaced by a negative indefinite. Both accounts, then,

²⁹Alternatively, the cumulative reading can be captured in other ways—this is just one possibility that arises if non-local QR is permitted. Two other possibilities are: through plural projection (Schmitt, 2019, 2020), through event semantics (Kratzer, 2002; Chatain, 2022a,b).

owe an explanation of the variation/cumulative ambiguity with plural existential subjects.

As we have seen, there are reasons to be skeptical about a simple-minded extension from the kind of cumulation that the cumulative inference analysis aims to capture to variation readings. If variation readings are really just cumulative inferences in disguise, we would expect them to be possible in the same range of cumulative scenarios as cumulation with overtly plural subjects, contrary to fact. Instead, we have to stipulate a restriction that differentiates variation readings from cumulation with overtly plural subjects. Furthermore, stipulating that singular indefinites can sometimes furnish a plurality of individuals introduces challenges that are even broader than reported in this section thus far. For instance, we find no other indicator that singular indefinites behave like a plurality (i.e., modification by *together*, (38-a), floated quantifiers, (38-b), plural pronouns or the effect of context via contextual covers).³⁰

- (38) a. #A student, together, made sure that every speaker had a ride.
 b. #A student each made sure that every speaker had a ride.

Therefore, while analyzing variation readings as cumulative inferences is an exciting hypothesis, the simple-minded approach that we outlined here faces serious challenges. The main drawback is that positing that a singular indefinite denotes a hidden plurality introduces unwanted predictions. Nevertheless, the approach has two major advantages. First, it ties together cumulation

³⁰We thank a reviewer for making this point. A reviewer also points out the following. If the singular indefinite can somehow provide a plurality, via the set of its witnesses/the elements in its restrictor, why do we not need to exhaust this set? For instance, assuming that there are 6 students that are salient in the discourse context. In contexts that don't license non-maximality for definite plurals (i-a) is bad if not all of the students are involved (Križ, 2015). However, under the exceptional scope reading of (i-b), nothing tells me that all of the students were involved, even in contexts that do not license non-maximality: the sentence is fine if only three of the students organized rides.

- (i) a. The students made sure that every professor had a ride.
 b. A student made sure that every professor had a ride.

We agree with the reviewer and, at this point in time, we do not have a satisfactory explanation of this observation. However, we would like to point out that this is only an issue if we assume that the plurality derived from the singular indefinite is akin to a singular *definite*, as we have been assuming. But this need not be the case; it is possible that the derived plurality is more like a plural indefinite. That is, in same context described above, there is no corresponding maximality inference in (ii).

- (ii) Students made sure that every professor had a ride.

and variation, providing a principled explanation of the predicate sensitivity of both phenomena: only certain embedding predicates have the inferential properties to license cumulation, which naturally accounts for why only those predicates permit variation readings. Second, it applies uniformly to variation readings with universal quantifiers and those with negative quantifiers. This makes the cumulative inference approach to variation readings an interesting hypothesis and an appropriate basis for further inquiry.

The important insight to take away is this: an adequate analysis of variation readings seems to reference the same inferential profile of embedding predicates as cumulation does. To this end, the crucial parallel with cumulation is the nature of the mechanism proposed in Harada 2023: it involves the inferential profile of the embedding predicate in a way that licenses certain inferences (making it inherently predicate sensitive). But does our analysis of variation readings actually have to invoke cumulation? What we have learned is maybe not. At this stage, then, a fully adequate development of the approach to variation readings is left to future work, but we believe that a better understanding of the inferential mechanism discussed in this paper is important for future research, and for resolving the puzzle outlined in this paper.

As things stand, we are left with the puzzle of why variation readings are sensitive to the inferential profile of the embedding predicates, just like cumulation, and more specifically, why they are stronger than the corresponding cases of cumulation.

3.6 Concluding remarks

The take-home message we have argued for is that, empirically, a clause embedding predicate seems to license variation readings (with universals and negative indefinites) if it licenses cumulation with a plural subject. We reached this conclusion in steps. First, we observed that variation readings with universal quantifiers and those with negative indefinites are correlated, insofar as they exhibit a similar predicate sensitivity. This calls into question analyses of variation readings that invoke non-local QR. Second, we observed that both kinds of variation readings and cumulation with plural subjects are connected insofar as they also exhibit the same predicate sensitivity:

there is a correlation between the degree to which an embedding predicate licenses cumulation and the degree to which it licenses variation. Regardless of how one ultimately chooses to capture this empirical generalization, this paper contributes to the emerging view that a more accurate characterization of the empirical generalization concerning locality constraints imposed by finite complement clauses requires a more fine-grained approach; particularly, one that is sensitive to the identity of the embedding predicate (Barker, 2022).

But what exactly is it that separates predicates that license variation readings from those that do not? As we pointed out, Barker (2022) proposed the following empirical generalization concerning variation readings: finite complement clauses of attitude and communication verbs are scope islands for universals. At this stage, we do not have a complete answer to the above question, but, as we have shown, we do believe that Barker’s hypothesis is on the right track, and we also believe that the local approach provides a reason for this generalization: the correlation between verb type (i.e., attitude/communication or not) and the ability to license cumulation.

Moreover, what is it about these ‘other’ predicates in our experiments that licenses cumulation/variation? In (39), we offer another observation which we believe may provide further insight into the empirical generalization underlying variation readings. The observation concerns the nature of the matrix subject DP. Some embedding predicates, like *make sure*, can readily take inanimate subjects, as illustrated in (39-a). Other embedding predicates, like *claim*, do not take inanimate subjects, as illustrated in (39-b).

- (39) a. A DNA test made sure that the suspect was found guilty.
 b. #A DNA test claimed that the suspect was guilty.

Finally, other embedding predicates—for example, some communication verbs—that on the surface do not take inanimate subjects have uses where they in fact can. We can illustrate this with the embedding predicate *reveal*, which is typically classified as a communication verb (Levin, 1993). In certain contexts though, *reveal* has a use where it does not really behave like a communication verb and, as a result, it can take an inanimate subject, as illustrated in (40-b). In those same

contexts, a variation reading seems to be more readily available as well, as illustrated in (40-c).

- (40) a. *[There were three suspects in a trial. The jury knew the first suspect was guilty because he failed a polygraph test. The jury knew the second suspect was guilty because of a DNA test. The jury knew the third suspect was guilty due to a verbal response test during questioning.]*
- b. The tests revealed that every suspect was guilty.
- c. A (reliable) test revealed that every suspect was guilty.

The takeaway: we believe that a more comprehensive account of variation readings may not necessarily be sensitive to the ‘word’ (the embedding predicate) itself, but also its meaning in the context created by the matrix subject DP. This could be tied to a particular inferential/semantic property that is licensed by the embedding predicate in certain contexts, but not others. What these observations show is that this inferential/semantic property is correlated not only with the ability to license cumulation, but also the ability to take inanimate subjects.³¹

These observations concerning the contextual factors that affect cumulation/variation also reinforce the experimental results presented throughout the paper, which showed a more gradient effect across embedding predicates. A complete account of variation readings should both explain the differences between classes of predicates, but also make sense of the gradiency across predicates. The proposal that these are underlyingly cumulative inferences that come about through the lexical meaning of the predicates *in a particular context* (which includes pragmatic factors as well) provides a plausible explanation for both.

³¹If this turns out to be the case, perhaps other embedding predicates, like attitude predicates, can be, in a sense, coerced by the context into having the relevant inferential/semantic property to license cumulation and variation readings. Along these lines, one promising avenue for future research builds on the observation that cumulativity is modulated by the question-under-discussion (QUD) (Haslinger, 2021).

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Part III

Discussion and Conclusions

4

Discussion

This dissertation has pursued a single question from three complementary angles: when a sentence licenses a variation reading—a reading under which the witness of an indefinite or the referent of a definite seems to vary with a lower universal—does that reading reliably diagnose wide-scope of the lower universal (in the sense of scope-shifting)? The short answer defended across the preceding chapters is that it does not, or at least not automatically. As the motto proposed in the introduction puts it: “All that varies is not scope.” The goal of this concluding discussion is to provide a synthesis of the main findings of the dissertation: draw together the empirical and theoretical threads developed across the three chapters, reflect on what their convergence reveals about the architecture of the grammar, and take stock of the limitations and open questions that remain.

4.1 The central argument and its structure

The point of departure for the dissertation was a class of data concerning the scope-taking capabilities of universal quantifiers (i.e., exceptional scope phenomena) and an exploration into the tools available to a semanticist to analyze these phenomena. The central observation is that the standard diagnostic for wide-scope universal force—the variation diagnostic—can sometimes be unreliable: it picks out any reading under which the value of some expression (i.e., the witness of an indefinite or referent of a definite) varies, without discriminating among the compositional pathways that could, in principle, deliver such a reading. Scope-shifting mechanisms are one possible pathway; pseudo-scope mechanisms are another. The central methodological contribution of the dissertation is to demonstrate that this diagnostic is systematically confounded in the two syntactic environments most central to the debate about the clause-boundedness of scope: relative clauses and finite complement clauses. To this end, one of the main contributions of the thesis is an empirical investigation into the locality constraints imposed by clause boundaries. The methodological strategy was to dig deeper into exceptional scope phenomena to see if we can identify any important empirical restrictions, to give us a better sense of what the underlying mechanism might be. The goal was to argue that it is not sufficient to assume that variation is symptomatic of scope-taking, rather, different compositional pathways can be responsible for variation in different environments.

The argument, as developed across the three chapters, proceeded in a stepwise fashion. Chapter 1 argued that varying definite readings in relative clauses—which have been analyzed as cases of non-local scope—can be derived through an alternative compositional pathway that does not involve scope-shifting at all: a functional interpretation of the definite DP, which denotes a function of type $\langle e, e \rangle$ from individuals to individuals. Chapter 2 extended the functional analysis to the internal composition of functional relative clauses and argued that a parsimonious analysis of functional definite DPs becomes available if, following Von Stechow et al. (2014), the notion of maximality invoked by the definite article is based on semantic strength. This allows for a unified analysis of principled and arbitrary variation readings. Moreover, this unified analysis also pro-

vides a principled account of how two different kinds of perceived variation (i.e., principled and arbitrary variation) can arise through a single mechanism. Chapter 3 established that variation readings in finite complement clauses are predicate sensitive in a way that cannot be straightforwardly attributed to non-local QR (or, more generally, non-local scope-shifting), and that the mechanism responsible for licensing these readings is better characterized in terms of the inferential profile of the embedding predicate—specifically, its capacity for licensing cumulative inferences. Thus, cumulative inferences provide another route through which variation can be captured without invoking scope-shifting.

It is important to emphasize that the three chapters do not argue that scope is necessarily clause-bounded, or that non-local scope never occurs. Rather, the claim is more modest: in both of the syntactic environments under consideration, variation readings are compatible with a pseudo-scope analysis, and the availability of that alternative analysis is independently motivated by data not easily accommodated under a scope-shifting approach. The net result is that variation data, on its own, cannot bear the argumentative weight it has been asked to carry in the recent literature on scope islands.

4.2 Constraining scope-taking out of relative clauses

4.2.1 The functional pseudo-scope account

The findings of Chapters 1 and 2 address the first syntactic environment: relative clauses. The motivation for the functional analysis in Chapter 1 comes not only from the fact that it can derive the varying definite reading without invoking non-local scope-shifting, but from a body of independent empirical evidence showing that the functional parse is required for a range of related data that a scope-shifting based account cannot accommodate. Once that evidence is taken into account, the varying definite reading does not necessitate any additional machinery beyond what is already independently needed.

A good deal of evidence for the functional parse comes not from the varying definite reading itself but from the behavior of structurally parallel relativized DPs in other contexts—contexts

where QR is either unavailable or generates the wrong truth conditions. Across several such cases in Chapter 1, a common thread emerges: variation of the referent of the definite arises where scope-shifting simply cannot be its source. The *vary* data makes this concrete: *the supervisor that each volunteer reported to varied*, on its intended reading, cannot be derived by scope-shifting the universal in the relative clause, since that configuration yields the wrong interpretation. Dependency readings, like *We revised the paper that each of us wrote*, present a different but convergent problem: scope-shifting again derives the wrong truth conditions. And varying definite readings without any embedded universal ('bare DPs') show that the universal force characteristic of these readings can arise even in the absence of a universal DP which can be scope-shifted. The specifics differ, but the conclusion is the same in each case: something other than scope-shifting is inducing variation of the referent of the definite. This convergence straightforwardly favors a functional analysis: if the functional parse is independently required for these other cases, the varying definite reading requires no additional mechanisms; the same ingredients that capture variation elsewhere extend to the core examples at no added theoretical cost. In contrast, an account invoking scope-shifting must be supplemented by separate analyses for each of these other cases. If variation is our diagnostic for wide-scope universal force, then these other instances of variation should be equally informative about the underlying mechanism delivering it; and what they collectively suggest is that the source of variation is the functional type of the DP itself, not the scope-taking behavior of the universal in the relative clause.

More generally, zooming out from the varying definite reading itself, the takeaway is that I have presented data that can help inform possible analyses of these readings, by extending the empirical gamut. Ideally, a parsimonious account should cover all cases of variation concerning relativized DPs, not just the varying definite reading that has been explored in the literature. Under this line of thought, the novel data points that are brought to the table can be viewed as desiderata for any mechanism that derives variation of definites.

Chapter 2 pushed the functional analysis a step further by examining the ambiguity between principled and arbitrary variation readings, two different kinds of variation observed with definites.

These two readings have sometimes been taken to reflect a genuine ambiguity at the level of the internal composition of the relative clause, with different internal structures generating the different meanings. The argument in Chapter 2 is that this is not the correct locus of the ambiguity. Both readings involve a functional parse of the relative clause that delivers a predicate of $\langle e, e \rangle$ functions. What distinguishes them is not the composition of the relative clause but the mechanism by which the definite article achieves reference to a function. This has two important implications. First, as pointed out above, this line of analysis explains two different kinds of perceived variation (i.e., principled and arbitrary variation) through a single mechanism. Second, it has implications for the theory of definiteness itself. More concretely, the appeal to strength-based maximality suggests that this view of definiteness deserves a more prominent role in the literature, as it has genuine utility in various empirical domains, including functional definite DPs. As a consequence, these kinds of definite DPs provide a novel argument, independent of the previous literature, for treating strength-based maximality as a more central notion underlying the semantics of the definite article.

4.2.2 Limitations

Chapter 1's functional analysis faces certain limitations. The claim that relative clauses are not necessarily scope islands is, at its core, a negative one. I argued that the data do not require positing non-local scope-shifting rather than arguing that non-local scope-shifting is impossible. The strongest version of the claim—that relative clauses are genuine scope islands for scope-shifting—would require showing that every case of varying definite readings can be derived through the functional analysis, and that scope-shifting analyses are either empirically inadequate or otherwise inferior in each case. Chapter 1 makes significant progress toward this goal but does not accomplish it entirely. As a result, the empirical conclusion that I drew was that cases of varying definite readings are *compatible* with the view that relative clauses are scope islands for universal quantifiers. In future work, it would be interesting to see if the conclusion drawn in that chapter can be strengthened to the stronger version: that relative clauses are, in fact, scope islands for scope-shifting. This would require further evidence and argumentation.

4.2.3 Future directions

Chapters 1 and 2 focused exclusively on functional interpretations of *definite DPs*. One avenue for future research could be to explore whether the functional approach extends to relative clauses headed by other kinds of expressions, like singular indefinites, plural indefinites, or other kinds of quantificational expressions (for example, DPs like *a supervisor that each volunteer reported to*). Exploring these kinds of questions would help to determine the availability of functional parses more generally, which I view as an important next step for the functional pseudo-scope account. Two other potential avenues for future work were briefly mentioned in Chapter 1. The first concerns the parallels between the data reported in that chapter and the wide-scope distributive readings reported by Chatain (2019). To account for the readings observed there, Chatain posited a covert universal operator which is similar in spirit to the one I posited in Chapter 1. That being said, the technical details of Chatain’s proposal differ from mine. It would be interesting to see if a unified analysis of these different phenomena is possible. Second, and relatedly, a reviewer for the manuscript in Chapter 1 points out that a connection in the literature could provide another potential justification for covert universal quantification: so called telescoping sentences (Sells et al., 1985; Roberts, 1989), as in (1).

- (1) a. Every chess set comes with a spare pawn. It is taped at the back. (Sells et al., 1985)
 b. Every actor came to the stage. He took his prize and made a speech.

The reviewer proposed that these kinds of sentence can be argued to provide evidence for a mechanism of covert universal quantification that could parallel what I argued for. Again, a detailed exploration of this connection would be a worthwhile endeavor for future research.

Finally, the theoretical conclusion of Chapter 2—that strength-based maximality, rather than unqualified uniqueness, is the operative notion in the semantics of functional definite DPs—suggests a broader research program in the theory of definiteness. If strength-based maximality is a fundamental notion for definiteness, as the findings here preliminarily suggest, then the implications extend well beyond the empirical domain of functional relative clauses. The present dissertation

has established a new argument for strength-based maximality from a novel empirical domain; the task of integrating that argument into a comprehensive theory of definiteness remains for future work.

4.3 Constraining scope-taking out of finite complement clauses

4.3.1 The predicate sensitivity of variation

The findings of Chapter 3 address the second syntactic environment: finite complement clauses. We presented the key empirical finding of Chapter 3 as the cumulation-variation correspondence: an embedding predicate licenses variation readings if it licenses non-local cumulation with plural subjects. This cumulation-variation correspondence is not an incidental observation but a substantive empirical generalization—one that constrains the space of possible analyses of variation readings. If variation were simply a matter of whether an embedded universal DP can take scope out of a finite complement clause, the identity of the embedding predicate would be irrelevant: locality constraints on scope-shifting are not generally sensitive to the lexical content of the embedding predicate, and accounts that posit this restriction (cf. Barker 2022) do so in an admittedly stipulative manner. The fact that predicate identity matters, and that it does so in a way that tracks an independently observable property of the predicate (i.e., its capacity to license cumulative inferences) constitutes a strong *prima facie* argument against an analysis invoking scope-shifting and in favor of one grounded in the inferential profile of the embedding predicate. More importantly, though, the findings from our acceptability rating experiments indicate that variation readings with universal quantifiers, variation readings with negative quantifiers, and non-local cumulation show the same predicate sensitivity. While variation readings with universal quantifiers and non-local cumulation might be amendable to a scope-shifting account, variation readings with negative quantifiers are not: no scope-shifting account is possible for variation readings with negative quantifiers, as scope-shifting the negative DP would derive the wrong truth conditions. This means that whatever route to variation is needed for negative quantifiers could also plausibly explain variation for universal quantifiers.

In contrast, the cumulative inference approach naturally explains predicate sensitivity because cumulative inferences are derived from the inferential profile of the embedding predicate, given a context. It also extends naturally to both types of embedded quantifier—universal and negative—in virtue of the inferential profile applying *mutatis mutandis* in each case: the same mechanism that licenses variation with universals does so with negative quantifiers as well. If instead, variation readings with universals were derived via scope-shifting and variation readings with negative quantifiers were derived by a distinct mechanism, the parallel predicate sensitivity in each case would be a striking and unexplained coincidence. Therefore, the more parsimonious conclusion is that both readings derive from the same mechanism, one that is inherently predicate sensitive. The cumulative inference analysis discussed in Chapter 3, following Harada (2023), is precisely such a mechanism: it is grounded in the lexical semantics of the embedding predicate insofar as it references its inferential profile. Furthermore, this mechanism interfaces with world knowledge and context in a way that licenses gradient acceptability, and applies uniformly to both types of embedded quantifiers.

One aspect of the experimental findings that deserves further comment here is the gradient nature of the predicate sensitivity effect. The acceptability ratings for variation readings lie in the middle of the Likert scale even for predicates that most robustly licensed them (according to my intuitions), and the effect size—while statistically significant—was relatively small. This could, in principle, be taken as a weakness of the experimental case. I would suggest, however, that the gradient pattern is not merely a methodological artifact but a substantive finding in its own right. It is consistent with the view that the inferential profile of an embedding predicate is not a categorical, lexically-encoded property, but one that emerges from the interaction of lexical meaning with context and pragmatic reasoning.¹ If cumulative inferences can, in principle, be licensed or

¹Two further findings of Philipp and Zimmermann (2025), who compare the availability of scope-shifted readings in English and German experimentally, converge with the position taken here. Even though they focused on different structural configurations than I focused on in this thesis, the parallels concerning the empirical patterns are at least worth mentioning. They report substantial variation in scope interpretation between speakers of the same language, and a rise in acceptability of the scope-shifted reading when the context biases towards it. Both are expected if the relevant readings are licensed through inference over world knowledge rather than fixed by grammar, since world knowledge varies across individuals and inferential support varies across contexts. Their results thus offer independent support for treating gradience as a substantive property of the phenomena of variation and/or scope-taking, rather than

suppressed by contextual factors—a proposal compatible with Haslinger (2021)’s suggestion that the QUD plays a role in licensing cumulative construals more generally—then the gradient pattern of predicate sensitivity follows naturally. Gradient acceptability is expected precisely because the inferential profile of a predicate is not fully determined by its lexical content alone. This connection between lexical semantics and contextual pragmatics is an area worth developing in future research.

4.3.2 Limitations

The most pressing limitation of Chapter 3 concerns the gap between the empirical generalization and its theoretical implementation. The cumulation-variation correspondence is robustly established by the experimental results, but the analysis offered in that chapter is explicitly preliminary. The simple-minded extension of the cumulative inference analysis to variation readings—treating them as cumulative inferences with a hidden plurality derived from the witnesses of the singular indefinite—faces serious challenges: for starters, it makes predictions about the behavior of singular indefinites that are not borne out. The analysis is therefore best understood as a research agenda for future work and an appropriate basis for further inquiry, rather than an adequate account. It establishes the right empirical connections but leaves the compositional implementation as an open problem.

Moreover, the experimental methodology of Chapter 3 also has limitations that deserve acknowledgment. While the experimental results were clear, they also reveal that intuitions about variation readings and readings involving non-local cumulation are subtle. For instance, sentences for which, in my intuition, a variation reading is inaccessible still received ratings within the middle of the Likert scale. Maybe this should not surprise us, because all the sentences in our experiments were grammatical sentences of English, and the intuitions are subtle and rely on having completely absorbed the context. We would expect a lot of noise simply because not every participant will be equally attentive to the provided contexts. But the variation in the results may also be indicative that whether or not a given embedding environment licenses variation or cumulation

as noise.

depends on how one enriches the context, meaning it might be part of the signal, rather than the noise. Future experiments could try different and more elaborate contextual manipulations to try to get better control over how participants may be filling pragmatic detail in the situations they are provided with. Finally, the gradient nature of the results, discussed above, is compatible with an analysis of variation readings where contextual factors play an important role in licensing these readings. To this end, more fine-grained experimental work—manipulating the discourse context more systematically and varying the nature of the QUD—would be needed to characterize the relative contributions of lexical semantics and context.

4.3.3 Future directions

The limitations just noted point toward several directions for future research. The most urgent, from a theoretical perspective, is the development of an adequate compositional analysis of variation readings that captures the cumulation-variation correspondence without positing that singular indefinites can denote hidden pluralities. As pointed out above, one promising direction involves exploring the correlation between those embedding predicates that license variation readings and those that can take inanimate subjects. This connection received only brief discussion in Chapter 3 but deserves more extended discussion, and more attention in future work. Chapter 3 observed that the embedding predicates that license variation readings tend to be those that also take inanimate subjects. I suspect that this is not a coincidence. There is reason to think that inanimate subjects correlate with a particular use of the embedding predicate in which its content concerns an objective, evidential relation between a state of affairs and a proposition, rather than a psychological attitude taken by an agent toward that proposition. *Make sure*, on this view, expresses a relation between a set of conditions and their consequent—a relation that is naturally ‘cumulative’ in the relevant sense—while *claim* expresses a kind of communicative act performed by an agent. Whether this lexical-semantic generalization can be made precise in a way that provides an explanation of *why* the cumulation-variation correspondence holds is a question for future work; but the observation at least points towards the possibility of a deeper explanation concerning which

predicates belong to which class. This raises another natural topic for future research. While the cumulation-variation correspondence (and the appeal to cumulative inferences) gives us a sense of which predicates license variation readings, it doesn't tell us *why* these predicates license variation readings. To this end, future work should try to identify the relevant semantic properties for variation readings, to provide a more explanatory account.

4.4 Towards a Unified Picture: pseudo-scope and its varieties

One of the goals of the introduction was to situate the individual findings of the three chapters within a broader theoretical picture—to show that they are not simply three disconnected empirical observations, but rather three reflections of a single, unifying insight. That insight is that the grammar is equipped with multiple compositional pathways capable of generating variation, and that these pathways do not always involve scope-shifting. The term 'pseudo-scope', introduced to cover this family of mechanisms, is deliberately theory-neutral and picks out numerous mechanisms by the feature they share: mimicking the semantic effect of scope-shifting without themselves involving a scope-shifting mechanism. In other words, the term 'pseudo-scope' is not meant to pick out any specific compositional implementations.

What is notable, in retrospect, is how different the pseudo-scope mechanisms identified in this dissertation are from one another. The cumulative inference mechanism is rooted in the inferential profile of the embedding predicate and involves cumulating individual contributions to derive variation. Functional interpretations are rooted in the meaning of the definite DP itself and involves the definite article selecting a function rather than an individual. These are not two instances of a single operation but two genuinely different ways in which variation can arise—and this variety is itself theoretically significant (not to mention other pseudo-scope mechanisms that have been proposed in the literature). It suggests that the grammar does not simply have one pseudo-scope mechanism in reserve, to be deployed when scope-shifting is unavailable. Rather, different construction types license variation through mechanisms that are tailored to their own compositional properties. This, in turn, reinforces the methodological point: the variation diagnostic can be unreliable

because there are several potential pseudo-scope mechanisms, and several potential compositional pathways, that one should keep in mind when analyzing variation readings.

4.5 Broader limitations and future directions

Finally, a broader limitation concerns the dissertation's focus on English. The three chapters draw entirely on data from English, and the theoretical conclusions—about the predicate sensitivity of variation, the availability of functional interpretations, and the semantics of the definite article—are proposed as claims about the grammar of English without systematic cross-linguistic comparison. That being said, there is no a priori reason to think that the pseudo-scope mechanisms identified here are restricted to English; functional interpretations of relativized DPs have been discussed across multiple languages, and cumulative predication is a pervasive phenomenon. At the same time, whether the observed predicate sensitivity of variation readings holds in other languages with different aspectual profiles, different verb classes, and different morphosyntactic properties is a genuinely open question. Similarly, it would be interesting to look at a cross-linguistic comparison concerning the availability of varying definite readings (and the range of other readings discussed in relation to relativized DPs). Sharvit (1999b), for instance, argues on the basis of Hebrew that the alternation between traces and resumptive pronouns conditions the availability of 'purely' functional/principled variation and varying definite/arbitrary variation readings, with resumptives licensing such readings only when the relative clause is embedded in an equative sentence. This bears directly on Chapter 2: if the availability of arbitrary variation is sensitive to the form of the relative clause-internal dependency, then languages with a productive resumption strategy offer clues about the internal composition of relativized DPs (something that English does not overtly provide).

The literature on scope rigidity bears on the central argument more directly still. It has long been observed that languages such as Japanese, Korean and Mandarin—and, in more restricted configurations, German, Dutch and Hungarian—resist scope-shifted readings that English permits (Kuroda 1970; Hoji 1985; Aoun and Li 1993; Szabolcsi 1997; Bobaljik and Wurmbrand 2012,

a.o.), with recent work arguing that rigidity is a property of particular configurations rather than of whole languages (though, see Philipp and Zimmermann 2025 for experimental evidence that there is no categorical cross-linguistic difference between English and German, but only a gradual one, with English more readily allowing for scope-shifted readings than German). This matters here because the argument developed in this dissertation makes a cross-linguistic prediction: if variation can arise through mechanisms that do not involve scope-shifting, variation readings should be attested even where non-local scope-shifting is independently unavailable. A scope-rigid language is therefore, in principle, a sharper testing ground for the pseudo-scope hypothesis than English is, precisely because the scope-shifting analysis is presumed to be unavailable there from the outset. Conversely, if the availability of variation readings tracked the availability of scope-shifting cross-linguistically, this would count as evidence against the present picture. A parallel point holds for Chapter 2: languages that morphologically distinguish types of definite article, German being the best-studied case (e.g. Schwarz 2009), offer a way to probe whether the strength-based definite is implicated in arbitrary variation readings.

Other directions for future work concern the broader taxonomy of pseudo-scope mechanisms and the conditions under which each is available. This dissertation has identified two potential mechanisms—cumulative inferences and functional interpretations—and has argued that each is operative in a distinct syntactic environment. But these are not the only pseudo-scope mechanisms the grammar employs. A more comprehensive taxonomy of pseudo-scope mechanisms, along with a principled account of which mechanisms are available in which environments, would be a significant contribution to the literature on quantifier scope.

In addition, and perhaps most directly relevant to the broader debate about locality constraints on quantifier scope, the findings of this dissertation call for a reconsideration of how the evidence for non-local scope is marshaled in the literature. If the variation diagnostic is confounded by pseudo-scope mechanisms in both relative clauses and finite complement clauses—the two environments most often cited as counterexamples to clause-boundedness—then the empirical case for non-local scope is considerably weakened. Of course, this does not show that clause-boundedness

is correct; it just shows that the evidence most often adduced against it is compatible with a grammar where clause-boundedness is a real constraint on scope-taking, and this constraint is supplemented by pseudo-scope mechanisms. Adjudicating between the two views would require finding empirical diagnostics that discriminate between them—properties of variation readings that would be surprising under a pseudo-scope account but expected under a scope-shifting account, or vice versa. Developing such diagnostics is one of the most important next steps for this line of research.

4.6 Conclusion

This dissertation began from a simple observation: the variation diagnostic is less reliable than has been assumed. This has important consequences since this diagnostic has served as a primary cue for investigating the scope-taking abilities of universal quantifiers. The three chapters developed this observation into a systematic empirical and theoretical argument that motivated the need to posit pseudo-scope mechanisms, instead of non-local scope-shifting mechanisms. The broader lesson is methodological as much as it is empirical: careful attention to the full range of compositional mechanisms that can produce variation is indispensable for making progress on the question of whether, and how, quantifier scope is constrained by syntactic locality (or other constraints).

Perhaps more importantly, the dissertation contributes to a growing body of work suggesting that the relationship between scope and variation is more nuanced than a simple equivalence. Variation is a surface phenomenon; scope is a theoretical construct. The two are connected, but the connection is mediated by a range of compositional mechanisms that can each, under the right conditions, produce variation without scope-shifting. Keeping this in mind—and remembering that all that varies is not scope—is the main takeaway that I am advocating for when it comes to future work on quantifier scope, variation, and this particular syntax-semantics interface.

Part IV

Appendix

Supplementary materials for Chapter 3

A.1 Experimental stimuli

All the experimental stimuli (including instructions) are available at the following OSF page
<https://osf.io/zgbmp/>

B.2 Model summaries

Predicting acceptability ratings for variation readings with negative quantifiers from acceptability ratings for variation readings with universal quantifiers

Cumulative Link Mixed Model fitted with the Laplace approximation

Formula: $\text{Response} \sim \text{VariationIndex} + (1 \mid \text{participant}) + (0 + \text{VariationIndex} \mid \text{participant})$

where ‘Response’ is the raw acceptability ratings for variation readings with negative quantifiers and ‘VariationIndex’ corresponds to the standardized mean acceptability ratings for each predicate from the experiment looking at variation readings with universal quantifiers

Random Effects			
Groups	Name	Variance	Std. Dev.
participant	(Intercept)	12.897	3.591
participant	VariationIndex	0.579	0.761

Table 1: Random effects for variation readings with universals and negative quantifiers. Number of groups: participant 40.

Coefficients					
	Estimate	Std. Error	z value	p value	
VariationIndex	0.635	0.175	3.62	0.00029	***

Table 2: Fixed effects for variation readings with universals and negative quantifiers. Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

Threshold Coefficients			
	Estimate	Std. Error	z value
1 2	−2.405	0.583	−4.13
2 3	−1.124	0.580	−1.94
3 4	−0.382	0.580	−0.66
4 5	0.975	0.580	1.68
5 6	2.674	0.583	4.58

Table 3: Threshold of coefficients for variation readings with universals and negative quantifiers.

Predicting acceptability ratings for cumulation with negative quantifiers from acceptability ratings for cumulation with universal quantifiers

Cumulative Link Mixed Model fitted with the Laplace approximation

Formula: $\text{Response} \sim \text{CumulationIndex} + (1 \mid \text{participant}) + (0 + \text{CumulationIndex} \mid \text{participant})$

where ‘Response’ is the raw acceptability ratings for cumulation with negative quantifiers and ‘CumulationIndex’ corresponds to the standardized mean acceptability ratings for each predicate from the experiment looking at cumulation with universal quantifiers

Random Effects			
Groups	Name	Variance	Std. Dev.
participant	(Intercept)	3.92	1.98
participant	CumulationIndex	1.46	1.21

Table 4: Random effects for cumulation with negative quantifiers and cumulation with universals. Number of groups: participant 39.

Coefficients					
	Estimate	Std. Error	z value	p value	
CumulationIndex	1.107	0.227	4.87	1.1×10^{-6}	***

Table 5: Fixed effects for cumulation with negative quantifiers and cumulation with universals. Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

Threshold Coefficients			
	Estimate	Std. Error	z value
1 2	−1.908	0.329	−5.79
2 3	−0.822	0.326	−2.52
3 4	−0.155	0.326	−0.48
4 5	0.774	0.327	2.37
5 6	1.948	0.331	5.89

Table 6: Threshold of coefficients for cumulation with negative quantifiers and cumulation with universals.

Predicting acceptability ratings for variation readings with universal quantifiers from acceptability ratings for cumulation with universal quantifiers

Cumulative Link Mixed Model fitted with the Laplace approximation

Formula: $\text{Response} \sim \text{CumulationIndex} + (1 \mid \text{participant}) + (0 + \text{CumulationIndex} \mid \text{participant})$

where ‘Response’ is the raw acceptability ratings for variation readings with universal quantifiers and ‘CumulationIndex’ corresponds to the standardized mean acceptability ratings for each predicate from the experiment looking at cumulation with universal quantifiers

Random Effects			
Groups	Name	Variance	Std. Dev.
participant	(Intercept)	6.69	2.59
participant	CumulationIndex	1.23	1.11

Table 7: Random effects for variation readings with universals and cumulation with universals. Number of groups: participant 40.

Coefficients					
	Estimate	Std. Error	z value	p value	
CumulationIndex	1.214	0.214	5.68	1.3×10^{-8}	***

Table 8: Fixed effects for variation readings with universals and cumulation with universals. Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

Threshold Coefficients			
	Estimate	Std. Error	z value
1 2	-3.2959	0.4285	-7.69
2 3	-1.8477	0.4208	-4.39
3 4	-1.0297	0.4187	-2.46
4 5	0.0402	0.4176	0.10
5 6	1.8603	0.4197	4.43

Table 9: Threshold of coefficients for variation readings with universals and cumulation with universals.

Predicting acceptability ratings for variation readings with negative quantifiers from acceptability ratings for cumulation with universal quantifiers

Cumulative Link Mixed Model fitted with the Laplace approximation

Formula: $\text{Response} \sim \text{CumulationIndex} + (1 \mid \text{participant}) + (0 + \text{CumulationIndex} \mid \text{participant})$

where ‘Response’ is the raw acceptability ratings for variation readings with negative quantifiers and ‘CumulationIndex’ corresponds to the standardized mean acceptability ratings for each predicate from the experiment looking at cumulation with universal quantifiers

Random Effects			
Groups	Name	Variance	Std. Dev.
participant	(Intercept)	13.42	3.66
participant	CumulationIndex	1.28	1.13

Table 10: Random effects for variation readings with negative quantifiers and cumulation with universals. Number of groups: participant 40.

Coefficients					
	Estimate	Std. Error	z value	p value	
CumulationIndex	0.736	0.226	3.26	0.0011	**

Table 11: Fixed effects for variation readings with negative quantifiers and cumulation with universals. Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

Threshold Coefficients			
	Estimate	Std. Error	z value
1 2	−2.472	0.596	−4.15
2 3	−1.175	0.593	−1.98
3 4	−0.422	0.592	−0.71
4 5	0.970	0.593	1.64
5 6	2.727	0.596	4.57

Table 12: Threshold of coefficients for variation readings with negative quantifiers and cumulation with universals.

Predicting acceptability ratings for variation readings with negative quantifiers from acceptability ratings for cumulation with negative quantifiers

Cumulative Link Mixed Model fitted with the Laplace approximation

Formula: Response ~ NegCumulationIndex + (1 | participant) + (0 + NegCumulationIndex | participant)

where ‘Response’ is the raw acceptability ratings for variation readings with negative quantifiers and ‘NegCumulationIndex’ corresponds to the standardized mean acceptability ratings for each predicate from the experiment looking at cumulation with negative quantifiers

Random Effects			
Groups	Name	Variance	Std. Dev.
participant	(Intercept)	13.69	3.70
participant	NegCumulationIndex	1.63	1.28

Table 13: Random effects for variation readings with negative quantifiers and cumulation with negative quantifiers. Number of groups: participant 40.

Coefficients					
	Estimate	Std. Error	z value	p value	
NegCumulationIndex	0.768	0.247	3.11	0.0018	**

Table 14: Fixed effects for variation readings with negative quantifiers and cumulation with negative quantifiers. Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

Threshold Coefficients			
	Estimate	Std. Error	z value
1 2	−2.516	0.602	−4.18
2 3	−1.203	0.599	−2.01
3 4	−0.435	0.598	−0.73
4 5	0.979	0.599	1.64
5 6	2.759	0.602	4.58

Table 15: Threshold of coefficients for variation readings with negative quantifiers and cumulation with negative quantifiers.

Acronyms

- **QR**: Quantifier **R**aising
- **LF**: Logical **F**orm
- **CVC**: Cumulation **V**ariation **C**orrespondence
- **CNPC**: Complex Noun **P**hrase **C**onstraint
- **DP**: Determiner **P**hrase
- **NP**: Noun **P**hrase

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